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Feature-based Propensities for Counterfactual Learning to Rank in Long-Tail Settings

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Abstract

Unbiased learning-to-rank (LTR) aims to learn ranking models from biased implicit feedback such as clicks. This thesis studies the effectiveness of prevalent propensity estimation methods in sparse datasets where query-document pairs are observed few times. We show that frequency-based propensity estimates can introduce bias in prevalent policy learning methods.

To address this issue, we propose feature-based propensity estimation, which generalizes across documents by estimating propensities based on query-document features. We theoretically show that feature-based propensity estimation can be performed under reasonable assumptions, and leads to decreased variance in propensity estimates. Experiments in semi-synthetic data show that neural feature-based propensity can recover logging policy propensities more effectively than frequency-based baselines and significantly improve performance of counterfactual LTR methods in diverse settings. We also investigate two-tower models for estimating position bias and relevance, offering a novel approach to bias and relevance estimation. This work provides theoretical and empirical support for feature-based propensity estimation and two-tower components as tools for improving unbiased LTR under sparse, long-tail data conditions.

Chapter 1

Introduction

Unbiased learning-to-rank (LTR) is characterized by the use of implicit feedback to optimize ranking models [26, 40, 76, 39]. A common source of implicit feedback in web search and recommendation is clicks. Clicks are well known to be a biased signal of user preference [57, 73, 72, 1]. This is because users are only able to click relevant items they have examined [18, 20, 58, 35]. As a result, factors which affect how often users will examine items, such as position, affect the number of clicks [57, 73, 72, 1]. In order to learn the underlying relevance of items, systems therefore have to learn to distinguish bias and relevance [13, 9, 8, 7, 17, 77, 4].

There are roughly two families of models which are applied for bias mitigation in LTR, click modelling, and counterfactual LTR [54]. Click-modelling is a family of approaches which attempts to predict clicks by training models which separate bias and relevance components. In recent years two-tower models have arisen as a popular click modelling approach, especially in industry settings [24, 33, 83]. These models work by optimizing two separate neural networks, also called *towers*. The first tower is referred to as the relevance tower, taking into account features which signify the relevance of a document for a given query, e.g., the document’s text, or classic LTR features such as BM25. The second tower is referred to as the bias tower and estimates bias based on features that are not indicative of relevance, but might affect clicks. This includes primarily the position in which the document is presented, but also contextual features, such as the user’s device (e.g., mobile or desktop) or the relative size of the presented item [16, 80]. As verified by Hager et al. [32], in standard conditions the bias tower is able to recover the ground-truth position bias. Two main issues with two-tower models are that they suffer from a lack of theoretical guarantees with respect to finding accurate relevance estimates, as well as that they can only be applied as a relevance estimate, which means they cannot directly learn to be fair or diverse [54, 32].

An alternative approach to unbiased LTR is counterfactual LTR, a method based on counterfactual estimation which aims to learn how the ranking model should have behaved to achieve the best reward based on logged data. In order to learn these counterfactual models, a debiased reward is constructed, which estimates user behavior through combining a user model with empirical estimates of user behavior. Their main advantage is that they provide an explicit causal interpretation: under the required assumptions, they optimize an unbiased estimate of the target policy reward. Additionally, they are able to learn a policy which can optimize for other objectives than relevance [55, 26, 68].

A traditional method to correct for biases is the use of inverse-propensity-scoring (IPS) estimators to correct position bias, as introduced by Joachims et al. [40] and Wang et al. [76]. This model works by inversely weighting the observed click data based on user examination probability. Under accurate propensity estimates the model is able to form a correct estimate of ranking metrics, with mathematical unbiasedness guarantees when propensities are correctly estimated. An issue with the IPS estimator is that it can suffer from high variance [55, 65, 19].

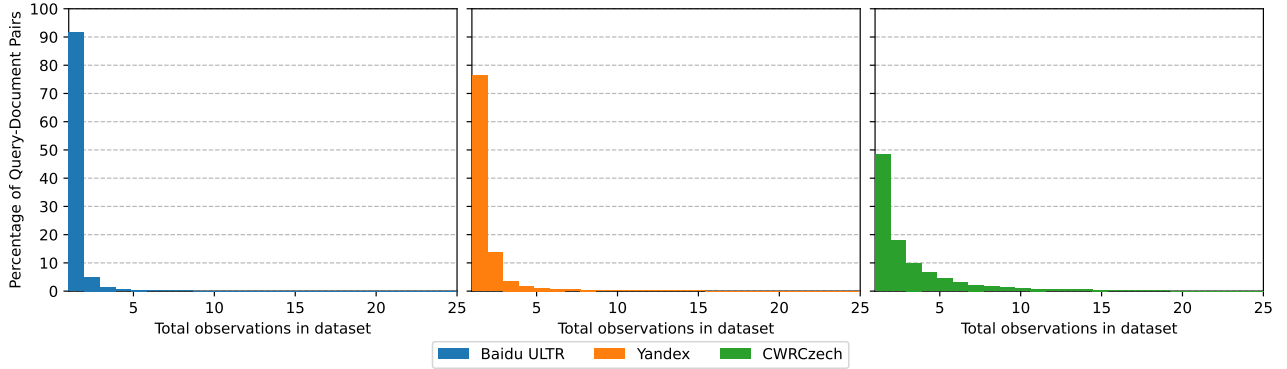


Figure 1.1: Query observation count distributions of three real-world web-search click datasets, showing that the query observation distributions are skewed towards single observations.

Recent work by Oosterhuis [55] has proposed a direct-method (DM) for policy learning for unbiased LTR, which works by learning a policy based on direct estimates of document relevance, trained using a bias-corrected regression loss. This estimate allows a model to learn from documents without observations, which is not possible for IPS. Furthermore, they introduce the doubly-robust (DR) estimator for unbiased LTR. This method combines inverse propensity estimation with the direct method by correcting relevance estimates with click-data informed IPS estimation. This allows the model class to have a much smaller variance and faster convergence, while retaining the mathematical unbiasedness conditions under correctly estimated propensities [55].

As discussed, unbiasedness of counterfactual LTR methods is conditional on having an accurate estimate of propensities. The components of propensities directly affect the effectiveness of debiasing, and accuracy of the final model. A widely used component of propensities in user models is the position bias, which indicates increase in click probability as a result of higher placement. Position bias is traditionally estimated through the application of intervention harvesting techniques [22, 3]. Recent work by Oosterhuis and de Rijke [57] has shown that propensities which use the policy placement probability in addition to position bias are more effective. This policy-based propensity is applied in the state-of-the-art IPS, DM, DR methods. Correct estimation of policy-based propensities relies on two core components: (i) bias in clicks as a result of displayed position; and (ii) the probability for the logging policy to place a document at each position [55].

An issue that arises with determining propensities is that in many industry settings, the logging policy may not be available, or be subject to changes over time [55, 32]. Historically, propensities have been estimated by looking at logging policy behavior during data gathering. The prevalent approach in this context is to estimate placement probabilities using an observed-frequency-based estimate [4, 72, 49, 55]. This estimate is commonly used as it is simple to construct and fast to compute.

A further issue that can arise with frequency-based propensity estimates is that for items with sparse observations, logging policy behavior estimates may be less precise. Long-tail query distributions naturally arise in many real-world settings, including web-search, and personalized search. The sparse nature of real-world click-datasets is illustrated in Figure 1.1, where we show that in three real-world datasets of web-search logs, a large portion of query-document pairs included in the dataset are only observed a single time [21, 85, 74]. In other settings, such as personalized search, including user information in queries can lead to highly unique query identifiers, making long-tailed query distributions inevitable. In all long-tail settings, it is infeasible to obtain accurate estimates of placement frequencies through these count-based methods, and consequently, it is equally infeasible to obtain accurate propensities, potentially resulting in bias for any propensity-based counterfactual LTR estimate.

In this work, we address this problem by investigating the estimation of propensity values by relying on features of query-document pairs. We theoretically analyze the limitations of current frequency-based estimation, as well as the resulting bias in LTR models. We propose the feature-based propensity estimator, which computes propensities based on query-document features, and can provide an accurate propensity estimate independent of observation frequencies.

In addition to improving placement estimates, we also investigate the other components of counterfactual LTR methods, which are position bias and relevance estimates. Traditionally, these have been estimated using separate intervention harvesting techniques and a debiased relevance estimate, meaning different models have to be tuned and optimized to form separate estimates [3, 55]. The two-tower model class provides a potential simplification to this paradigm, as the optimized bias tower offers an estimate of position bias, and the relevance tower serves as a debiased relevance estimate. In this work, we study the extent to which neural click-modelling approaches can be applied for position bias and relevance estimation.

To summarize, we investigate the estimated components of policy-learning methods for counterfactual LTR. We investigate issues that currently arise when computing propensities, and propose the use of feature-based propensity estimation. We also consider how the representational capacities of two-tower models can be applied for estimating position bias and document relevance in counterfactual LTR. Our work addresses the following two core research questions:

Does feature-based propensity estimation lead to more effective learning to rank in settings with long-tail query distributions?

How can two-tower models be used in conjunction with doubly robust estimation for more effective learning-to-rank methods?

To investigate this, we tackle the following research questions:

- RQ1:** What are the theoretical challenges that long-tail query distributions pose for propensity estimation in counterfactual estimation?
- RQ2:** Under what assumptions can propensity estimation be performed based on query-document features?
- RQ3:** How effectively can the feature-based propensity estimation method recover the true logging policy propensities in semi-synthetic experiments?
- RQ4:** To what extent can feature-based propensity estimation improve the performance of counterfactual learning to rank methods in semi-synthetic experiments?
- RQ5:** Can the position bias and document relevance for doubly-robust estimation be estimated using two-tower models for increased performance in semi-synthetic click data?

We first investigate the prevalent frequency-based propensity estimation, and find that it leads to bias that is inversely proportional to the number of samples taken (**RQ1**). Following this, we show that if we assume that the logging policy is based on features, we can design feature-based propensity estimation methods, with favorable unbiasedness conditions (**RQ2**). We then demonstrate that both clustering methods and direct neural propensity estimation can recover the logging policy propensities more effectively in synthetic data (**RQ3**). After that, we evaluate performance on semi-synthetic data, and find that feature-based IPS methods can significantly improve performance of counterfactual LTR methods (**RQ4**). Finally, we investigate the effectiveness of two-tower models for estimating bias and relevance for counterfactual LTR, and show that both the position bias and relevance estimates can effectively be applied in counterfactual LTR (**RQ5**). Our contributions are the following:

- A theoretical analysis of propensity estimation in unbiased LTR, showing consistent bias in frequency-based propensity estimation.
- An extension of propensity estimation that allows for the use of generalized estimation based on query-document features with favorable unbiasedness conditions.
- A systematic study of the effectiveness of feature-based propensity estimation.
- A novel approach for bias and relevance component estimation with two-tower models for counterfactual LTR.
- A comparative study of bias and relevance estimation techniques for counterfactual LTR.

The remainder of this work is structured as follows. Initially, we discuss relevant existing work on click-based and unbiased LTR as well as logging policy estimation in related fields. We then formalize learning from user interactions, counterfactual LTR and two-tower models. Following this, we theoretically analyze the bias in propensity estimation, and show under what assumptions we can perform propensity estimation based on features. We then perform semi-synthetic experiments to study the effectiveness of feature-based propensity in recovering oracle propensities, as well as the effect of feature-based propensity estimates on performance of counterfactual LTR in different dataset settings. Finally, we evaluate the effectiveness of estimating position bias and relevance using two-tower models for counterfactual LTR in semi-synthetic data.

Chapter 2

Related Work

2.1 Counterfactual Learning-to-Rank

The field of counterfactual LTR considers the learning of policies from clicks by capturing interaction biases and optimization methods which can correct for them [59, 71, 40, 76]. A traditional approaches for learning a debiased policy is IPS [55, 77, 72, 14, 52, 39]. This method relies on the ratio between taken actions of the policy we are currently optimizing and the logging policy used during training to compute a propensity score [81, 55]. Work in counterfactual LTR focuses on capturing different types of interaction bias. A traditional component of this bias is position bias, which is measured using EM optimization [4], intervention harvesting [3, 22], or randomization [2]. Work by Oosterhuis and de Rijke [56] generalized the IPS estimator to correct for item-selection bias of the logging policy, showing that this is an important bias factor for top- k ranking settings. Further work by Oosterhuis and de Rijke [57] proposed an IPS method which considers the updating of the logging policy during gathering of data.

Counterfactual LTR also allows for optimizing for different objectives, including fairness and safety. Works by Singh and Joachims [68], Morik et al. [51], Yadav et al. [79] apply IPS to optimize fair distributions of exposure over items. Jagerman et al. [36] consider safe deployment of LTR models by bounding model performance. Gupta et al. [25] propose the learning of safe counterfactual LTR models by minimizing exposure-based risk.

A key issue in the practical application of IPS-based methods is that they may suffer from high variance [55]. Recent works have attempted to alleviate these issues by proposing new optimization techniques. Work by Wang et al. [75] introduced Ratio-Propensity-Scoring (RPS), which applies weights to pairs of clicked and non-clicked items by their ratio between the propensities, introducing a bias-variance trade-off. Work by Oosterhuis [55] has proposed the DM and DR methods for LTR. The DR method combines IPS estimates with a regression model, leading to lower variance and improved sample complexity [27, 28]. Recent work by Liang et al. [45] proposes correcting logging policy behavior with document similarity to improve effectiveness in sparse-data settings.

Several limitations remain in counterfactual LTR when the assumptions underlying the propensity estimates are violated. First, if the user model that is assumed is not correct or estimated parameters are inaccurate, unbiasedness cannot be guaranteed [56, 72, 40, 55, 28]. Furthermore, the reward estimates are subject to variance, and may not be correct in sparse data settings [25, 54]. To be unbiased, an important challenge for the field is to optimize the learned propensities effectively in sparse-data settings. While a large amount of research has investigated effectively learning positional bias [4, 3, 22, 2], to the best of our knowledge, in the field of LTR, there has been no research into learning propensities in user models which apply item-selection bias of the logging policy, despite being applied in state-of-the-art approaches for LTR [56, 57, 45, 55]

2.2 Logging Policy Estimation

As described in the previous section, including the logging policy behavior in propensities is a relatively new concept in the field of counterfactual LTR. Theoretically, if the logging policy that gathered the data is available, we can apply it for debiasing immediately. However, in many settings, especially in industry, the logging policy is not available due to changing systems or business rules may prevent storage. When working with published industry datasets, it is also a common practice not to publish the logging policy that was used to generate the data, as is the case with many unbiased LTR datasets [85, 74, 21, 44, 48, 60]. In practice, it may also be easier to estimate logging policies, as it does not require any parsing or storage setup.

The challenge of estimating logging policies is not limited to LTR, and has previously been studied in the field of off-policy estimation in reinforcement learning [81, 14, 50, 46, 69]. Work by Raghu et al. [62] demonstrated the importance of effectively estimating logging policies, showing that large errors arise when behavior is not estimated correctly. They show that by applying simple clustering techniques for policy estimation, effective behavior estimation can be performed, leading to real-world performance increases. Works by Kato et al. [42], Narita et al. [53] have applied logging policy estimation in different settings, and found that estimating the logging policy can improve over the true logging policy by providing improved stability. Zhang et al. [81] investigates uncertainty-aware reweighting for propensities, based on observed frequency of documents, leading to more robust logging policy estimates.

An important challenge for logging policy estimation in LTR is the size of the action space. Current logging policy estimation methods treat placements for each query-document pair individually [55]. This leads to sparsity issues, where logging policy estimation is difficult due to low information. A similar issue is found in counterfactual estimation for recommender systems, which often have many possible actions [64, 66, 5, 67]. Recent work in reinforcement learning has considered reducing the size of action spaces through modelling embeddings of items for placements [70, 11, 65]. Saito and Joachims [65] propose a marginalized IPS estimator that reduces variance in large action spaces by learning an embedding over placements. By representing actions for placements of items with an embedding that captures it, they reduce the size of the action space, and thereby reduce variance. While an embedding needs to be generated for the field of reinforcement learning, a source of embeddings already exists for LTR, as query-document similarity features are already applied as the input of the target policy.

To summarize, there is a clear need for logging policy estimation methods in real-world unbiased LTR settings. Evidence from adjacent fields suggests that improved logging policy estimation can have positive effects. Recent approaches have shown that data sparsity issues from large placement spaces can be effectively resolved by applying embeddings of placements. These works provide a strong foundation for improved propensity estimation for unbiased LTR through improved logging policy estimation.

2.3 Two-Tower Models

In addition to counterfactual estimators, click models form a second major family of approaches for learning from biased interaction data [17, 54]. Rather than correcting observed rewards through propensity weighting, click models aim to model the click-generation process directly by decomposing clicks into relevance and bias components [17, 63, 24]. The position-based model is a traditional click model that models position and document relevance [63]. Work by Guo et al. [24] proposed a neural version of the PBM for LTR, referred to as the two-tower model.

The two-tower model is a popular approach in industry settings, combining strong empirical performance with a simple yet scalable implementation [33, 83, 43]. Work by Yan et al. [80]

investigates additive assumptions and explores alternatives for different user behavior settings. Work by Zhuang et al. [84], Wu et al. [78] investigates how additional context can be used to model bias towers. Hager et al. [29] compares pointwise inverse-propensity-scoring, which learns a direct relevance score, with two-tower models that separately model relevance and bias. Work by Hager et al. [31] provides a comparison of two-tower models with neural implementations of other click models, showing alternative user behavior patterns can be effectively modeled for improved performance on real-world data.

Recent work on two-tower models has focused on identifiability as a condition for the effectiveness of two-tower models in separating bias and relevance [32, 15, 23]. Work by Hager et al. [32] finds that two-tower models recover underlying bias effectively when user behavior is correctly specified. Fris and Hager [23] investigates identifiability and specifies a method for diagnosing identifiability for model-dataset combinations.

Two-tower models provide an important related line of work for counterfactual LTR, since they study whether relevance and presentation bias can be estimated jointly from clicks. This is closely related to the position-bias and relevance-estimation components required by DM and DR methods [55]. However, while prior work has studied two-tower models as click models and as alternatives to pointwise IPS [29], their use as component estimators within counterfactual LTR methods remains underexplored.

Chapter 3

Background

In this section, we provide a theoretical background to serve as a framework for further analysis. We start with learning from user interactions in general, and then define the problem definition. We then consider the unbiased learning methods for policy learning that are applied in this work. Finally, we detail two-tower models.

3.1 Problem Definition

In this work, we investigate learning of preference based on clicks for a document d based on a position (referred to as rank) $k \in \{1, 2, \dots, k\}$, at which d is displayed. We rely on a dataset of clicks based on user-issued queries q , upon which a ranking by the logging policy π_0 is generated for a set of documents. Preference is then based on the joint features of the query and document, also referred to as a query-document pair. To align with previous work and for brevity, our notation focuses on a single query, which can be easily extended to multiple queries or contexts [55, 40, 57]. The probability of user preference for document d for rank k is defined as $R_d = P(R = 1 \mid d)$. In the following section, we detail the user-model which is applied in this work to describe the probability of a click $C \in \{0, 1\}$ on ranked documents in terms of bias and relevance components.

3.1.1 User Behavior Assumptions

We follow the user model of Agarwal et al. [2] and further derived by Vardasbi et al. [72], which is commonly applied in LTR settings [28, 55, 2, 36, 40, 56, 57, 58, 68, 72, 75, 76]:

$$P(C = 1 \mid d, k) = \alpha_k P(R = 1 \mid d), \quad (3.1)$$

where $\alpha_k \in [0, 1]$ is the positional bias, the increase in click probability as result of a specific position. Position bias is commonly modeled as a result the examination assumption, which states that users are only able to click on items that they have observed [18, 77, 63]. Formally, position bias is a combination of observation at a rank, and click-through rates conditional on observation at a rank:

$$\alpha_k = P(O = 1 \mid k)P(C = 1 \mid R = 1, O = 1, k) = P(C = 1 \mid R = 1, k). \quad (3.2)$$

Some work also applies trust bias, which models clicks that occur despite irrelevance [55, 2]. In this work we do not apply trust bias, defining ϵ_k^- to be exactly equal to zero, this simplifies writing, and aligns with other related works [40, 76, 77]. Importantly, theory can easily be re-extended in future work by adding a trust bias component following [55].

To summarize, we apply a user model for click probability that captures observing the item at a specific rank, the probability of clicking on a relevant item we observe, and the relevance of the specific item.

3.1.2 Definition of the Learning-to-Rank Goal

We have now defined how we expect clicks to be generated based on relevance and bias. Based on this generation, we can now define the LTR goal. Suppose a set of documents $D = \{d_1, d_2, \dots, d_N\}$ that we wish to rank, and a ranking of size $K : y = [y_1, y_2, \dots, y_K]$. We then define a logging policy π , with $\pi(k | d)$ being the probability that a specific document d is placed at rank k by the logging policy. Ranking models then generally attempt to maximize the probability that a user clicks on an item that they prefer. Metrics for ranking are then generally a sum of item relevances, weighted by a set of weights ω_k , which depend on item positions [55]:

$$\mathcal{R}(\pi) = \mathbb{E}_{y \sim \pi} \left[\sum_{k=1}^K \omega_k R_{y_k} \right] = \sum_{d \in D} R_d \sum_{k=1}^K \pi(k | d) \omega_k. \quad (3.3)$$

A commonly used metric for LTR is discounted cumulative gain (DCG) [38], which is a weighted measure of the sum of the item relevance. The DCG is commonly applied in a normalized way, leading to normalized discounted cumulative gain (NDCG) [38, 55]:

$$DCG@K(y) = \sum_{k=1}^K \frac{R_{y_k}}{\log_2(k+1)}, \quad NDCG@K(y) = \frac{DCG@K(y)}{\max_{y'} DCG@K(y')}. \quad (3.4)$$

This metric can then be applied as a measure of ranking quality across the dataset. Oosterhuis [55] propose the expected click preference (ECP) metric, which relies on the user behavior assumptions, by applying bias weights $\omega_k = \alpha_k$:

$$\omega_d = \mathbb{E}_{y \sim \pi} [\omega_{k(d)}] = \mathbb{E}_{y \sim \pi} [\alpha_{k(d)}] = \sum_{k=1}^K \pi(k | d) \alpha_k. \quad (3.5)$$

As described in Oosterhuis [55], this metric corresponds to the number of items that are both clicked and preferred in the assumed user behavior model:

$$\begin{aligned} R(\pi) &= \sum_{d \in D} \omega_d R_d = \sum_{d \in D} P(R=1 | d) \sum_{k=1}^K \pi(k | d) P(C=1 | R=1, k) \\ &= \sum_{d \in D} \mathbb{E}_{y \sim \pi} [P(C=1, R=1 | d, k)] = \sum_{d \in D} P(C=1, R=1 | d). \end{aligned} \quad (3.6)$$

3.2 Position Bias Estimation in Unbiased Learning-to-Rank

The position bias, α_k , is the probability of observing a click at position k , as defined in Eq. 3.2. As discussed, unbiased LTR methods such as IPS and DR rely on the position bias estimate for debiasing. Therefore, it is important to accurately estimate the position bias for each rank $\hat{\alpha}_k$.

A very basic method of position bias estimation includes the use of the mean click-through rate at a specific position. Although easy to compute and optimize, this method is naive, as

it ignores the effect of logging policy confounding, which means that logging policies generally place more relevant items in higher positions. As a result, this method will generally result in an overestimation of the position bias for higher positions, unless collection is performed over a set of randomly ranked items.

To estimate position bias in an unbiased way from a dataset of clicks and positions, we can use a process referred to as intervention harvesting [3]. Intervention harvesting methods rely on co-occurrence of query-document pairs at different positions, which allows us to estimate a difference in click rate based on positions. Methods rely on click ratios between adjacent positions (adjacent pair), positions and a fixed rank (pivot rank), or all ranking positions (all pairs) [3, 30]. We refer to Agarwal et al. [3] for an explanation of prevalent intervention harvesting methods.

3.3 Inverse Propensity Scoring Estimation in Unbiased Learning-to-Rank

A traditional method for unbiased off-policy learning is IPS. We will now discuss how IPS can be defined for unbiased LTR, following Oosterhuis [55]. First, we define the propensity ρ_d which measures how well clicks on document d (at rank k) reflect relevance under the logging policy used to gather data π_0 :

$$\rho_d = \mathbb{E}_{y \sim \pi_0}[\alpha_{k(d)}] = \sum_{k=1}^K \pi_0(k | d) \alpha_k. \quad (3.7)$$

If the position bias and logging policy are unknown, the propensity can be estimated using an estimated logging policy $\hat{\pi}_0$ and position bias $\hat{\alpha}$:

$$\hat{\rho}_d = \sum_{k=1}^K \hat{\pi}_0(k | d) \hat{\alpha}_k. \quad (3.8)$$

In previous work, a clipping parameter $\tau \in (0, 1]$ is introduced for variance reduction [40, 69, 55]. In this work, we do not apply a clipping parameter, as we specifically focus on the effect of incorrectly estimating bias. We define the expected position weight per item based on the expected exposure under the target policy π , as well as its estimator:

$$\omega_d = \mathbb{E}_{c, y \sim \pi}[\omega_{k(d)}] = \mathbb{E}_{c, y \sim \pi}[\alpha_{k(d)}] = \sum_{k=1}^K \pi(k | d) \alpha_k. \quad (3.9)$$

$$\hat{\omega}_d = \sum_{k=1}^K \pi(k | d) \hat{\alpha}_k. \quad (3.10)$$

We can then define the IPS method, as introduced by Oosterhuis and de Rijke [57]. This inversely weights the click probability with the propensity to correct for the correlation between user preference and click probability [56, 55].

$$\hat{\mathcal{R}}_{\text{IPS}}(\pi) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} (c_i(d)). \quad (3.11)$$

Where we have a dataset of N click observations over our document set D , with a binary click signal $c_i(d) \in \{0, 1\}$ for each document observation.

Bias of the IPS Estimator We detail the bias of the IPS estimator and its gradient. Work by Oosterhuis [55] considered estimating propensities as described in Eq. 3.8, estimating the logging policy π_0 and position bias α_k separately. We reiterate previous work, which has shown that these estimators are unbiased if ρ_d and α_k are correctly estimated [55, 40, 72]. First of all, the IPS estimator (Eq. 3.11) has the following bias [55]:

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d. \quad (3.12)$$

In Appendix A, we reiterate the proof of this bias, we also show that the IPS estimator is unbiased when the position bias and propensity are correctly estimated:

$$((\forall d \in D, \rho_d = \hat{\rho}_d) \wedge (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\hat{R}_{\text{IPS}}] = R(\pi_\theta). \quad (3.13)$$

Gradient of the IPS Estimator A relevant factor for IPS estimation is whether the gradient over individual samples leads to the true reward. We derive the gradients of the reward (Eq. 3.6) and IPS estimator (Eq. 3.11) as follows:

$$\nabla_\theta \hat{R}_{\text{IPS}}(\pi_\theta) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} c_i(d) \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_\theta \pi_\theta(k \mid d), \quad (3.14)$$

$$\nabla_\theta R(\pi_\theta) = \sum_{d \in D} R_d \sum_{k=1}^K \alpha_k \nabla_\theta \pi_\theta(k \mid d). \quad (3.15)$$

Given the gradient, we can express the bias of the gradient of the IPS estimator with respect to the true reward as follows:

$$\mathbb{E}[\nabla_\theta \hat{R}_{\text{IPS}}] - \nabla_\theta R(\pi_\theta) = \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\alpha}_k - \alpha_k \right) \nabla_\theta \pi_\theta(k \mid d). \quad (3.16)$$

We provide the derivation of these gradients in Appendix A, and show that the gradient of the IPS estimator is unbiased when the position bias and propensity are correctly estimated:

$$((\forall d \in D, \rho_d = \hat{\rho}_d) \wedge (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\nabla_\theta \hat{R}_{\text{IPS}}] = \nabla_\theta R(\pi_\theta). \quad (3.17)$$

3.4 Direct Method and Doubly Robust Optimization

Research by Oosterhuis [55] has adapted and extended the DM and DR estimation for use in unbiased LTR.

3.4.1 The Direct Method in Learning-to-Rank

The DM relies on regression to estimate performance, based on the weight estimate $\hat{\omega}_d$ and the regression estimate $\hat{R}_d$ per document d . The DM estimate of $\mathcal{R}(\pi)$ is then:

$$\hat{\mathcal{R}}_{DM}(\pi) = \sum_{d \in D} \hat{\omega}_d \hat{R}_d. \quad (3.18)$$

The relevance estimate for DM can be any estimator that estimates document relevance based on an input document. For this [55] propose the use of a loss that corrects for trust bias, and ignores items for which no relevance estimate was taken:

$$\widehat{\mathcal{L}}(\hat{R}) = -\frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{1}{\hat{\rho}_d} \left(c_i(d) \log(\hat{R}_d) + (\hat{\alpha}_{k(d_i)} - c_i(d) \log(1 - \hat{R}_d)) \right). \quad (3.19)$$

This loss can then be used to train any multilayer perceptron (MLP) model for relevance prediction. This estimator has the following unbiasedness conditions [55]:

$$((\forall k, \alpha_k = \hat{\alpha}_k) \wedge (\forall d, \pi_0(d) = \hat{\pi}_0(d))) \longrightarrow \mathbb{E}_{c, y \sim \pi_0}[\widehat{\mathcal{L}}(d)] = \mathcal{L}(d). \quad (3.20)$$

Notably, these unbiasedness conditions are the same as the unbiasedness condition of the IPS estimator. The general unbiasedness conditions for $\widehat{\mathcal{R}}_{DM}(\pi)$ are correct estimation of $\hat{\alpha}$ and correct estimation of relevance values $\hat{R}_d$ for every document, where we use Eqs. 3.2 and 3.9:

$$((\forall k, \hat{\alpha}_k = \alpha_k) \wedge (\hat{R}_d = R_d \forall d \in D)) \longrightarrow \widehat{\mathcal{R}}_{DM}(\pi) = \mathcal{R}(\pi). \quad (3.21)$$

As noted by Oosterhuis [55], we see that finding exactly values for $\hat{R}_d$ requires a correct relevance for every document, and may introduce bias when applied as a standalone objective. To combat this bias, while still being able to apply the regression estimator, Oosterhuis [55] propose the DR estimator.

3.4.2 Doubly Robust Estimation

DR estimation is the combination of IPS methods with DM for optimization, allowing for favorable unbiasedness conditions [19]. The general DR formula consists of three components, the DM, IPS estimator, and a control variate (CV):

$$\widehat{\mathcal{R}}_{DR}(\pi) = \widehat{\mathcal{R}}_{DM}(\pi) + \widehat{\mathcal{R}}_{IPS}(\pi) - \widehat{\mathcal{R}}_{CV}(\pi). \quad (3.22)$$

Oosterhuis [55] propose the following CV, which has the same expected value as DM when IPS is unbiased, and a high covariance with IPS:

$$\widehat{\mathcal{R}}_{CV}(\pi) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \hat{\alpha}_{k(d_i)} \hat{R}_d. \quad (3.23)$$

This theoretically satisfies the requirement of having the same expected value as the DM when $\hat{\pi}_0$ is correct and clipping has no effect, as derived in Theorem B.2:

$$(\forall d, \hat{\pi}_0(d) = \pi_0(d)) \longrightarrow \mathbb{E}_{y \sim \pi_0}[\widehat{\mathcal{R}}_{CV}(\pi)] = \widehat{\mathcal{R}}_{DM}(\pi). \quad (3.24)$$

The DR estimator then follows from the definition of IPS and the CV as follows:

$$\begin{aligned} \widehat{\mathcal{R}}_{DR}(\pi) &= \widehat{\mathcal{R}}_{DM}(\pi) + \widehat{\mathcal{R}}_{IPS}(\pi) - \widehat{\mathcal{R}}_{CV}(\pi) \\ &= \widehat{\mathcal{R}}_{DM}(\pi) + \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \left(c_i(d) - \hat{\alpha}_{k(d_i)} \hat{R}_d \right). \end{aligned} \quad (3.25)$$

The DR estimator reweighs the difference between the click signal $c_i(d)$ and the predicted click probability, $\hat{\alpha}_{k(d_i)} \hat{R}_d$ is reweighted with an IPS estimate [55]. DR has the following bias by Theorem B.3:

$$\mathbb{E}_{c, y \sim \pi_0}[\widehat{\mathcal{R}}_{DR}(\pi)] - \mathcal{R}(\pi) = \sum_{d \in D} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d + \left(\frac{\mathbb{E}_{y \sim \pi_0}[\hat{\alpha}_{k(d)}]}{\hat{\rho}_d} - 1 \right) \hat{\omega}_d \hat{R}_d. \quad (3.26)$$

Furthermore, in Corollary B.5 we show that if the bias parameters are correctly estimated, the bias can be described as:

$$(\forall k, \hat{\alpha}_k = \alpha_k) \longrightarrow \mathbb{E}_{c,y \sim \pi_0}[\hat{\mathcal{R}}_{DR}(\pi)] - \mathcal{R}(\pi) = \sum_{d \in D} \omega_d \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right) (R_d - \hat{R}_d). \quad (3.27)$$

Finally, we have the following unbiasedness conditions, by Theorem B.6:

$$(\hat{\alpha}_k = \alpha_k \forall k \wedge (\forall d \in D, (\hat{\rho}_d = \rho_d \vee \hat{R}_d = R_d)) \longrightarrow \mathbb{E}_{c,y \sim \pi_0}[\hat{\mathcal{R}}_{DR}] = \mathcal{R}_\pi. \quad (3.28)$$

The DR estimator has favorable unbiasedness conditions compared to DM and IPS, being unbiased when either Relevance estimates or IPS estimates are unbiased, and has lower variance [55].

3.5 Two-Tower Models

Two-tower models can be used to predict the probability of a click c for a query-document pair d observed at rank k . The most common type of two-tower model used is the additive two-tower model [83, 80, 33, 24]. We follow the simplified notation of Hager et al. [32], where we can define the commonly used additive two-tower model as follows:

$$P(C = 1|d, k) = \sigma(\theta_k + \gamma_d), \quad (3.29)$$

where θ_k is the output of our bias tower for position $k \in \{1, 2, \dots\}$, γ_d is our relevance estimate for a query-document pair $d \in D$, where we use $\sigma(\cdot)$ to denote the sigmoid function.

User Behavior Assumptions The two-tower model relies on two core assumptions: the conditional interdependence of clicks and document features, and the negligibility of clicks on irrelevant items. In this work, applying two-tower models is straightforward, as both of these assumptions directly align with the click probability definition in Eq. 3.1.

Optimization The bias and relevance tower are jointly optimized the bias through the use of the binary cross-entropy loss as follows:

$$\mathcal{L}(\theta, \gamma) = -\mathbb{E}_{(d,k,c) \sim P(D,K,C)} [c \log(\theta_k \cdot \gamma_d) + (1 - c) \log(1 - (\theta_k \cdot \gamma_d))]. \quad (3.30)$$

This objective is defined with respect to the true data distribution, which is often unknown. In practice, we therefore optimize using the empirically observed data as follows [32]:

$$\hat{\mathcal{L}}(\theta, \gamma | D) = -\frac{1}{N} \sum_{(d,k,c) \in \mathcal{D}} [c \log(\theta_k \cdot \gamma_d) + (1 - c) \log(1 - (\theta_k \cdot \gamma_d))]. \quad (3.31)$$

3.6 Comparison Between Counterfactual Policy Learning and Two-Tower Models

Counterfactual policy learning methods and two-tower models provide two different approaches to learning from biased click feedback. Counterfactual methods, such as IPS, DM, and DR, correct the objective by reweighting observed clicks using estimates of position bias and logging propensities. Their main advantage is that they provide an explicit causal interpretation: under the required assumptions, they optimize an unbiased estimate of the target policy reward. Additionally, the learning of a policy is useful, as it allows the model to learn policies that are

inherently stochastic, as well as include fairness or diversity objectives [68, 51, 79]. However, counterfactual LTR methods do require correct propensity estimation, and may suffer from high variance.

The main advantage of two-tower models is practical: they can be optimized with standard supervised learning objectives and are straightforward to integrate into neural ranking pipelines [32, 80]. This makes two-tower models a flexible and powerful approach. An issue with two-tower models is that relevance estimates may be confounded when observational logs do not provide enough information to separate relevance from position bias [32, 23].

Thus, the two approaches trade off causal interpretability and theoretical guarantees against modeling flexibility and practical integration.

Chapter 4

RQ1: Propensity Estimation in Long-Tail Query Distributions

The standard IPS estimator assumes that exact values are available for propensities and position effects. Previous work by Oosterhuis [55] specifies that unbiasedness is achieved when both quantities are estimated correctly. However, to characterize the error introduced in practice, we should analyze how the chosen estimators for propensities affect the resulting IPS estimate. In this section, we investigate the bias of the frequency-based propensity estimator, answering **RQ1**: *what theoretical challenges arise for propensities in long-tail query distributions?*

We first specify the frequency-based sample mean estimator applied for propensity estimation. We then show that an unbiased propensity estimator does not lead to an unbiased IPS estimate due to Jensen’s Inequality. We specify the bias of the feature-based propensity estimator and its bias. We show how this bias affects unbiasedness of IPS and the trust-bias regression estimator for DM and DR estimates. Finally, we specify DR estimator under frequency-based propensity estimates, showing that under frequency-based propensity estimates, a simple decomposition of DR into IPS and DM based on observations exists.

4.1 The Frequency-based Propensity Estimator

We first define the frequency-based propensity estimator $\tilde{\rho}_d$, which is commonly used to estimate the propensity, $\hat{\rho}_d$ [55]. This estimator works by taking the sample mean with respect to the logging policy, and using this to weight the estimated position bias:

$$\tilde{\rho}_d := \frac{1}{N} \sum_{i=1}^N \hat{\alpha}_{k_i(d)}. \quad (4.1)$$

The position bias $\hat{\alpha}_k$ can be given, or estimated using any method. In this section, we investigate the frequency-based propensity estimator, and will assume that the position bias is correctly estimated. Given the i.i.d. assumption conditional on documents, we can write the expected value of the frequency-based propensity estimator for document d as follows:

$$\mathbb{E}_{\pi_0}[\tilde{\rho} \mid d] = \mathbb{E}_{\pi_0} \left[\frac{1}{N} \sum_{i=1}^N \hat{\alpha}_{k_i(d)} \mid d \right], \quad (4.2)$$

$$= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\pi_0} [\hat{\alpha}_{k_i(d)} \mid d]. \quad (4.3)$$

Under the i.i.d. assumption with respect to d , this estimator is unbiased estimate of the logging policy:

$$\mathbb{E}_{\pi_0}[\alpha_{k_i(d)} \mid d] = \rho_d, \quad \bar{\rho}_d := \mathbb{E}_{\pi_0}[\tilde{\rho}_d \mid d] = \mathbb{E}_{\pi_0}[\hat{\alpha}_{k_i(d)} \mid d]. \quad (4.4)$$

Comparing the two equations, we can see that this estimator is unbiased with respect to ρ iff $\hat{\alpha}_k$ is correctly estimated for all ranks:

Lemma 4.1. *Under correct position bias estimation, the frequency-based propensity estimate is unbiased w.r.t. ρ_d .*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \implies \bar{\rho}_d = \rho_d. \quad (4.5)$$

As a result, we have found that $\tilde{\rho}_d$ provides an unbiased estimate of the logging policy, which leads to an unbiased propensity estimate conditional on correct position bias estimation.

An issue that arises when applying the propensity estimator is that IPS applies the reciprocal of the propensity (see Eq. 3.11). By Jensen's Inequality, the expected value of an unbiased estimator is not the estimated value when it is part of a convex function. Since the function $\frac{1}{x}$ is convex, we have that:

$$\frac{1}{\mathbb{E}_{\pi_0}[\tilde{\rho}_d \mid d]} = \frac{1}{\rho_d} \leq \mathbb{E}_{\pi_0} \left[\frac{1}{\tilde{\rho}_d} \mid d \right]. \quad (4.6)$$

Therefore, applying the propensity estimate $\frac{1}{\tilde{\rho}_d}$ may introduce an upward bias in the IPS weight. In the following section, we approximate this bias for the frequency-based propensity estimator.

4.2 Bias of the Frequency-based Propensity Estimate Under Jensen's Inequality

To quantify the bias introduced by applying feature-based propensity estimation, we further estimate the expected propensity under the logging policy. We can approximate the bias of the propensity estimator by applying the Taylor expansion for the first moment:

$$\mathbb{E}[f(X)] \approx f(\mu_X) + \frac{f''(\mu_X)}{2} \sigma_X^2.$$

Here, f is the applied approximation function with respect to random variable X , where μ_X , σ_X are the mean and standard deviation. We apply $X = \tilde{\rho}_d$ as a random variable, and consider the function $f(x) = \frac{1}{x}$. First, we compute the derivatives:

$$f(X) = \frac{1}{X}, \quad f'(X) = -\frac{1}{X^2}, \quad f''(X) = \frac{2}{X^3}.$$

Filling in $X = \hat{\rho}_d$, as well as $\mu_X = \bar{\rho}_d$, which follows from Eq. 4.4:

$$f(\mu_X) = f(\bar{\rho}_d) = \frac{1}{\bar{\rho}_d}, \quad f''(\mu_X) = f''(\bar{\rho}_d) = \frac{2}{\bar{\rho}_d^3}.$$

We can now apply these in the Taylor expansion to get the expected value:

$$\mathbb{E} \left[\frac{1}{\hat{\rho}_d} \right] \approx \frac{1}{\bar{\rho}_d} + \frac{\sigma_{\hat{\rho}_d}^2}{\bar{\rho}_d^3}. \quad (4.7)$$

We observe that the approximation results in the actual value plus a weighted residual factor that is a result of the variance (squared standard deviation) of the estimated propensity. This bias is a general result for any unbiased propensity estimator applied in this setting. To quantify

the deviation under the frequency-based propensity estimator, we can then consider the variance of the frequency-based propensity estimator $\tilde{\rho}_d$:

$$\sigma_{\tilde{\rho}_d}^2 = \mathbb{V}_{k \sim \pi_0}(\tilde{\rho}_d \mid d) = \frac{1}{N^2} \sum_{i=1}^N \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d) = \frac{1}{N} \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d). \quad (4.8)$$

Here, we apply the fact that samples from the logging policy are i.i.d. across impressions conditional on d . We can combine this with the expected value of the propensity estimate (Eq. 4.7) to get an approximation under the frequency-based propensity estimator:

$$\mathbb{E}\left[\frac{1}{\tilde{\rho}_d}\right] \approx \frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3}. \quad (4.9)$$

We observe that the overestimation of the propensity under a frequency-based propensity estimate is the variance of the logging policy weighted inversely by the number of observations. In the following sections we apply this quantified bias to the IPS and the trust-bias corrected regression estimate used in DM and DR to quantify the bias.

4.3 Bias of Inverse Propensity Scoring Under the Sample Mean

We can now derive the bias of the IPS estimator with our specified propensity estimation for our set of documents $\mathcal{D}$. We assume the position bias is correctly estimated for simplicity of notation.

Theorem 4.2. *The bias of the IPS estimator (Eq. C.13) under a frequency-based propensity estimate can be described as follows:*

$$\mathbb{E}_{c, y \sim \pi_0}[\hat{R}_{IPS}(\pi)] - R(\pi) \approx \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\bar{\rho}_d} \hat{\omega}_d - \omega_d + \frac{\rho_d \hat{\omega}_d}{\bar{\rho}_d^3} \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N} \right) R_d. \quad (4.10)$$

Proof. Follows from applying Eq. 4.9 to Theorem A.1,

$$\mathbb{E}_{c, y \sim \pi_0}[\hat{R}_{IPS}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d \quad (4.11)$$

$$= \sum_{d \in \mathcal{D}} \left(\rho_d \hat{\omega}_d \cdot \mathbb{E}_{\pi_0} \left[\frac{1}{\hat{\rho}_d} \right] - \omega_d \right) R_d \quad (4.12)$$

$$\approx \sum_{d \in \mathcal{D}} \left(\rho_d \left(\frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3} \right) \hat{\omega}_d - \omega_d \right) R_d \quad (4.13)$$

$$= \sum_{d \in \mathcal{D}} \left(\left(\frac{\rho_d}{\bar{\rho}_d} + \frac{\rho_d \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3} \right) \hat{\omega}_d - \omega_d \right) R_d \quad (4.14)$$

$$= \sum_{d \in \mathcal{D}} \left(\left(\frac{\rho_d}{\bar{\rho}_d} \hat{\omega}_d - \omega_d \right) + \frac{\rho_d \hat{\omega}_d}{\bar{\rho}_d^3} \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N} \right) R_d. \quad (4.15)$$

□

Here, we clearly see two components that contribute to bias: The first component is the error in the position bias estimate, specifically in how the error in position bias affects the propensity estimate $\bar{\rho}_d$, and under the current logging policy $\hat{\omega}_d$. The second component is

a combination of a fixed value based on a reweighting of relevance, expected exposure, and propensity, and a dynamic value which is the variance of the logging policy π_0 for document d , divided by the number of observations for document d . To further analyze this component, we can show the bias under correctly estimated position:

Corollary 4.3. *The bias of IPS under a frequency-based propensity estimate when position bias is correctly estimated can be described as follows:*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \implies \mathbb{E}_{c, y \sim \pi_0} [\hat{R}_{IPS}(\pi)] - R(\pi) \approx \sum_{d \in \mathcal{D}} \frac{\omega_d R_d}{\rho_d^2} \frac{\mathbb{V}_{k \sim \pi_0}(\alpha_{k_i(d)} \mid d)}{N}. \quad (4.16)$$

Proof. Follows from applying Lemmas 4.1 and A.2 to Theorem 4.2, □

This result shows that the bias under a frequency-based propensity estimate is a combination of a fixed value based on a reweighting of relevance, expected exposure, and propensity, and a dynamic value which is the variance of the logging policy π_0 for document d . As a result, we have derived an asymptotic unbiasedness condition: As $N \rightarrow \infty$, we have $\frac{1}{N} \rightarrow 0$, therefore, document bias vanishes as the number of samples goes to infinity. Note that asymptotic unbiasedness relates directly to the number of observations for each individual document. In addition, our bias is directly related to the variance (or randomness) of our logging policy, meaning that a more random logging policy will introduce a larger bias under the sample mean propensity estimator. For any document where the logging policy is not deterministic, we find that a bias proportional to the number of samples for that document exists. As a result, in long-tail query distributions where we have few observations for each document, our frequency-based propensity estimate can lead to a large amount of bias in the IPS reward estimate.

4.4 Bias of the Trust-Bias Corrected Regression Estimator

Work by Oosterhuis [55] introduces a regression loss that corrects for trust bias, and ignores items which have not been observed (Eq. 3.19). This method also requires a propensity estimate, and is therefore also biased under the frequency-based propensity estimate:

Theorem 4.4. *The expected value of the trust-bias corrected relevance loss under a frequentist logging policy estimate with correctly estimated position bias can be described as follows:*

$$\begin{aligned} \mathbb{E}_{c, y \sim \pi_0} [\hat{\mathcal{L}}(\hat{R})] - \mathcal{L}(\hat{R}) &\approx - \sum_{d \in \mathcal{D}} \left(\left(\frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3} \right) \left(\rho_d R_d \log(\hat{R}_d) \right. \right. \\ &\quad \left. \left. + (\bar{\rho}_d - \rho_d R_d) \log(1 - \hat{R}_d) \right) - \left(R_d \log(\hat{R}_d) + (1 - R_d) \log(1 - \hat{R}_d) \right) \right). \end{aligned} \quad (4.17)$$

Proof. Follows applying Eq. 4.9 to Theorem D.1. from Oosterhuis [55].

$$\mathbb{E}_{c, y \sim \pi_0} [\hat{\mathcal{L}}(\hat{R})] = - \sum_{d \in \mathcal{D}} \frac{1}{\bar{\rho}_d} \left(\rho_d R_d \log(\hat{R}_d) + (\mathbb{E}_{k \sim \pi_0}[\hat{\alpha}_{k_i(d)}] - \rho_d R_d) \log(1 - \hat{R}_d) \right) \quad (4.18)$$

$$\approx - \sum_{d \in \mathcal{D}} \left(\frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3} \right) \left(\rho_d R_d \log(\hat{R}_d) + (\bar{\rho}_d - \rho_d R_d) \log(1 - \hat{R}_d) \right) \quad (4.19)$$

$$\approx - \sum_{d \in \mathcal{D}} \left(\frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N \bar{\rho}_d^3} \right) \left(\rho_d R_d \log(\hat{R}_d) + (\bar{\rho}_d - \rho_d R_d) \log(1 - \hat{R}_d) \right). \quad (4.20)$$

□

Looking at the result, we observe the same two components again. First a direct result of position bias estimation error in the propensity, where the bias becomes smaller when the position bias is estimated more correctly. Furthermore, we again have the sample variance bias, which is rescaled by the squared inverse propensity, as well as the error in relevance estimation. We can make the bias effects more explicit by considering the bias of the loss objective under correct position bias estimation:

Corollary 4.5. *The bias of IPS under a frequency-based propensity estimate when position bias is correctly estimated can be described as follows:*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \implies \left(\mathbb{E}[\hat{\mathcal{L}}(\hat{R})] - \mathcal{L}(\hat{R}) \approx - \sum_{d \in \mathcal{D}} \frac{\mathbb{V}_{k \sim \pi_0}(\alpha_{k_i(d)} \mid d)}{N \rho_d^2} \left(R_d \log(\hat{R}_d) + (1 - R_d) \log(1 - \hat{R}_d) \right) \right). \quad (4.21)$$

Proof. Follows from applying Lemmas 4.1 and A.2 to Theorem 4.2, □

Once again, we observe an error that is a result of a combined term of the variance of the logging policy and the number of available samples for the document. This may result in an unbiased estimator when no variance is present, or we have an infinite number of samples for document d . Notably, this bias when applied in the DM or DR estimator is expressed at the level of the loss, not necessarily at the level of the trained relevance model or the resulting DM estimator. In particular, as optimization is approximate, the learned model may not exactly reproduce the document-level bias present in the loss. Nevertheless, reducing bias in the loss target improves alignment with the target loss, and is expected to lead to an improved result.

4.5 The Doubly-Robust estimator under the frequency-based propensity estimate

We now investigate the application of the frequentist logging policy estimate with the DR estimator. Here, it is important to first denote that in practice, the propensity estimator is applied with an indicator function, to ensure division by zero does not occur, as well as to correctly apply the covariate [55]. This means, that if our document is never shown in the top- k of our ranker, the propensity estimate $\hat{\rho}_d$ as well as the inverse propensity estimate $\mathcal{R}_{IPS}(\pi)$ are defined as 0. We write this using an indicator function based on the count of top- k observations $1_{N_{topk}=0}$.

Theorem 4.6. *The DR estimator given a frequency-based propensity estimate can be written as follows:*

$$\mathcal{R}_{DR}(\pi) = 1_{N_{topk}=0} \cdot \mathcal{R}_{DM}(\pi) + \mathcal{R}_{IPS}(\pi). \quad (4.22)$$

Proof. Follows from applying the frequency-based propensity estimator (Eq. 4.1) to the definition of DR (Eq. 3.25)

$$\mathcal{R}_{DR}(\pi) = \mathcal{R}_{DM}(\pi) + \mathcal{R}_{IPS}(\pi) - \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \hat{\alpha}_{k_i(d)} \hat{R}_d \quad (4.23)$$

$$= \mathcal{R}_{DM}(\pi) + \mathcal{R}_{IPS}(\pi) - \sum_{d \in D} \hat{\omega}_d \frac{\frac{1}{N} \sum_{i=1}^N \hat{\alpha}_{k_i(d)}}{\frac{1}{N} \sum_{i=1}^N \hat{\alpha}_{k_i(d)}} \hat{R}_d \quad (4.24)$$

$$= \mathcal{R}_{DM}(\pi) + \mathcal{R}_{IPS}(\pi) - 1_{N_{topk} > 0} \cdot \sum_{d \in D} \hat{\omega}_d \hat{R}_d \quad (4.25)$$

$$= \mathcal{R}_{DM}(\pi) + \mathcal{R}_{IPS}(\pi) - 1_{N_{topk} > 0} \cdot \mathcal{R}_{DM}(\pi) \quad (4.26)$$

$$= 1_{N_{topk}=0} \cdot \mathcal{R}_{DM}(\pi) + (1 - 1_{N_{topk}=0}) \cdot \mathcal{R}_{IPS}(\pi). \quad (4.27)$$

□

The important insight that is gained from this analysis is that the current DR estimator does combine IPS and the DM to combine aspects of the two approaches. However, this combination is performed in a strict way, with the direct method estimation only being applied in the scenario where we have no samples at all, and as soon as we have a single sample, the IPS estimate is used. With this, we offer a novel non-trivial condition in which DR is reduced to IPS, which was not yet discussed in Oosterhuis [55]: when a document has one or more observations and frequency-based propensity estimation is used, and no clipping is applied, DR and IPS are equivalent. This is a non-trivial setting, as this is the setting which is applied in Oosterhuis [55] (they do apply clipping, but clipping only has an effect for documents with low propensity scores).

This approach of relying IPS from a single sample may become problematic when we have a propensity estimation method which performs poorly with few samples, which is the case for the frequency-based propensity estimate.

4.6 Conclusion

To conclude, in this chapter we have formulated an answer to **RQ1**: Sparse observations pose significant challenges to existing propensity estimation methods in unbiased LTR in settings where no clipping is applied. We show that bias of both IPS and DM are directly dependent on the sample variance, and the number of observations for a specific query-document pair when the frequency-based propensity estimates are used. As a result, long-tail query distributions are likely to have poor propensity estimations. Finally, we show that in DR estimates with frequency-based propensity estimates that rely on the the same dataset for propensity estimation, the reward estimate relies directly on either DM or IPS estimates based on the amount of top-k observations for each document. As a result, DR can also be directly adversely affected by sparse observations under frequency-based propensity estimation.

This chapter has provided a clear view on the errors that can arise when query observations are sparse. This clearly illustrates that existing methods may not be effective in long-tailed query distributions. In the following chapters, we explore propensity estimation based on features, to minimize bias introduced in reward estimates for counterfactual LTR due to limited observations.

Chapter 5

RQ2: Feature-Based Propensity Estimation

In the previous section, we have shown that the frequency-based propensity estimates applied in previous work result in a consistent bias. In this chapter, we answer **RQ2**: *under what assumptions can propensity estimation be performed based on query-document features*. We first explore how we can construct a propensity estimator that can generalize across query-document pairs by predicting propensities based on document features. We do so by first defining the feature based propensity, and then showing that this requires that the logging policy is determined by the features. After this, we formally analyze the bias of a feature-based grouping approach, showing how a larger effective sample size can reduce bias in IPS estimation. Finally, we detail how to generalize feature-based propensity estimation to (neural) propensity estimation models.

5.1 Assumptions for Feature-Based Propensity Estimation

The feature-based propensity estimator estimates the propensity of a document based on its features. Formally, we can define a propensity estimate that is based on features for the set of documents D as:

$$\rho_x := \mathbb{E}_D[\rho_D \mid x_D = x]. \quad (5.1)$$

In counterfactual LTR, we apply propensity estimation for specific documents, therefore, we define the feature-based propensity estimator for document d as ρ_{x_d} :

$$\rho_{x_d} := \mathbb{E}_d[\rho_d \mid x_d]. \quad (5.2)$$

The goal of the feature-based propensity estimator is therefore to estimate the behavior of the document-based propensity based on the set of features x_d for document d .

We will first show that unbiasedness of this estimator requires that the logging policy is determined by the features, first showing that the alternative results in a biased estimator. We assume a set of documents for a single query, where each has the same feature (A), but both have different propensities, as shown in Table 5.1. Supposing that we have an unbiased estimator for propensities based on features and documents, we will then get the following propensities for feature-based and document based propensity estimates:

$$\mathbb{E}[\rho_d \mid x_d = A] = 0.5, \quad \mathbb{E}[\rho_d \mid d = d_1] = 0.9, \quad \mathbb{E}[\rho_d \mid d = d_2] = 0.1.$$

From these results, we observe that propensity scores on a feature level are not aligned with the propensity scores at a document level. To illustrate how this results in a biased IPS estimator,

Document ID	Relevance Label	Propensity ρ_d	Feature
d_1	1	0.9	A
d_2	0	0.1	A

Table 5.1: Example set of documents for a single query, where propensity is not tied to features.

Document ID	Relevance Label	Propensity ρ_d	Feature
d_1	1	0.2	A
d_2	0	0.2	A
d_3	1	0.8	B
d_4	0	0.8	B

Table 5.2: Example set of documents for a single query, where propensity is tied to features.

we can examine the bias of the IPS estimator, where we assume that all position bias estimates α_k are correct, following Corollary A.3:

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right) R_d.$$

We can now fill in our set of estimated propensities using different propensity estimates. We find that when we use $\hat{\rho}_{x_d}$ with this set of documents, we have a biased estimator, while $\hat{\rho}_d$ gives us an unbiased result:

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) = \left(\frac{\rho_{d_1}}{\hat{\rho}_{d_1}} - 1 \right) = \frac{0.9}{0.9} - 1 = 0,$$

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) = \frac{\rho_{d_1}}{\hat{\rho}_A} - 1 = \frac{0.9}{0.5} - 1 = 0.8.$$

By providing a counterexample, we have now proven that when logging policy can behave differently for other documents with the same features, propensities are not guaranteed to be equivalent, or more formally:

$$\neg(\forall(k, d) \in K \times D, \pi_0(k \mid d) = \pi_0(k \mid x_d)) \not\rightarrow (\forall d, \rho_d = \rho_{x_d}). \quad (5.3)$$

This issue stems from the fact that we currently have not guarantee about logging policy behavior with respect to features, and therefore we do not require identical behavior for the same set of features. To resolve this, we take the assumption that the propensity is exactly explained by the features, in other words, the historical logging policy is determined by the document features exactly, or:

Assumption 5.1. *The logging policy used to gather data is exactly determined by the document features:*

$$\forall(k, d) \in K \times D, \pi_0(k \mid d) = \pi_0(k \mid x_d). \quad (5.4)$$

Under this assumption, we can exactly compute document propensities using feature-based propensities (the proof of this statement is shown in Appendix C):

$$\forall(k, d) \in K \times D, \pi_0(k \mid d) = \pi_0(k \mid x_d), \longrightarrow (\forall d, \rho_d = \rho_{x_d}). \quad (5.5)$$

To illustrate how this changes the computation, we assume a new set of documents and features in Table 5.2. Here, note that the equivalence of propensity for d_1 and d_2 as well as d_3 and d_4 is a direct result of Assumption 5.1, as $\rho_{d_1} = \rho_A = \rho_{d_2}$ and $\rho_{d_3} = \rho_B = \rho_{d_4}$. Assuming an effective estimator, we get the following propensities:

$$\begin{aligned}\mathbb{E}[\rho_d | x_d = A] &= 0.2, & \mathbb{E}[\rho_d | d = d_1] &= \mathbb{E}[\rho_d | d = d_2] = 0.2, \\ \mathbb{E}[\rho_d | x_d = B] &= 0.8, & \mathbb{E}[\rho_d | d = d_3] &= \mathbb{E}[\rho_d | d = d_4] = 0.8.\end{aligned}$$

We can now recompute the bias of the IPS estimator, following Corollary A.3:

$$\begin{aligned}\mathbb{E}_{c, y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) &= \left(\frac{\rho_{d_1}}{\hat{\rho}_{d_1}} - 1 \right) + \left(\frac{\rho_{d_3}}{\hat{\rho}_{d_3}} - 1 \right) = \frac{0.2}{0.2} + \frac{0.8}{0.8} - 2 = 0, \\ \mathbb{E}_{c, y \sim \pi_0} [\hat{R}_{\text{IPS}}(\pi)] - R(\pi) &= \left(\frac{\rho_{d_1}}{\hat{\rho}_A} - 1 \right) + \left(\frac{\rho_{d_3}}{\hat{\rho}_B} - 1 \right) = \frac{0.2}{0.2} - 1 + \frac{0.8}{0.8} - 1 = 0.\end{aligned}$$

Similarly, we can compute the gradients, following Corollary A.8:

$$\begin{aligned}\mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) &= \sum_{d \in D} R_d \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right) \sum_{k=1}^K \nabla_{\theta} \pi_{\theta}(k | d), \\ \mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) &= \left(\frac{\rho_{d_1}}{\hat{\rho}_{d_1}} - 1 + \frac{\rho_{d_3}}{\hat{\rho}_{d_3}} - 1 \right) \sum_{k=1}^K \nabla_{\theta} \pi_{\theta}(k | d) = 0, \\ \mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) &= \left(\frac{\rho_{d_1}}{\hat{\rho}_A} - 1 + \frac{\rho_{d_3}}{\hat{\rho}_B} - 1 \right) \sum_{k=1}^K \nabla_{\theta} \pi_{\theta}(k | d) = 0.\end{aligned}$$

We observe that the gradients are exactly unbiased in expectations, mean that feature level propensities can be used to compute an unbiased estimate of document propensities, in our example. We will now generalize the feature-based propensity estimation to IPS as a whole.

Suppose a set of document features corresponding to a document x_d , and a feature-based document propensity ρ_{x_d} , we can construct a feature-based IPS estimator:

$$\hat{\mathcal{R}}_{\text{Feature-IPS}}(\pi) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_{x_d}} c_i(d). \quad (5.6)$$

Importantly, under under Eq. 5.5, we can show that the feature-based IPS estimator is unbiased, and has unbiased gradients. All proofs and derivations are given in Appendix C. First, we have the bias of feature-based IPS:

$$\mathbb{E}_{c, y \sim \pi_0} [\hat{R}_{\text{Feature-IPS}}(\pi)] - R(\pi) = \sum_{d \in D} \left(\hat{\omega}_d - \omega_d \frac{\rho_d}{\hat{\rho}_{x_d}} \right) R_d. \quad (5.7)$$

We also show that the feature-based IPS estimator is unbiased when the position bias and propensity are correctly estimated, and we have a logging policy that is determined by features:

$$((\forall d \in D, \rho_d = \hat{\rho}_{x_d}) \wedge (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow (\mathbb{E}[\hat{R}_{\text{Feature-IPS}}] = R(\pi_{\theta})).$$

We also derive the bias of the gradient, in order to ensure that the estimator can be optimized correctly:

$$\mathbb{E}[\nabla_{\theta} \hat{R}_{\text{Feature-IPS}}] - \nabla_{\theta} R(\pi_{\theta}) = \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_{x_d}} \hat{\alpha}_k - \alpha_k \right) \nabla_{\theta} \pi_{\theta}(k | d).$$

Here, we can again show that the gradient of the feature-based IPS estimator is unbiased with respect to the true reward when the position bias and feature-based propensity are correctly estimated, and our logging policy is determined by our features exactly:

$$((\forall d \in D, \rho_d = \hat{\rho}_{x_d}) \wedge (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow (\mathbb{E}[\nabla_{\theta} \hat{R}_{\text{Feature_IPS}}] = \nabla_{\theta} R(\pi_{\theta})).$$

This means that we can construct an unbiased IPS estimator based on document features. This allows us to generalize unbiased IPS with estimated propensities to the feature level, and construct estimators which can generalize to unseen documents based on their features. In the following section, we propose two model classes for feature-based propensity estimation, an unsupervised clustering method, as well as a general supervised learning approach. For both classes, we also consider the bias introduced. The feature-based propensity can be simply applied and extended to DM and DR methods by applying Assumption 5.1 to their definitions.

5.2 Motivation: Grouping and Effective Sample Size

To define clustering methods, we first illustrate how we can define a similarity between query-document pairs. Under the feature-based logging policy used to gather data π_0 , assume the existence of two documents d and d' with similar feature vectors,

$$\|x_d - x_{d'}\|_2 \leq \varepsilon. \quad (5.8)$$

Under the continuity of π_0 , if we assume that the logging policy representations of our feature vectors share a Lipschitz constant L , we can bound the difference in logging policy, following an approach by Hager et al. [32]:

$$|\pi_0(k \mid x_d) - \pi_0(k \mid x_{d'})| \leq L\|x_d - x_{d'}\|_2. \quad (5.9)$$

This example bound illustrates that documents with similar feature representations can have similar logging probabilities, and as feature overlap increases, the difference in logging policy can be smaller. In contrast to document-specific propensities, where parameters may vary arbitrarily across documents, feature-based propensities enforce a structural coupling between documents that share similar feature support. In this section, we do not explicitly assume Lipschitz continuity, but use it as a motivation for applying clustering methods to group propensities of documents with similar features.

Specifically, we define a grouping function g , which is able to group documents together based on their similarity. Given our set of documents D , for each document $d \in D$ grouping function g defines a group G_d based on the features, with $G_d \subseteq D$. After grouping is completed, we can pool position-bias values from documents in G_d and use these to estimate a shared feature-based propensity. The grouped propensity estimator $\rho_{G(x_d)}$ can be written as:

$$N_{G_d} = \sum_{d' \in G_d} N_{d'}, \quad \hat{\rho}_{G(x_d)} := \frac{1}{N_{G_d}} \sum_{d' \in G_d} \sum_{i=1}^{N_{d'}} \hat{\alpha}_{k(d'_i)}, \quad (5.10)$$

where N_d is the number of observations for document d , and N_{G_d} is the number of observations for documents inside the group associated with document d . Importantly, here, we are able to use the samples of different queries, meaning that even if N_d associated with our document is small, it is still possible to have many samples if other documents in the group do have more samples. Following this, we can define the *effective sample size* at x_d as:

$$N_{\text{eff}(d)} = N_{G_d}, \quad (5.11)$$

where the samples are from the entire set of observations. This allows us to simplify the grouping-based propensity estimator as follows:

$$\hat{\rho}_{G(x_d)} = \frac{1}{N_{\text{eff}(d)}} \sum_{i=1}^{N_{\text{eff}(d)}} \alpha_{k_i(x_d)}. \quad (5.12)$$

Under the same i.i.d. assumptions as the frequency-based estimator, the variance of this sample mean estimator can be described as:

$$\mathbb{V}_{d \sim p(d)}(\hat{\rho}_{G(x_d)}) = \frac{1}{N_{\text{eff}(d)}} \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid x_d). \quad (5.13)$$

Under Assumption 5.1, we can rewrite this from x_d to d as logging policy behaviour is equivalent:

$$\mathbb{V}_{d \sim p(d)}(\hat{\rho}_{G(x_d)}) = \frac{1}{N_{\text{eff}(d)}} \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d). \quad (5.14)$$

We now further derive the bias of the feature-based IPS estimator in order to obtain an expression analogous to the document-based case. We apply the variance Eq. 5.14 to the expected value of the propensity estimate (Eq. 4.7) to get an approximation under the clustering feature-based propensity estimator:

$$\mathbb{E}\left[\frac{1}{\rho_{G(x_d)}}\right] \approx \frac{1}{\bar{\rho}_d} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)} \mid d)}{N_{\text{eff}(d)} \bar{\rho}_d^3}. \quad (5.15)$$

Here, we use that $\mu_X = \bar{\rho}_{G(x_d)}$ and under Assumption 5.1, $\bar{\rho}_d$ and $\bar{\rho}_{G(x_d)}$ are equivalent. This feature-based estimate can then be applied to estimate the bias of the feature-based propensity estimator following the same proof structure as Section 4.3

Theorem 5.1. *Under correct position bias estimation, the bias of the IPS estimator with feature-grouping based propensities can be approximated as:*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \implies \mathbb{E}_{c, y \sim \pi_0}[\hat{R}_{\text{Feature-IPS}}(\pi)] - R(\pi) \approx \sum_{d \in \mathcal{D}} \frac{\omega_d R_d}{\rho_d^2} \frac{\mathbb{V}_{k \sim \pi_0}(\alpha_{k_i(d)} \mid d)}{N_{\text{eff}(d)}}. \quad (5.16)$$

Proof. Follows from applying Eqs. 5.14 and 5.15 to Theorem C.2. $\square$

We can now compare Theorems 4.2 and 5.1, where we can see that while both are biased with the sample variance, this feature-based IPS estimator is scaled by the sample variance of $N_{\text{eff}(d)}$, which is expected to be much larger in non-trivial cases. As a result, while both may be unbiased in infinite data per document, this feature-based IPS estimator will have less variance, and may therefore be less bias. In practice, designing a feature-based propensity estimator that is entirely unbiased is difficult, as generalizing across items could introduces a bias-variance tradeoff. Furthermore, grouping methods rely on the existence of related documents, and can therefore still suffer from a lack of representation in specific areas of the feature space. In the following section, we resolve this by proposing a feature-based propensity estimation method which does not rely on observed documents directly.

5.3 Supervised Learning for Feature-Based Propensity Estimation

As a further extension to feature-based propensity estimation, we propose propensity estimation that relies entirely on the features, removing the dependency on the number of documents in a specific area of the feature space. We formalize this by defining the estimator f which computes propensities based only on the set of features:

$$f(x_d) \approx \rho_{x_d}. \quad (5.17)$$

This estimator can be any general function approximator, which maps from the set of features $\mathcal{X}$ to a real-valued output $\mathbb{R}$. We can then define our feature-based propensity estimator exactly as the output of this mapping:

$$\hat{\rho}_{x_d} := f(x_d). \quad (5.18)$$

We train estimator f using the samples from a train dataset of click logs by treating the placement of documents (and the associated propensity under the position bias) as a supervised learning task. In Section 6.2, we propose optimization for feature-based propensity estimation based on an MSE-based regression loss.

Because the supervised propensity estimator is not based on a document-specific count or a local neighborhood average, its bias is no longer directly controlled by N_d or $N_{\text{eff}}(x_d)$. Instead, the quality of the estimator depends on how well f approximates the oracle propensity function.

5.4 Conclusion

In this chapter, we have formulated an answer to **RQ2**, specifying the assumption required for feature-based propensity estimation: A logging policy for data gathering that is determined by features. In addition to this, we specified two general frameworks for propensity estimation based on features: a clustering approach, which relies on the grouping of documents based on feature similarity, as well as a supervised approach, which directly predicts propensities from document features. In the following chapter, we propose approaches to apply this in practice, as well as how to apply two-tower click models for position bias and relevance estimation.

Chapter 6

Method

In this section, we present our methods for improving counterfactual LTR through the application of feature-based propensity estimation and neural-click models for bias and relevance estimation. First, we describe our proposed methods for data-based logging policy estimations and their features. Then, we describe how to apply two-tower models for position bias and relevance estimation in policy optimization methods.

6.1 Unsupervised Learning for Effective Document Sample Grouping

In Section 5.2, we have shown that feature-based propensity estimation can reduce variance by increasing the effective sample size through grouping documents with similar feature representations, specifying the requirement of a function g which maps documents to groups. In this section, we detail the three clustering method used in this work to group documents.

Distance-Threshold Grouping The first grouping strategy is thresholding based directly on Euclidean distance. For a given threshold ε and a specific document d , we assign document d' to group G_d iff we have that:

$$\Delta(x_d, x_{d'}) \leq \varepsilon.$$

This basic distance method allows us to group documents that lie within a certain radius of one another. Smaller values of ε produce smaller and more homogeneous groups, while larger values of ε produce larger groups and therefore stronger pooling of observations. This distance thresholding can also be applied through other distance measures, such as cosine similarity.

K-Nearest-Neighbor Grouping The second strategy is k-nearest neighbors (KNN) grouping. Here, each document is connected to its k nearest neighbors under the euclidean distance function Δ .

The group G_d is then defined as the k nearest documents d' for document d . Small values of k keep the grouping close to the original document-specific estimator, while larger values of k increase the amount of smoothing and therefore the number of observations that contribute to each estimate.

k-Means Clustering The third strategy is k -means clustering. Unlike the previous two methods, k -means does not define groups through pairwise neighborhoods, but through a global partition of the feature space. Given a chosen number of clusters M , the algorithm [6, Ch. 9]

learns centroids $\mu_1, \dots, \mu_M$ and assigns each document to its nearest centroid:

$$c(x_d) = \arg \min_{m \in \{1, \dots, M\}} \|x_d - \mu_m\|_2^2.$$

The resulting group is then

$$G_d = \{d' \in D : c(x_{d'}) = c(x_d)\}.$$

k -Means therefore assigns groups such that each group is a cluster of documents with similar feature values. Note that the generated groups are still document-specific in our definition, but all documents within a cluster will have the same group assignment. The main hyperparameter is the number of clusters M : fewer clusters create larger groups with stronger variance reduction, while more clusters move the estimator closer to the document-level case.

6.2 Neural Propensity Estimation

In this section, we consider how we can approximate propensities directly from document features. We do so in two stages. First, we estimate the position bias $\hat{\alpha}_k$ for each position k using a position bias estimator. Then, we train a neural network to predict the position bias for each document using the document features as input. Given that many previous methods have considered the estimation of position bias from rank [3, 77], in this section we focus on defining a novel neural method for propensity estimation based on logged data, while assuming the existence of an appropriate position bias estimate.

6.2.1 Regression-Based Neural Propensity Estimation

Following Chapter 5, we have the feature based-propensity estimate defined as follows:

$$\rho_{x_d} = \mathbb{E}_{\pi_0}[\alpha_k].$$

Important to note for this estimator is that we estimate without explicit knowledge of the logging policy and without conditioning on the rank k . In order to do so, we define a neural network $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ which takes only document features as input. We wish to optimize the set of parameters θ of this network such that it produces an estimate $\hat{\rho}_d := f_\theta(x_d)$ that approximates the oracle propensity ρ_d .

We propose optimizing f_θ using supervision from the estimated position bias $\hat{\alpha}_k$ by minimizing the mean squared error (MSE) loss.

$$\mathcal{L}(f_\theta) = \mathbb{E}_{\pi_0} \left[(f_\theta(x_d) - \hat{\alpha}_k)^2 \right], \quad (6.1)$$

where k is used only to generate the target $\hat{\alpha}_k$ and is not provided as input to the network. This ensures that the neural network purely learns a distribution over position biases. As we do not have access to the true distributions of the data, we can rely on empirically observed data as follows:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \sum_{d=1}^D (f_\theta(x_d) - \alpha_{k(d_i)})^2.$$

Convergence of Mean Squared Error Propensity Estimator

An important feature of the loss to verify is whether our estimator will converge to the average of $\hat{\alpha}_k$ at all ranks at which it has been observed.

$$f(x_d) \approx \mathbb{E}[\alpha_k \mid x_d] = \sum_{k=1}^K \alpha_k \pi_0(k \mid x_d).$$

To verify this, we investigate the gradient of our MSE loss, which can be described as follows:

$$\nabla_{\theta} \mathcal{L}(f_{\theta}(x_d), \alpha_{k(d_i)}) = 2(f_{\theta}(x_d) - \alpha_{k(d_i)}) \nabla_{\theta} f_{\theta}(x_d).$$

If we then take the expected value of this loss, we get the following:

$$\mathbb{E}[\nabla_{\theta} \mathcal{L}] = \mathbb{E}[2(f_{\theta}(x_d) - \alpha_{k(d_i)}) \nabla_{\theta} f_{\theta}(x_d)].$$

Recall that $\alpha_{k(d_i)}$ is an i.i.d. sample from $P(k \mid x_d)$ and as a result, we can write:

$$\mathbb{E}[\alpha_{k(d_i)} \mid x_d] = \mathbb{E}[\alpha_{k(d)} \mid x_d].$$

Which gives us the following when conditioning on a specific document:

$$\mathbb{E}_{x_d}[\nabla_{\theta} \mathcal{L}] = \mathbb{E}_{x_d}[2(f(x_d) - \mathbb{E}[\alpha_{k(d)} \mid x_d]) \nabla_{\theta} f(x_d)].$$

From here, we can see that this loss converges to zero if we have that our function approximates the distribution of position bias under the set of document features:

$$f(x_d) = \mathbb{E}[\alpha_{k(d)} \mid x_d].$$

This means that in expectation, our estimator $f(x_d)$ will converge to the feature based propensity ρ_{x_d} .

6.3 Two-Tower Models for Bias and Relevance Estimation

In the previous sections, we have assumed the position bias to be given or approximated using an appropriate estimator. In this section, we consider how the two-tower model can be applied for bias and relevance estimation. We first propose an alternative model formulation for two-tower models to align better with use in position bias and relevance estimation. Following this, we discuss how we apply the bias and relevance towers for counterfactual LTR.

6.3.1 Multiplicative Two-Tower Model

In this work, we consider independently applying bias and relevance towers. Position bias and relevance can be considered as logit scores separately, which can be used to investigate parameter sets [32, 23]. Technically, the sigmoid can be applied to reach a score which generally represent the position bias probabilities captured by the model. However, this is not exactly the same, as in the additive model, addition is applied before the sigmoid. To investigate a formulation which applies the sigmoid before combination, we propose the multiplicative two-tower model:

$$P(C = 1 \mid d, k) = \sigma(\theta_k) \cdot \sigma(\gamma_d). \quad (6.2)$$

Notably, instead of addition for position bias and relevance tower, we apply multiplication in this model. This change is applied as combination is performed in probability space rather than in logit space.

6.3.2 Normalization

Supposing that we have an optimized two-tower model for a given dataset, we have a bias value for each rank input θ_k , which is a logit value that represents click probability. An issue that remains is that both additive and multiplicative two-tower models represent position bias values up to a multiplicative or additive constant depending on the model, meaning that although relative values are fixed, all values can shift by an arbitrary constant [32]. To ensure consistency, we normalize all estimates such that the output logits are equal to 0 for the first position. The resulting position bias estimates $\hat{\alpha}_k$ can then be used directly in the computation of propensities and reweighting.

For the regression estimate, we do not need to apply any normalization. We can directly apply it in the DM and DR estimator formulations. After the sigmoid is applied, the relevance tower produces a value between 0 and 1 for each input, which can be used as the relevance estimate for each query-document pair.

6.4 Summary

We briefly summarize the way in which we apply changes to estimation in doubly robust estimation. We reiterate the DR estimator:

$$R_{DR} = R_{DM} + R_{IPS} - R_{CV} = \quad (6.3)$$

$$\sum_{d \in D} \hat{\omega}_d \hat{R}_d + \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} c_i(d) - \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \hat{\alpha}_{k(d_i)} \hat{R}_d. \quad (6.4)$$

Here, we have three interventions which we consider. The first of these interventions is to apply a feature-based propensity estimator. This estimator can be directly applied in the IPS component by replacing $\hat{\rho}_d$ with $\hat{\rho}_{x_d}$. It can also be applied in the bias-corrected relevance loss (see Eq. 3.19), as this includes a propensity estimate. The second intervention we can perform is to alter the position bias estimation method, which affects $\hat{\alpha}$ directly, and $\hat{\rho}_d$ (or $\hat{\rho}_{x_d}$), $\hat{\omega}_d$ and $\hat{R}_D$ indirectly as a component (See Eqs. 3.7, 3.10, 3.19, and 6.1). The positional bias can then be set as the sigmoid of the output logits of the bias tower. Finally, we can perform relevance estimation through the relevance tower of the two-tower model, providing an alternate method for a debiased relevance estimate. Each intervention can also be applied to IPS and DM separately, with the note that as IPS does not contain a regression estimate, this intervention cannot be applied.

In the following, we validate our proposed approaches for feature-based propensity estimation and the application of click models for doubly robust estimation. We address three research questions to verify our theoretical motivations. First, we verify that feature-based propensity estimation techniques can recover the logging policy propensities in a semi-synthetic setting (**RQ3**). Second, we investigate to what extent the feature-based propensity estimation method can improve performance of counterfactual LTR methods in semi-synthetic datasets (**RQ4**). Finally, we investigate to what extent we can apply position bias and relevance estimated using two-tower models to improve the performance of doubly robust estimation for unbiased LTR (**RQ5**).

Chapter 7

Experimental Setup

In this section, we detail the general setup of our experiments, including datasets, logging policy setting, and significance settings.

Datasets In our synthetic simulations, we apply traditional learning-to-rank datasets with expert-annotated relevance labels between 0-4. Specifically, we apply the MSLR-WEB30K [60] dataset from the Bing search engine (31,531 queries and 136 dimensional feature vectors), as well as the Istella-S [48] dataset (33,018 queries and 220 dimensional feature vectors), and the Yahoo! Webscope [12] dataset (29,921 queries and 699 dimensional feature vectors). We use the official training, validation, and testing splits. We apply query normalization of $\log p(x) = \ln(1 + |x|) \odot \text{sign}(x)$ following previous work [32, 61]. All experiments are conducted in a top-25 ranking setting.

Logging Policy Our theoretical results in Chapter 4 have shown that the variance of the logging policy has an effect on the bias of propensity estimation. To simulate this variance, we need to model different levels of stochasticity in the logging policy. To do so, we reuse the epsilon-greedy logging policy proposed in Hager et al. [32]. With probability T , we sample a uniform ranking of documents. Otherwise, we rank deterministically based on logging policy scores. We refer to the probability T as the policy’s *temperature*. A policy with $T = 0$ never swaps a document across positions, whereas a policy with $T = 1$ uniformly swaps documents across all positions. In all base experiments, we apply a temperature setting of 0.5, which means that the model displays a deterministic ranking sorted by relevance estimates of the logging policy in 50% of samples, and a uniformly random ranking in the other samples.

Oracle Propensity Baseline

In our results, in addition to the standard frequency-based method, we compare to an oracle propensity baseline. This oracle is computed by evaluating the known logging policy directly: for each query-document pair and displayed position, we use the exact exposure probability induced by the data-generation policy instead of estimating it from observed clicks. Formally, given a greedy logging policy, the oracle propensity for a query-document pair is computed as:

$$\rho_d = \sum_{k=1}^K \pi_0(k \mid d) \alpha_k,$$

where $\pi_0(k \mid d)$ is the probability of observing document d at position k under the logging policy. In a deterministic case, this is then simply the result of ranking the logging policy relevance estimates. The IPS weights are therefore formed from the exact propensities that

generated the data. Applying the oracle propensity is equivalent to having access to the model used during data gathering, and is a theoretically fully unbiased approach.

Under the epsilon-greedy logging policy used in this thesis to simulate stochastic or changing logging policy, this becomes a mixture of the deterministic and random exposure:

$$\rho_{\text{true}}(d) = (1 - T) \sum_{k=1}^K \pi_0(k | d) \alpha_k + T \frac{1}{K} \sum_{k=1}^K \alpha_k,$$

where T is the policy temperature. The IPS weights are therefore formed from the exact propensities that generated the data. Applying the oracle propensity is equivalent to having access to the model used during data gathering, and is a theoretically fully unbiased approach.

Metrics The primary purpose of evaluation in this work is to measure the effectiveness of models in optimizing ranking policies in semi-synthetic data. Our primary metric is the NDCG metric, as described in Section 3.1. Although the ECP metric aligns better with user behavior assumptions as described by Oosterhuis [55], the metric depends on the position bias, which may be unknown. As a result we cannot apply the ECP in settings where we don’t know the actual bias values. The NDCG metric is general, and does not require any estimation, allowing for effective comparison between methods. In settings where we simulate using the true-position bias, we do report the ECP metric for comparative purposes.

Core results for feature-based propensity estimation across sample sizes are based on six independent runs. All other results are based on three independent runs.

Models Our policy optimization models are adapted from Oosterhuis [55] and two-tower models are adapted from Hager et al. [32]. We use the same general policy optimization network across all experiments: a feed forward neural network (2 layers, 32 neurons and ELU activations). Policy and model Optimization was done using mini-batch gradient-based learning using AdamW. We learn up to 100 epochs, with early stopping after three epochs of no improvement in validation loss. Our setup is implemented in Jax [10], Flax [34] and Rax [37].[Stan: Todo: Add the repository link.](#) Experiments were carried out on 2 AMD Rome 7H12 CPUS.

Chapter 8

RQ3: logging policy Estimation with Feature-based Propensities

In this chapter, we investigate **RQ3**: *Can feature-based propensities effectively recover logging policy propensities?*. In order to compare our methods to the existing frequency-based propensity methods, we conduct experiments with a synthetic dataset. This means that we will use a synthetic logging policy to collect data, and measure the alignment of this logging policy with our estimation methods. This allows us to investigate how well our proposed logging policy estimation is able to capture logging policy behavior. In our experiments, we will consider both the mean and variance of our estimator. Here, it is important to consider that variance has a large effect in our estimation, as the individual propensity estimates made for each document are the estimates which are used throughout optimization. As a result, variance in propensity estimates becomes a bias in our reward estimate. It remains important to consider the bias, as it indicates whether our estimates are consistently over- or underestimating the propensity values of a position.

We simulate our results on MSLR-WEB30K. To increase computational efficiency and following work by Hager et al. [32] and Fris and Hager [23] we truncate the number of documents per query to the 25 most relevant documents. This ensures that the logging policy is able to rank relevant documents, allowing for sufficient difference between different ranks.

8.1 Frequency-Based Propensity Estimation

First, we consider the frequency-based propensity estimation baseline, which has been applied in previous work when the original logging policy was not available. In Chapter 4, we have shown that bias is higher when fewer observations exist. We will now empirically investigate how effectively this method estimates propensity with varying numbers of observations for our frequency-based estimator. In Figure 8.1, we compare the estimated propensities to the oracle propensities for different numbers of observations. With only one observation, the estimator is highly inaccurate: the large standard deviation indicates that many estimates deviate substantially from the oracle values. As the number of observations increases, the standard deviation decreases and the estimates align more closely with the oracle propensities. This confirms the theoretical motivation that frequency-based propensity estimation performs worse in low-observation settings.

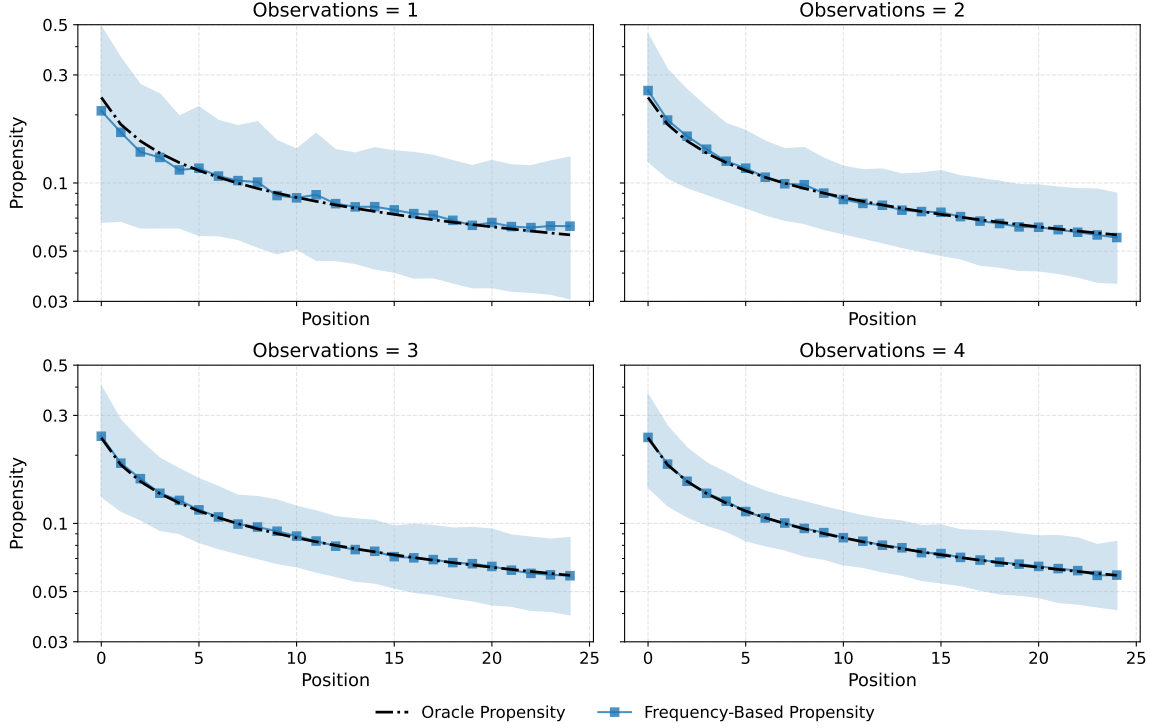


Figure 8.1: Frequency-based propensity estimation compared to the oracle propensity values for different amounts of observations of each query-document pair. Shown for from 1 to 4 observations.

8.2 Increasing Effective Samples Through Clustering Based on Feature Similarity

As shown in Section 6.2, when we assume the logging policy relies on features, we can use clustering based on feature similarity as a method to reach a larger effective sample size. This allows us to use all grouped values to make a propensity estimate, which can improve accuracy in sparse query settings. In this section, we investigate how methods for sample grouping can decrease variance in propensity estimation, by investigating how well methods can recover underlying propensities.

Setup We first rely on the definition of the neighborhood for a specific document, following Eq. 5.8. Here, we require a definition of distance, as well as a threshold value ϵ . We apply thresholding based on Euclidean distance between feature vectors, where a stricter threshold groups only query-document pairs with smaller feature-space distances. In our implementation, we apply KNN with a large number of neighbors (50), and then apply thresholding to these neighbor groups, in order to reduce computational cost.

Results The resulting mean differences and standard deviations are shown for between 1 and 8 observations in Figure 8.2. We observe that with increasing group thresholds, the deviation from the oracle propensity is increased, while the average standard deviation is reduced. For the frequency-based estimator, the standard deviation is high with few observations and decreases as more observations become available. In contrast, larger Euclidean grouping thresholds produce flatter curves. This reveals a clear bias-variance trade-off: less strict grouping reduces variance across all observation counts, but increases bias, while stricter grouping limits bias at the cost of smaller variance reduction.

The bias-variance trade-off that we observe can be explained through the distribution of

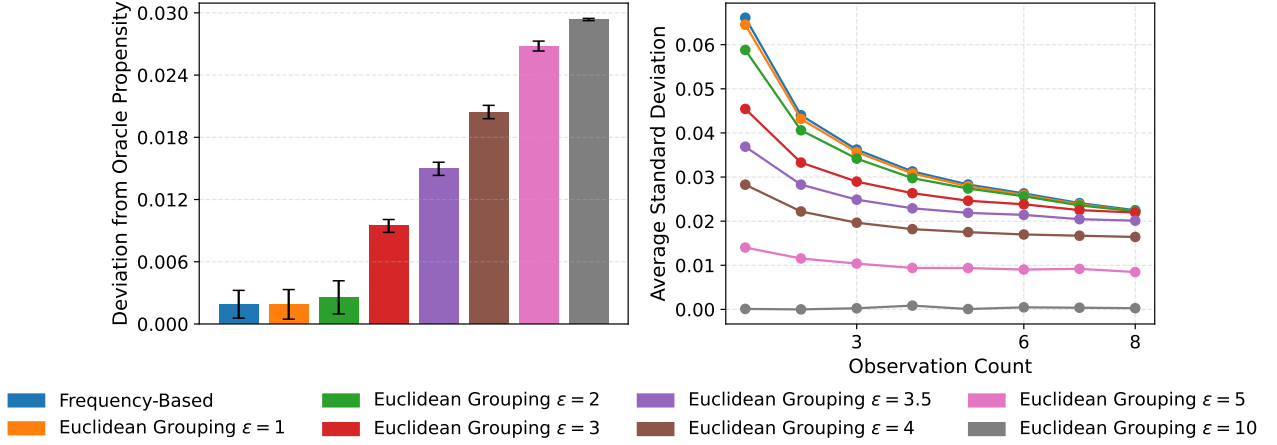


Figure 8.2: Frequency-based propensity estimation compared to Euclidean-distance grouped propensities with different threshold values.

samples. By sampling features in a larger neighborhood we gain observations, but lose specific information. Therefore, applying a grouping factor introduces smoothing of our distribution, which may lead to a biased propensity estimate. The smoothing introduced by Euclidean grouping is shown in Figure 8.3. We observe that although variance is reduced generally, especially for cases where we have a single observation, the grouping of values introduces a bias towards the center of logits, which remains consistent even as the number of observations increases.

To further investigate grouping methods, we also applied k -Means clustering and k -nearest neighbors, as well as a cosine-similarity-based grouping. Here, we found the same bias-variance trade-off. When using a smaller number of clusters, standard deviations are reduced at the cost of increased bias. As results are very similar to those for euclidean grouping, we include these results in Appendix C.

Overall, we observe that clustering methods can lead to a reduction in variance. However, the methods we evaluated also introduce a structural increase in bias. In addition to this, clustering methods are very dataset-specific, requiring users to tune this bias-variance trade-off per dataset, depending on the number of samples and feature distributions. Therefore, it may be useful to consider feature-based propensity that does not rely on grouping based on features.

8.3 Neural Feature-Based Propensity Estimation

In this section, we investigate whether supervised feature-based propensity estimation can further reduce variance while limiting bias.

Setup We perform minimal hyperparameter tuning to ensure generalizability and a fair comparison to untuned methods. Specifically, we run a grid search over the number of layers 2, 3, 4, neurons 64, 128, 256, and dropout 0.0, 0.1, 0.2, 0.3. The final model is a feed-forward neural network with 4 layers, 128 neurons, no dropout, and ELU activations. We optimize using AdamW [47] with a learning rate of 0.005, training for at most 100 epochs with early stopping after 3 epochs without validation improvement. We apply the same truncation as in Section 8.2.

Results We compare the MLP regressor to the best-performing grouping-based methods in Figure 8.4. The regressor performs competitively, with an average deviation of about 0.013 from the oracle propensity. Its standard deviation is substantially reduced and remains nearly

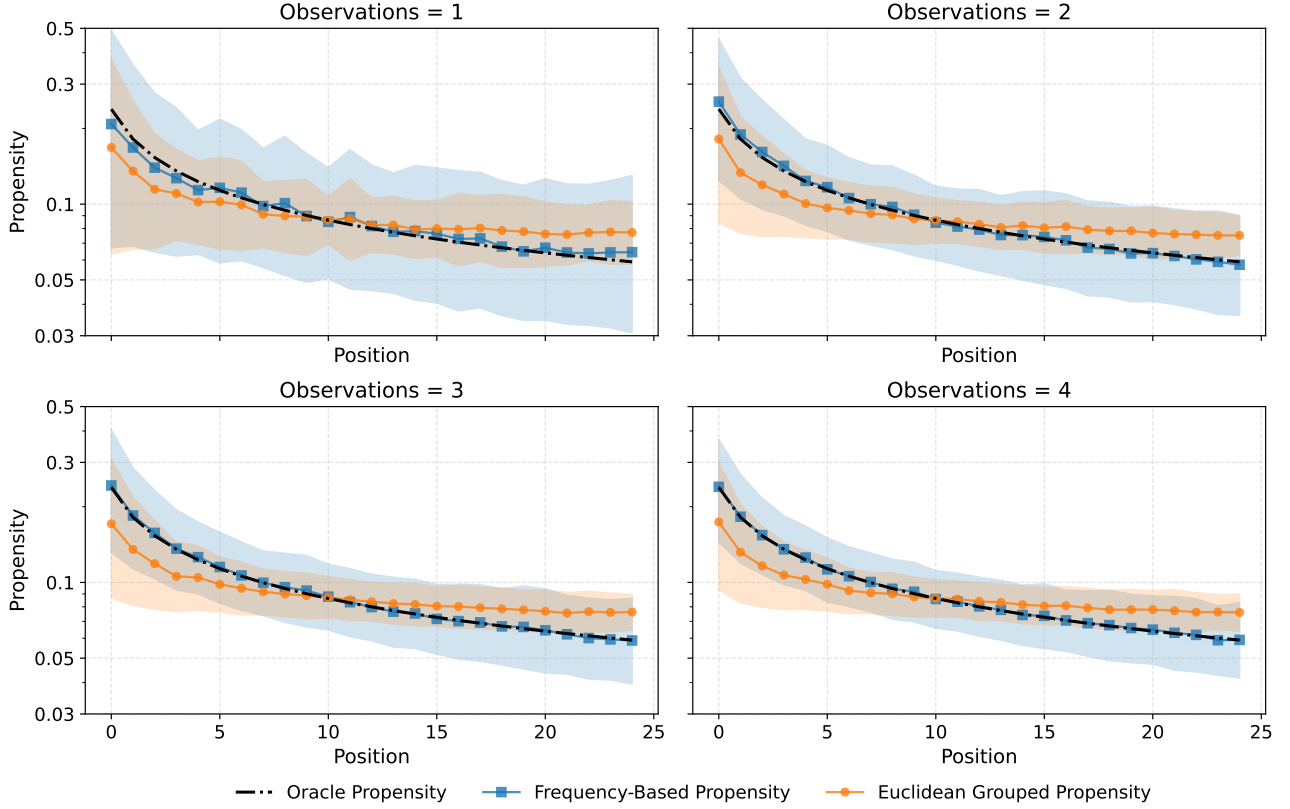


Figure 8.3: Frequency-based propensity estimation compared to Euclidean-distance grouped propensities with a threshold of $\epsilon = 3.5$ for different numbers of observations across positions. Shaded areas show standard deviations.

constant as the number of observations changes. The low variance for documents with few observations suggests that the model generalizes across documents and can estimate propensities accurately from features alone.

We further examine the distribution across positions in Figure 8.5. For documents with one or two observations, the feature-based estimator has much lower standard deviation than the frequency-based estimator across all positions, indicating that it predicts placement well even with limited observations. However, its mean is slightly below the true average at low positions and above it at high positions, suggesting a structural bias. This likely results from the MSE loss, which encourages conservative predictions and discourages very large estimates. As the number of observations increases, the variance of the frequency-based and regression-based estimates becomes more similar, although the feature-based method generally remains less variable.

8.4 Conclusion

To conclude, we answer **RQ3** in the affirmative: Feature-based propensity estimators can offer large improvements in variance, while only marginally increasing the bias, recovering the oracle propensity more effectively. We observe that the clustering-based propensity estimates introduce a clear bias-variance trade-off, with hyperparameters having a large effect on the outcome. We find that an MLP trained for propensity estimation can drastically reduce variance, while only introducing a small amount of bias. This is a positive signal for application of feature-based propensity estimation for counterfactual LTR, which we investigate in the following chapter.

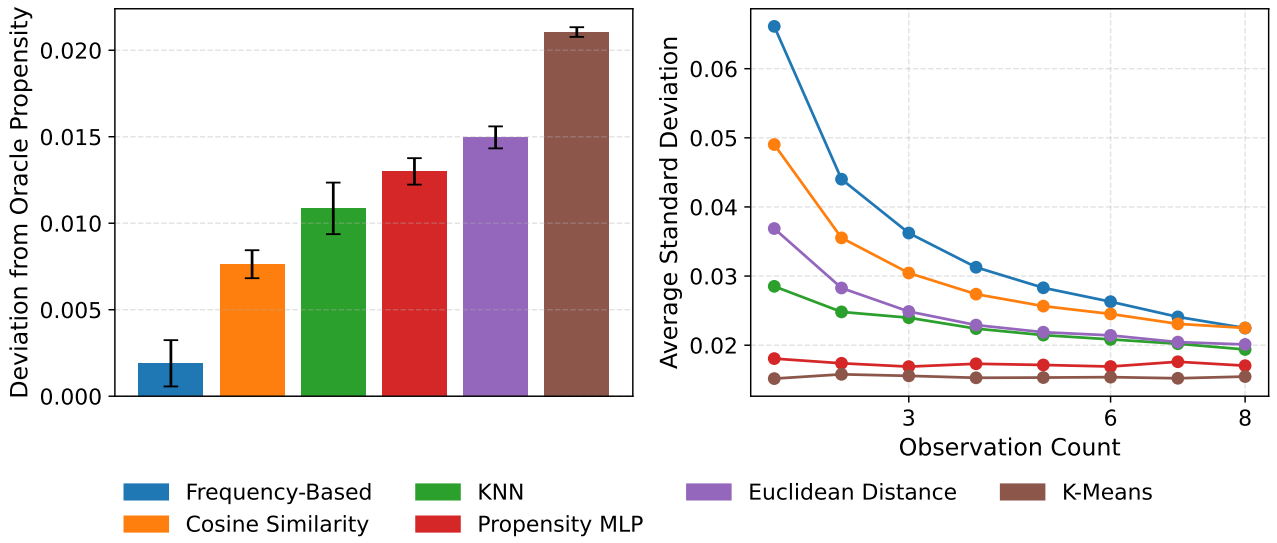


Figure 8.4: Mean and standard deviation of propensities estimated using frequency-based, clustering-based methods and supervised-learning based MLP methods.

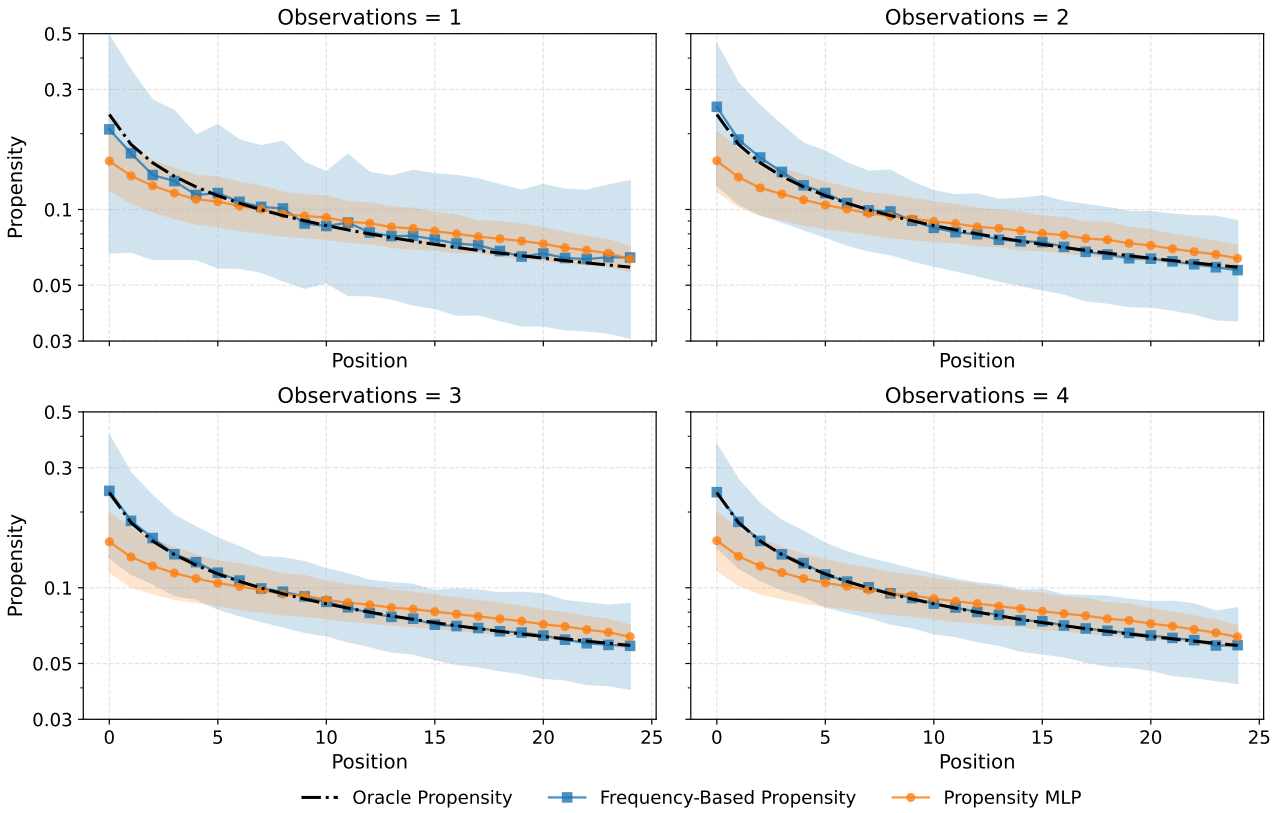


Figure 8.5: Frequency-based propensity estimation compared to propensities estimated using a feature-based MLP for different numbers of observations across positions. Shaded areas show standard deviations.

Chapter 9

RQ4: Feature-Based Propensity Estimation in Semi-Synthetic Data

In this section, we investigate how improvements in propensity estimation translate to performance on semi-synthetic datasets. We aim to answer **RQ4**: *to what extent feature-based propensity estimation can improve the performance of counterfactual LTR methods in semi-synthetic datasets*. We study several dataset settings to evaluate effectiveness of feature-based propensity estimation. We first investigate different sample sizes, then different logging policy settings, and finally, performance in a sparse observation setting.

9.1 Performance of Feature-Based Propensities Across Sample Sizes

We first evaluate how model performance changes as the amount of logged data increases. We consider sample sizes ranging from $N = 10^3$ to $N = 10^6$ on the Istella-S, MSLR-WEB30K, and Yahoo! datasets. For each dataset, we compare the learned models against two reference points: the performance of the logging policy and the performance of an oracle model trained directly on relevance labels.

Before evaluating ranking performance, we inspect the amount of document-level observation in the logged data. The setting with a varied amount of samples is also applied in related work, where we sample uniformly over the fixed set of queries. This sampling method introduces a specific distribution in the amount of observations for items implicitly as more samples lead to relatively more observations per document. In Figure 9.1, we show the proportion of documents that are observed only once. Across datasets, we observe that for sample sizes up to roughly 50,000 sessions, more than half of the documents have only a single observation. This indicates that the smaller sample settings represent highly sparse observational regimes.

To further characterize the logged data, we also report the average number of observations per document in Figure 9.2. As expected, the average number of observations increases with the number of sampled sessions. For all datasets, by approximately 500,000 sessions, documents have multiple observations on average. Together, these two visualizations show that our experimental setting covers both sparse and dense logging regimes. This allows us to compare frequency-based and feature-based propensity estimation in both sparse and dense observational frequencies.

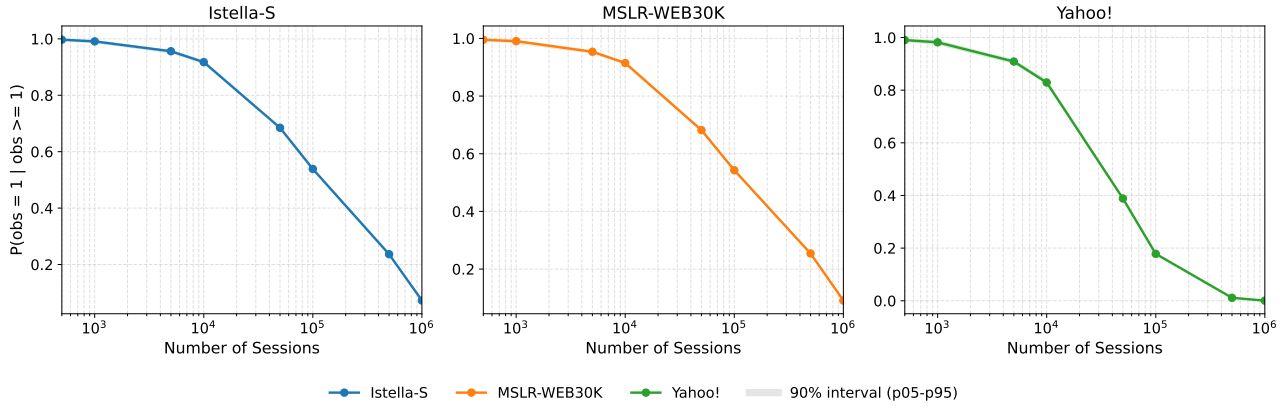


Figure 9.1: Single-observation proportion in the training data when generating different number of sessions for the Istella-S, MSLR-WEB30K, and Yahoo! datasets.

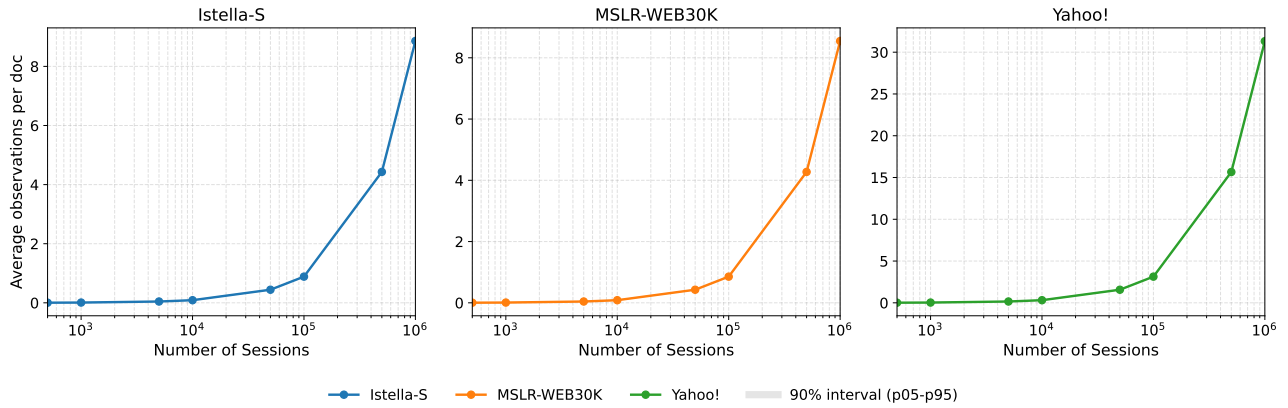


Figure 9.2: Average number of observations per document in the training data when generating different number of sessions for the Istella-S, MSLR-WEB30K, and Yahoo! datasets.

9.1.1 Performance of Feature-Based Propensity Estimation for Inverse Propensity Scoring

We first evaluate whether feature-based propensity estimation improves IPS-based learning-to-rank performance. The results are shown in Figure 9.3, where we compare NDCG scores across different sample sizes. We observe a clear improvement in NDCG scores when applying the propensity MLP rather than the existing frequency-based methods across all datasets. For all datasets, we observe performance improvements between $N = 10^3$ and $N = 10^4$. When looking at the MSLR-WEB30K and Yahoo! datasets, we observe performance improvements across all sample settings, consistently improving performance by $\approx 2\%$ for MSLR-WEB30K and $\approx 1.5\%$ for Yahoo!. For Istella-S, when taking very large numbers of samples, we observe that the difference in performance becomes smaller. This aligns with the expectation that frequency-based methods perform worse when we have few observations, but can recover performance when we take many samples.

When comparing the propensity MLP with the oracle propensity method, we observe that they are very similar in performance, with the propensity MLP even performing marginally better in many cases. This shows that the feature-based propensity method can improve consistently over frequency-based methods, performing at a similar level as the oracle propensity method. At first, this may appear surprising, since the oracle propensities are theoretically unbiased with respect to the logging policy. However, this does not necessarily imply that they produce the best empirical learning signal for a finite logged sample. The sampled dataset can deviate from the expected logging distribution, and a learned propensity model may provide a

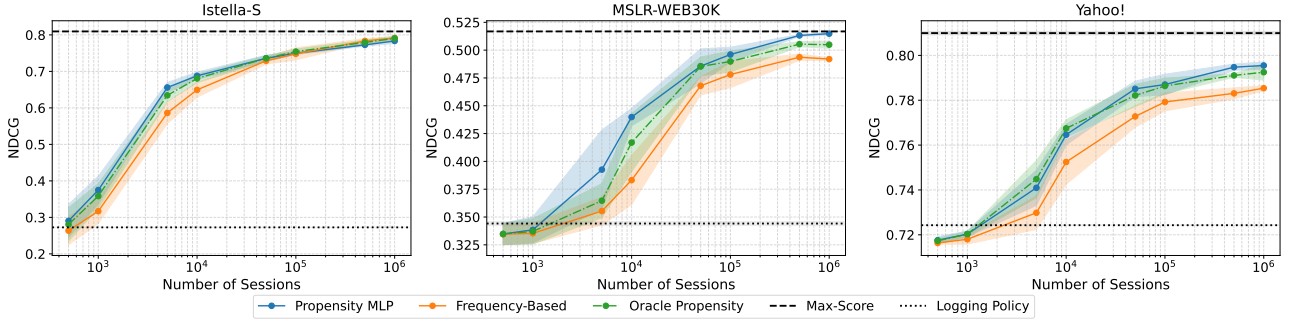


Figure 9.3: Performance of IPS estimation with different propensity estimation methods on NDCG at different sample size, shown for Istella-S, MSLR-WEB30K, and Yahoo!. Curves show means over random seeds; shaded bands indicate one standard deviation.

Table 9.1: IPS results for propensity estimation methods across sample counts. Columns are grouped by **sample count**, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-sample-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	N = 10 ³		N = 10 ⁴		N = 10 ⁵		N = 10 ⁶	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.337 (0.011)	0.455 (0.015)	0.417 (0.026) $\triangle$	0.590 (0.049) $\triangle$	0.490 (0.009) $\triangle$	0.711 (0.013) $\triangle$	0.505 (0.003) $\blacktriangle$	0.734 (0.004) $\blacktriangle$
	Frequency-Based	0.335 (0.009)	0.453 (0.012)	0.383 (0.022)	0.620 (0.266)	0.478 (0.012)	0.693 (0.017)	0.492 (0.001)	0.714 (0.003)
	MLP	0.338 (0.012)	0.457 (0.016)	0.440 (0.008)$\blacktriangle$	0.671 (0.128)$\triangle$	0.496 (0.007)$\blacktriangle$	0.719 (0.009)$\blacktriangle$	0.515 (0.001)$\blacktriangle$	0.746 (0.002)$\blacktriangle$
Istella-S	Oracle Propensity	0.358 (0.034) $\blacktriangle$	0.448 (0.047) $\blacktriangle$	0.681 (0.012) $\blacktriangle$	0.971 (0.020) $\blacktriangle$	0.754 (0.007)	1.094 (0.010)	0.791 (0.006)	1.151 (0.010)
	Frequency-Based	0.317 (0.034)	0.390 (0.045)	0.649 (0.020)	0.916 (0.033)	0.748 (0.017)	1.083 (0.026)	0.792 (0.003)	1.152 (0.006)
	MLP	0.375 (0.038)$\blacktriangle$	0.471 (0.053)$\blacktriangle$	0.688 (0.010)$\blacktriangle$	0.983 (0.018)$\blacktriangle$	0.750 (0.004)	1.085 (0.005)	0.783 (0.007) $\triangledown$	1.140 (0.011) $\triangledown$
Yahoo!	Oracle Propensity	0.720 (0.001) $\blacktriangle$	0.762 (0.002) $\blacktriangle$	0.767 (0.004) $\blacktriangle$	0.844 (0.006) $\blacktriangle$	0.786 (0.004) $\blacktriangle$	0.874 (0.005) $\blacktriangle$	0.792 (0.003) $\blacktriangle$	0.885 (0.005) $\blacktriangle$
	Frequency-Based	0.718 (0.002)	0.758 (0.004)	0.752 (0.010)	0.818 (0.017)	0.779 (0.004)	0.863 (0.006)	0.785 (0.001)	0.874 (0.002)
	MLP	0.720 (0.001)$\triangle$	0.762 (0.002)$\triangle$	0.765 (0.005)$\triangle$	0.839 (0.008)$\triangle$	0.787 (0.005)$\blacktriangle$	0.874 (0.007)$\blacktriangle$	0.795 (0.002)$\blacktriangle$	0.888 (0.002)$\blacktriangle$

smoother or more stable reward signal during neural network optimization. These factors may explain why the propensity MLP can occasionally outperform the oracle propensity baseline in practice.

Finally, we evaluate the significance levels of these results for $N = 10^3$, $N = 10^4$, $N = 10^5$, and $N = 10^6$. The results of this evaluation are shown in Table 9.1. Here, we find that results are consistently significant, with significant improvements in performance at $N = 10^4$ for all models. In addition, we find that significant performance increases are found for almost all sample sizes for MSLR-WEB30K and Yahoo!. Notably, we do find a significant decrease in performance for the MLP propensity estimator, although performance differences are very marginal and remain within one standard deviation of one another.

Overall, we conclude that for IPS, feature-based propensity estimation can lead to significantly improved performance. We find that this performance increase is consistent across different sample sizes. We find that the proposed feature-based propensity estimation method can even outperform the oracle propensity in many settings, which emphasizes the effectiveness. We note that the level of improvement is dataset-dependent, with limited improvement on the Istella-S dataset, especially in large sample sizes.

9.1.2 Performance of Feature-Based Propensity Estimation for the Direct Method and Doubly Robust estimation

We have shown that propensity estimation can offer effective improvements for policy optimization using IPS. Since both the DM and DR estimators also rely on propensity estimates, our theoretical analysis suggests that these methods may benefit from the same feature-based

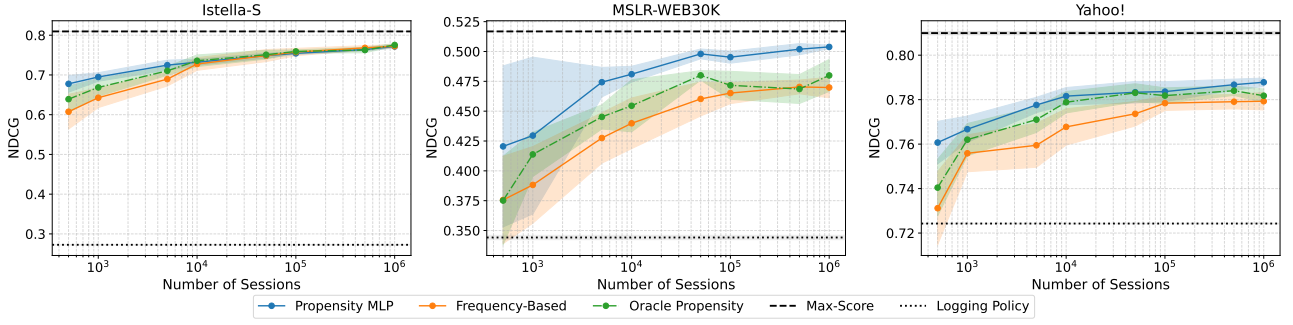


Figure 9.4: Performance of the DM with different propensity estimation methods on NDCG at different sample size, shown for Istella-S, MSLR-WEB30K, and Yahoo!. Curves show means over random seeds; shaded bands indicate one standard deviation.

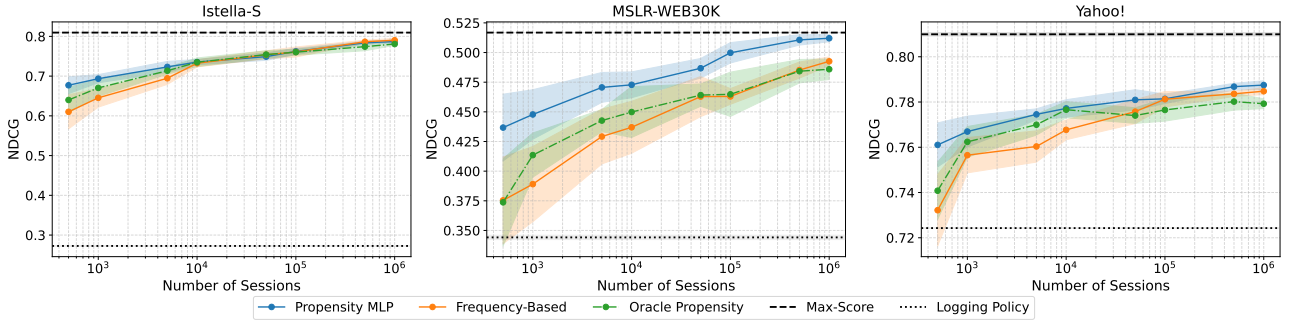


Figure 9.5: Performance of the DR estimator with different propensity estimation methods on NDCG at different sample sizes, shown for Istella-S, MSLR-WEB30K, and Yahoo!. Curves show means over random seeds; shaded bands indicate one standard deviation over random seeds.

approach. In this section, we evaluate whether the improvements observed for IPS also extend to DM and DR based optimization for unbiased LTR.

The results are shown in Figures 9.4 and 9.5. For both DR estimation and the DM, we observe a clear improvement in NDCG scores when using the propensity MLP instead of the existing frequency-based method. We observe that differences are largest in small sample settings, especially between 500 and 5000. A possible explanation for this is that the use of the propensity MLP improves the training objective of the regression model, allowing it to be more accurate with fewer samples. Similar to the evaluation on IPS, we find that improvements are larger on MSLR-WEB30K and Yahoo!, with especially large improvements on MSLR-WEB30K. For the Yahoo! dataset.

We find that performance of the oracle propensity model can behave erratically, even when compared to the frequency-based model for DR, although these differences are within each other’s standard-deviation bands. Similarly, we observe that performance on MSLR-WEB30K fluctuates for the MLP method, where the standard-deviation bands overlap. A likely explanation for this phenomenon is that when few samples are taken, the optimization method for the regression model can optimize to a different optimum, producing poor relevance estimates that do not generalize to the rest of the dataset.

When comparing the propensity MLP with the oracle propensity method, we again observe that they are very similar in performance for both DM and DR, with the propensity MLP even performing better in some settings.

We also study the statistical significance of the results for DM and DR. For DM, these results are shown in Table 9.2. Here, we again find that the largest differences occur at lower sample counts, while the methods become more similar as the number of sessions grows. For MSLR-WEB30K and Yahoo!, MLP remains consistently stronger than the frequency-based

Table 9.2: DM results for propensity estimation methods across sample counts. Columns are grouped by **sample count**, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-sample-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	N = 10 ³		N = 10 ⁴		N = 10 ⁵		N = 10 ⁶	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.414 (0.019) ^{$\triangle$}	0.589 (0.031) ^{$\triangle$}	0.455 (0.022) ^{$\blacktriangle$}	0.657 (0.029) ^{$\blacktriangle$}	0.472 (0.012) ^{$\triangle$}	0.684 (0.013) ^{$\blacktriangle$}	0.480 (0.013) ^{$\triangle$}	0.697 (0.018) ^{$\blacktriangle$}
	Frequency-Based	0.388 (0.032)	0.538 (0.055)	0.440 (0.021)	0.629 (0.029)	0.465 (0.009)	0.666 (0.010)	0.470 (0.009)	0.674 (0.013)
	MLP	0.430 (0.066)	0.617 (0.105)	0.481 (0.007)^{$\blacktriangle$}	0.696 (0.010)^{$\blacktriangle$}	0.495 (0.005)^{$\blacktriangle$}	0.720 (0.006)^{$\blacktriangle$}	0.504 (0.001)^{$\blacktriangle$}	0.733 (0.002)^{$\blacktriangle$}
Istella-S	Oracle Propensity	0.669 (0.022) ^{$\blacktriangle$}	0.946 (0.040) ^{$\blacktriangle$}	0.736 (0.015)	1.060 (0.025)	0.759 (0.004)	1.099 (0.006)	0.775 (0.003) ^{$\triangle$}	1.127 (0.005) ^{$\blacktriangle$}
	Frequency-Based	0.643 (0.025)	0.902 (0.044)	0.728 (0.016)	1.047 (0.026)	0.758 (0.009)	1.096 (0.014)	0.772 (0.004)	1.120 (0.007)
	MLP	0.695 (0.011)^{$\blacktriangle$}	0.993 (0.020)^{$\blacktriangle$}	0.732 (0.008)	1.055 (0.011)	0.754 (0.003)	1.091 (0.005)	0.772 (0.004)	1.121 (0.007)
Yahoo!	Oracle Propensity	0.762 (0.007) ^{$\blacktriangle$}	0.829 (0.013) ^{$\blacktriangle$}	0.779 (0.005) ^{$\blacktriangle$}	0.861 (0.007) ^{$\blacktriangle$}	0.782 (0.003) ^{$\triangle$}	0.866 (0.007) ^{$\triangle$}	0.782 (0.003)	0.869 (0.005)
	Frequency-Based	0.756 (0.008)	0.818 (0.015)	0.768 (0.008)	0.840 (0.014)	0.778 (0.003)	0.860 (0.006)	0.779 (0.004)	0.863 (0.007)
	MLP	0.767 (0.006)^{$\triangle$}	0.836 (0.011)^{$\triangle$}	0.782 (0.004)^{$\blacktriangle$}	0.864 (0.007)^{$\blacktriangle$}	0.784 (0.004)^{$\triangle$}	0.869 (0.008)^{$\triangle$}	0.788 (0.002)^{$\blacktriangle$}	0.878 (0.004)^{$\blacktriangle$}

Table 9.3: DR results for propensity estimation methods across sample counts. Columns are grouped by **sample count**, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-sample-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	N = 10 ³		N = 10 ⁴		N = 10 ⁵		N = 10 ⁶	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.414 (0.019) ^{$\triangle$}	0.589 (0.031) ^{$\triangle$}	0.450 (0.021) ^{$\triangle$}	0.650 (0.029) ^{$\triangle$}	0.465 (0.019)	0.669 (0.030)	0.486 (0.009)	0.697 (0.015) ^{$\triangledown$}
	Frequency-Based	0.389 (0.032)	0.539 (0.054)	0.437 (0.022)	0.627 (0.031)	0.463 (0.007)	0.670 (0.009)	0.493 (0.004)	0.715 (0.004)
	MLP	0.448 (0.021)^{$\triangle$}	0.645 (0.034)^{$\triangle$}	0.473 (0.011)^{$\blacktriangle$}	0.685 (0.015)^{$\blacktriangle$}	0.500 (0.009)^{$\blacktriangle$}	0.723 (0.013)^{$\blacktriangle$}	0.512 (0.003)^{$\blacktriangle$}	0.740 (0.007)^{$\blacktriangle$}
Istella-S	Oracle Propensity	0.670 (0.021) ^{$\blacktriangle$}	0.949 (0.038) ^{$\blacktriangle$}	0.735 (0.011)	1.061 (0.018)	0.760 (0.008)	1.099 (0.013)	0.781 (0.003) ^{$\blacktriangledown$}	1.131 (0.007) ^{$\blacktriangledown$}
	Frequency-Based	0.645 (0.023)	0.907 (0.042)	0.732 (0.009)	1.057 (0.016)	0.761 (0.013)	1.103 (0.020)	0.790 (0.005)	1.149 (0.007)
	MLP	0.694 (0.010)^{$\blacktriangle$}	0.991 (0.017)^{$\blacktriangle$}	0.735 (0.007)	1.062 (0.010)	0.762 (0.008)	1.107 (0.012)	0.787 (0.004)	1.146 (0.005)
Yahoo!	Oracle Propensity	0.762 (0.007) ^{$\blacktriangle$}	0.830 (0.012) ^{$\blacktriangle$}	0.777 (0.004) ^{$\blacktriangle$}	0.859 (0.005) ^{$\blacktriangle$}	0.776 (0.005)	0.857 (0.009)	0.779 (0.002) ^{$\blacktriangledown$}	0.860 (0.004) ^{$\blacktriangledown$}
	Frequency-Based	0.756 (0.008)	0.818 (0.014)	0.768 (0.004)	0.843 (0.008)	0.781 (0.003)	0.866 (0.005)	0.785 (0.002)	0.873 (0.002)
	MLP	0.767 (0.007)^{$\blacktriangle$}	0.837 (0.013)^{$\blacktriangle$}	0.777 (0.004)^{$\triangle$}	0.858 (0.006)^{$\triangle$}	0.781 (0.002)	0.867 (0.003)	0.787 (0.002)^{$\triangle$}	0.876 (0.003)

estimator, whereas for Istella-S performance increases are not as pronounced.

The significance testing for the DR estimator with different propensity estimators is shown in Table 9.3. Here we observe the same general pattern: gains from the MLP are clearest in lower-data settings, while differences shrink as the number of available sessions increases, with high performance on MSLR-WEB30K and less pronounced increases on Istella-S.

From these results, we conclude that for DM and DR, feature-based propensity can lead to significantly improved performance. We find that this performance increase is consistent across different sample sizes. Looking at all optimization, we find that for all modelling methods performance when comparing them for dataset and sample sizes, which shows that the improving Feature-based propensity estimation is relevant across methods, and while the effect size is highly dependent on the dataset setting, positive effects are observed across settings when applying feature-based propensity estimation.

To further verify the effectiveness of feature based propensity estimation, in the following sections we study the effects of different amounts of randomness in the logging policy for data gathering, as well as how models perform when we vary only the amount of the dataset seen, and constrain observations to a single sample.

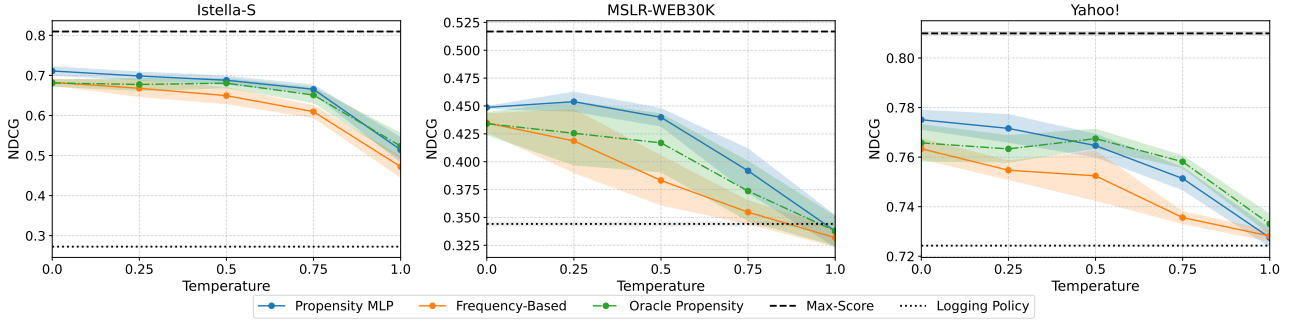


Figure 9.6: Performance of IPS estimation with different propensity estimation methods on NDCG in the real-target experiments at $N = 10^4$ sessions, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the logging policy temperature increases. Curves show means over random seeds; shaded bands indicate one standard deviation over random seeds.

9.2 Performance of Feature-Based Propensity Estimation for Counterfactual Learning-to-Rank Under Different Logging Policies

In previous sections, we have shown that the propensity estimator can lead to consistent performance improvements across sample sizes under a logging policy that shows a random ranking in 50% of settings. Following the theoretical motivation, the variance of the logging policy is an important factor, as it determines the magnitude of the bias introduced for frequency-based propensity estimation. This suggests that differences between feature-based and frequency-based propensity estimation may vary with different levels of randomness in the logging policy. In this section, we test this by evaluating performance of LTR methods with different levels of randomness through varying the temperature setting.

We evaluate results for IPS, DM, and DR in the $N = 10^4$ setting with policy temperatures $T \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. We show results for the NDCG metric, and again compare with an oracle and a single logging policy baseline, which is the $T = 0.5$ setting.

9.2.1 Performance of Feature-Based Propensity Estimation Under Different Logging Policies

We first investigate the results for IPS, which are shown in Figure 9.6. Here, we observe performance of feature-based IPS methods is consistently higher than frequency-based IPS methods. Differences are smallest in the fully deterministic ($T = 0.0$) and random ($T = 1$) settings. The limited improvement in the deterministic setting can be explained by the fully unbiased frequency-based IPS estimate. If the presented ranking is always the same, the frequency-based estimate is correct from the first sample (and as variance of the logging policy is 0, there is no bias in Theorem 4.2). The unbiasedness of frequency-based IPS estimation in the deterministic logging policy setting is also observed in the equivalence in performance of the oracle propensity baseline setting and the frequency-based estimate. Notably, the feature-based propensity model still outperforms the frequency-based estimate and oracle across all dataset settings. This could be explained by a more stable optimization when applying the feature-based propensity estimate, or be a result of a smoothing of propensities. In the fully random setting, we observe that performance of propensity estimation methods is very similar across datasets. We expect that this is a result of limited headroom for improvement by estimating the propensities well (as the correct convergence would mean equal placement probability across positions).

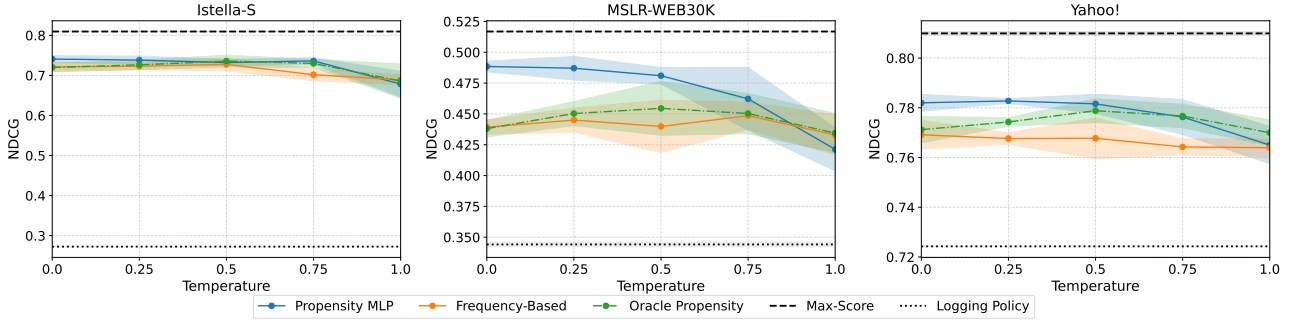


Figure 9.7: Performance of the Direct Method with different propensity estimation methods on NDCG in the real-target experiments at $N = 10^4$ sessions, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the logging policy temperature increases. Curves show means over random seeds; shaded bands indicate one standard deviation over random seeds.

When we compare performance across all different temperature settings for IPS, we observe that general performance decreases with higher temperature settings for frequency-based and oracle settings. When using the oracle propensity, however, this decrease is much less pronounced, as and performance remains more stable. This suggests that feature-based propensity estimation is more robust under different sampling settings, and can be applied effectively for diverse logging policy settings.

9.2.2 Performance of the Direct Method and Doubly Robust Estimation Under Different Logging Policies

We also study the performance of the propensity-estimation methods under different logging policy temperatures for DM and DR, as shown in Figures 9.7 and 9.8. Overall, we observe performance patterns that are similar to the IPS setting, although the differences between propensity-estimation methods are less pronounced. In particular, the improved performance of feature-based propensity estimation across temperature settings is also observed for MSLR-WEB30K and Yahoo!. For Istella-S, however, differences between propensity-estimation methods remain minimal across all settings. This is likely because the logging policy has a limited effect in this dataset, as the oracle propensity setting also does not outperform the frequency-based method. Furthermore, for MSLR-WEB30K, the feature-based propensity method is outperformed by the frequency-based method under the fully random logging policy setting, although the standard-deviation bands overlap. In addition to the figures in this section, we also provide the result tables for IPS, DM, and DR, including significance tests, which are shown in Appendix E.

Overall, we have found that the feature-based propensity estimator performs consistently across different levels of randomness in the logging policy. We find effects are strongest in settings where some randomness is present, but the logging policy is not fully random. This is a common setting in real-world scenarios, as we might expect practitioners to balance policy exploration and good downstream rankings. Overall, we observe robust performance for feature-based propensity, finding that performance is less affected by randomness than other propensity estimators. Differences are found to be largest in the IPS model, but also clear in DM and DR settings.

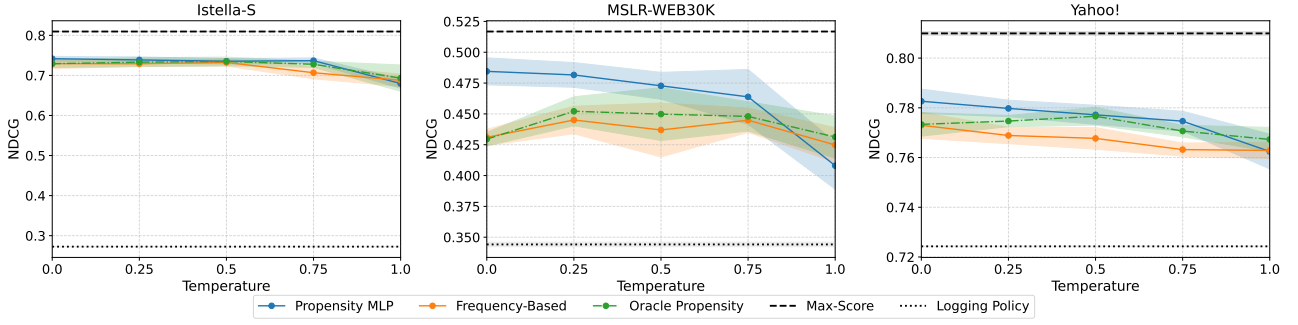


Figure 9.8: Performance of the Doubly-Robust estimator with different propensity estimation methods on NDCG in the real-target experiments at $N = 10^4$ sessions, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the logging policy temperature increases. Curves show means over random seeds; shaded bands indicate one standard deviation over random seeds.

9.3 Performance of Feature-Based Propensity Estimation in Sparse Data Settings

As discussed in Section 9.1, in several evaluation settings from previous work, only the sample size is used as a parameter, which means that an increase in data and an increase in observations per document are highly correlated. In this section, we investigate the effect of only increasing the number of queries in our dataset, while keeping the observation distribution sparse.

To do so, we constraint the sampling of queries in the training and validation set to sampling without replacement, meaning that each query-document pair can only be sampled once. To increase dataset coverage under the top-25 ranking setting, we randomly split queries into subsets of 25 documents, and apply each as a separate set of query-document pairs. We evaluate effectiveness of IPS, DM and DR under the three propensity estimators, while increasing the percentage of query-document sets selected (we apply $N \in \{10\%, 25\%, 50\%, 75\%, 100\%\}$). The test set remains unchanged in this setting.

9.3.1 Performance of Propensity Estimation Methods for Inverse Propensity Scoring in Sparse Data

We first investigate results for IPS, as shown in Figure 9.9. Here we observe a strong and consistent increase in performance for feature-based propensity estimation across all methods, with especially large improvements in MSLR-WEB30K and Yahoo!. Comparing the performance of frequency-based estimation and the oracle propensity baseline method, we observe a large disjoint, with frequency-based methods being consistently outperformed across session sizes. This confirms that frequency-based propensity estimation performs poorly when few observations are taken per document, and suggests that these issues are independent of dataset size. The results indicate that for IPS estimation, feature-based propensity estimation can effectively resolve poor propensity estimation issues in sparse data settings, even outperforming the oracle propensity in many cases.

9.3.2 Performance of Propensity Estimation Methods for Direct Method and Doubly Robust in Sparse Data

We now consider the results for DM and DR in the sparse distributional setting, shown in Figures 9.10 and 9.11. For DM, performance generally improves across all methods, although the differences are much smaller than for IPS. Standard-deviation bands often overlap, and performance is marginally worse for MSLR-WEB30K in the high-session setting. Differences between

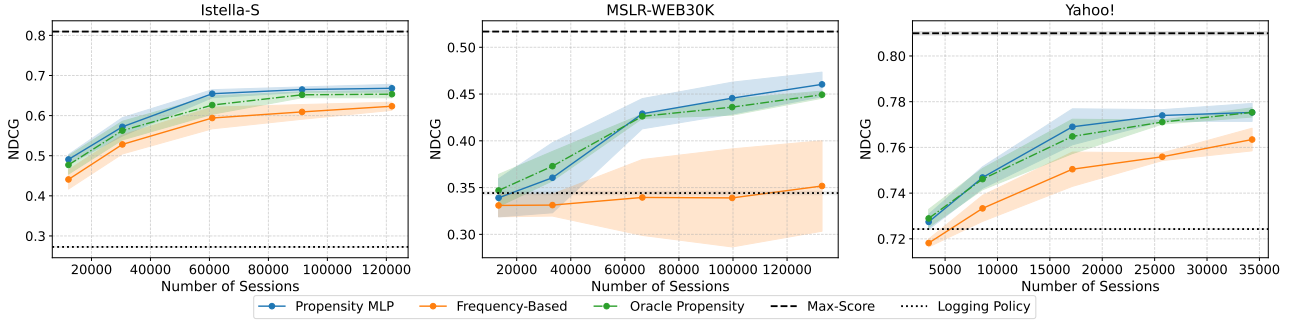


Figure 9.9: Performance of IPS estimation with different propensity estimation methods on NDCG for the disjoint-query-chunk experiments, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the number of available sessions increases. The final point in each curve corresponds to 100% of the available sessions, meaning that no more (splitted) queries can be sampled. Curves show means over random seeds; shaded bands indicate one standard deviation.

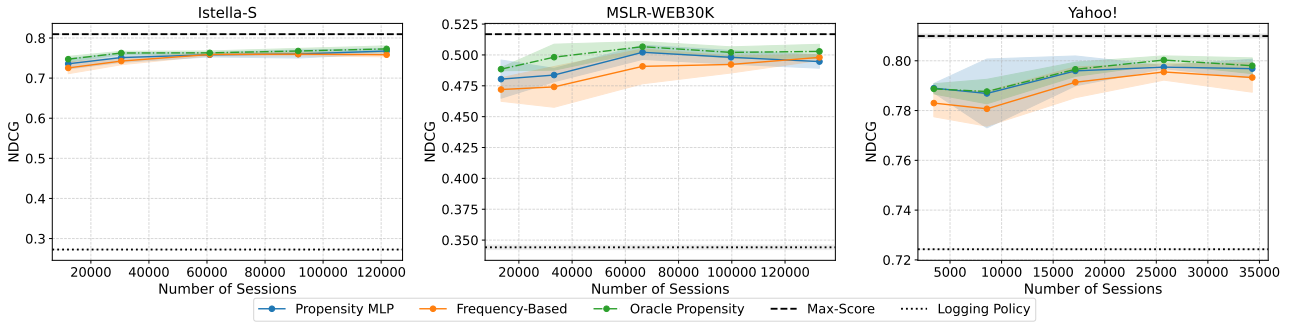


Figure 9.10: Performance of the Direct Method with different propensity estimation methods on NDCG for the disjoint-query-chunk experiments, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the number of available sessions increases. The final point in each curve corresponds to 100% of the available sessions, meaning that no more (splitted) queries can be sampled. Curves show means over random seeds; shaded bands indicate one standard deviation.

the oracle propensity method and frequency-based methods are also small, although models using frequency-based propensity estimates perform consistently better. These results suggest that feature-based propensity estimation can improve DM, but its effect appears smaller than for IPS.

Looking at the results for DR in Figure 9.11, we observe a counterintuitive pattern: performance decreases as the number of samples increases. This can be explained by the theoretical analysis of the DR estimator under the frequency-based propensity estimate in Section 4.5. In the current DR optimization approach, the IPS estimate is used whenever data are available, while the DM is used only when no data are available. With a limit of a single observation, the IPS estimate from that observation may be worse than the estimate produced by the relevance model in the DM. Consequently, as a larger part of the dataset is estimated through IPS, performance may decrease.

This behavior becomes clearer when comparing DM, DR, and IPS under the frequency-based propensity estimate, as shown in Figure 9.12. Here, DM and DR start at similar performance, but DR gradually converges to the exact performance of IPS. This exposes a clear failure mode of DR estimation which, to the best of our knowledge, has not yet been identified for the DR estimator in LTR. In addition, in Figure 9.11, when performance decreases, the frequency-based propensity estimate sometimes outperforms the oracle and feature-based propensities. A possible explanation could be that increased propensity accuracy may adversely affect weighting when the correction itself is an incorrect signal.

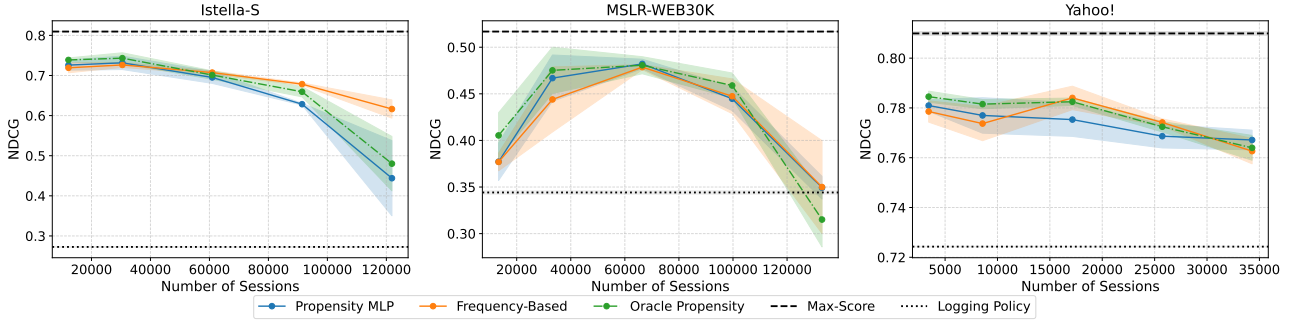


Figure 9.11: Performance of the Doubly-Robust estimator with different propensity estimation methods on NDCG for the disjoint-query-chunk experiments, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the number of available sessions increases. The final point in each curve corresponds to 100% of the available sessions, meaning that no more (splitted) queries can be sampled. Curves show means over random seeds; shaded bands indicate one standard deviation.

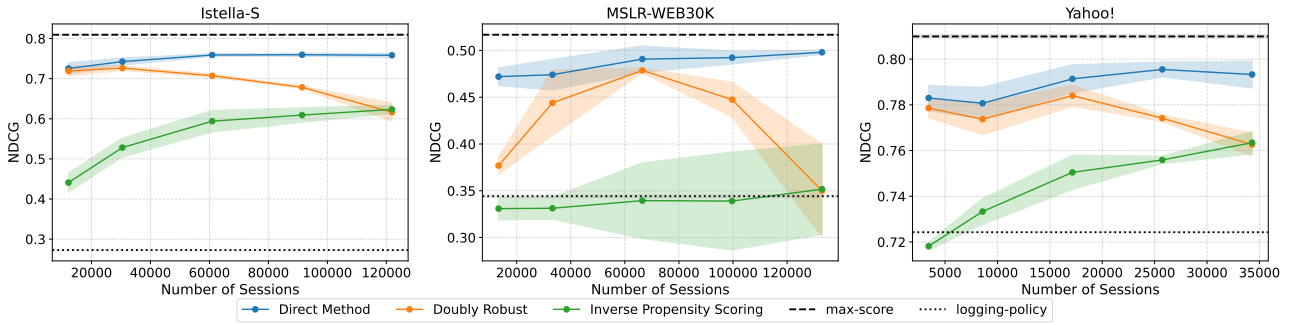


Figure 9.12: Comparison of IPS, DM, and DR with frequency-based propensity estimates on NDCG for the disjoint-query-chunk experiments, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the number of available sessions increases. The final point in each curve corresponds to 100% of the available sessions, meaning that no more (splitted) queries can be sampled. Curves show means over random seeds; shaded bands indicate one standard deviation.

9.4 Conclusion

We summarize the findings of this chapter, and formulate an answer to **RQ4**. Initially, we investigated performance of feature-based propensity estimation across different sample sizes. Here, we found large performance increases over frequency-based across evaluated counterfactual LTR methods, finding improvements across most sample sizes and dataset settings, and outperforming even the full information oracle propensity in many settings. Following this, we studied the effect of varying the logging policy, where we again found strong performance increases for feature-based propensity estimation, with especially large improvements for IPS. Finally, we investigated a sparse data setting, where each query-document pair was observed at most once. Here, we found strong improvements for feature-based propensity estimation methods across all dataset sizes for IPS. In addition, we identified an issue in doubly-robust estimation, where increasing the number of samples can lead to a decrease in performance in the sparse sample setting. To conclude, we answer **RQ4** in the affirmative, feature-based propensity estimation can significantly and consistently outperform existing propensity across dataset sizes, logging policies and in sparse settings.

Chapter 10

RQ5: Two-Tower Models for Counterfactual Learning-to-Rank

In this chapter, we investigate **RQ5**: *whether position bias and document relevance for policy optimization methods in LTR can be effectively estimated using two-tower models*. We split this research question into three separate components:

1. Can the bias tower of an optimized two-tower model be effectively applied as position bias estimate in policy optimization methods?
2. To what extent can the optimized relevance tower be applied as a relevance target for the DM and DR estimation?
3. Can jointly estimating the position bias and relevance estimate for DR estimation improve performance in semi-synthetic data?

We answer each of these subquestions in sequence in dedicated sections. We again apply synthetic datasets, with the same logging policy settings as described in Chapter 7. For each section, we study both the widely used additive two-tower model, as well as the proposed multiplicative two-tower model architecture.

10.1 Position Bias Estimation for Counterfactual Learning-To-Rank Using Two-Tower Models

We first study the application of the two-tower model bias tower for counterfactual LTR methods, where we compare results to an oracle position bias, as well as baseline position bias estimation based on intervention harvesting [22, 3]. We evaluate performance of counterfactual LTR models trained using IPS, DM, and DR on MSLR-WEB30K, Yahoo!, and Istella-S for $N \in \{10^4, 10^5\}$ with frequency-based propensity estimates, as shown in Table 10.1. When looking at the results, we observe that differences across all methods are generally small, with performance differences compared to the best method being non-significant for most models. We do observe that adjacent chain and pivot one performance appear to be generally lower, with results often being significantly worse than the best method, and the best result in only a single setting. When comparing the results for the two-tower models, we observe the additive two-tower model performs well, with results being significantly worse in only a single case, and performing similar or better than other models in all other settings. For the multiplicative two-tower model performance appears less stable, with some results having very high standard deviation ($N = 10^5$ for DM on MSLR-WEB30K and Istella-S). In other settings, performance

Table 10.1: Comparison of position bias estimation methods for counterfactual LTR under the frequency-based propensity setting on MSLR-WEB30K, Yahoo!, and Istella-S. Rows are grouped by dataset, and columns are grouped first by counterfactual learning objective and then by sample count. Cells show mean (SD) NDCG. Bold marks the best non-oracle model within each comparison block. Significance markers compare each other model with this best model using a paired two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Bias Source	IPS		DM		DR	
		N = 10 ⁴	N = 10 ⁵	N = 10 ⁴	N = 10 ⁵	N = 10 ⁴	N = 10 ⁵
MSLR-WEB30K	True Bias ^{oracle}	0.397 (0.013)	0.483 (0.011)	0.451 (0.007)	0.464 (0.012)	0.446 (0.008)	0.462 (0.010)
	Add. Two-Tower Bias	0.397 (0.011)	0.483 (0.005)	0.421 (0.017) $\triangledown$	0.470 (0.005)	0.421 (0.009)	0.469 (0.012)
	Mul. Two-Tower Bias	0.396 (0.006) $\triangledown$	0.429 (0.005) $\blacktriangledown$	0.415 (0.015)	0.408 (0.080)	0.418 (0.012)	0.408 (0.024)
	Adjacent Chain	0.414 (0.010)	0.345 (0.001) $\blacktriangledown$	0.328 (0.008) $\blacktriangledown$	0.373 (0.012) $\blacktriangledown$	0.408 (0.029)	0.336 (0.006) $\blacktriangledown$
	CTR	0.389 (0.025)	0.481 (0.002)	0.440 (0.013)	0.472 (0.002)	0.434 (0.014)	0.468 (0.010)
	Global All Pairs	0.350 (0.020) $\blacktriangledown$	0.486 (0.003)	0.362 (0.022) $\blacktriangledown$	0.474 (0.009)	0.346 (0.015) $\blacktriangledown$	0.471 (0.016)
	Pivot One	0.411 (0.006)	0.396 (0.024) $\triangledown$	0.328 (0.008) $\blacktriangledown$	0.400 (0.025) $\triangledown$	0.411 (0.030)	0.365 (0.038)
Yahoo!	True Bias ^{oracle}	0.744 (0.004)	0.776 (0.001)	0.768 (0.005)	0.777 (0.002) $\triangledown$	0.767 (0.002)	0.780 (0.004)
	Add. Two-Tower Bias	0.733 (0.001)	0.777 (0.002)	0.742 (0.004)	0.769 (0.001) $\blacktriangledown$	0.742 (0.002) $\triangledown$	0.779 (0.002)
	Mul. Two-Tower Bias	0.728 (0.003)	0.770 (0.011)	0.741 (0.001)	0.783 (0.001)	0.739 (0.004)	0.770 (0.007)
	Adjacent Chain	0.723 (0.006) $\triangledown$	0.750 (0.005) $\triangledown$	0.715 (0.002) $\triangledown$	0.741 (0.003) $\blacktriangledown$	0.729 (0.006) $\triangledown$	0.751 (0.003) $\blacktriangledown$
	CTR	0.739 (0.011)	0.782 (0.005)	0.763 (0.010)	0.779 (0.006)	0.764 (0.009)	0.774 (0.004)
	Global All Pairs	0.718 (0.002)	0.776 (0.003)	0.723 (0.002) $\triangledown$	0.774 (0.008)	0.719 (0.007) $\triangledown$	0.775 (0.003) $\triangledown$
	Pivot One	0.719 (0.004) $\triangledown$	0.776 (0.003)	0.723 (0.010) $\triangledown$	0.768 (0.005) $\triangledown$	0.721 (0.008) $\blacktriangledown$	0.775 (0.005)
Istella-S	True Bias ^{oracle}	0.651 (0.017)	0.743 (0.020)	0.719 (0.013) $\triangle$	0.754 (0.005)	0.728 (0.007)	0.755 (0.016)
	Add. Two-Tower Bias	0.635 (0.005)	0.742 (0.011)	0.689 (0.018)	0.756 (0.010)	0.701 (0.014)	0.751 (0.012)
	Mul. Two-Tower Bias	0.622 (0.020)	0.611 (0.016) $\blacktriangledown$	0.676 (0.014)	0.463 (0.216)	0.682 (0.023)	0.615 (0.021) $\triangledown$
	Adjacent Chain	0.573 (0.038)	0.547 (0.037) $\blacktriangledown$	0.234 (0.016) $\blacktriangledown$	0.545 (0.050) $\triangledown$	0.635 (0.024)	0.487 (0.023) $\blacktriangledown$
	CTR	0.590 (0.005) $\triangledown$	0.722 (0.006) $\triangledown$	0.700 (0.017)	0.759 (0.009)	0.695 (0.027)	0.737 (0.024)
	Global All Pairs	0.453 (0.022) $\blacktriangledown$	0.752 (0.010)	0.428 (0.071) $\triangledown$	0.762 (0.005)	0.379 (0.114) $\triangledown$	0.764 (0.007)
	Pivot One	0.568 (0.022) $\triangledown$	0.743 (0.005)	0.234 (0.016) $\blacktriangledown$	0.754 (0.001)	0.631 (0.025)	0.753 (0.004)

is comparable to the additive two-tower model, although performance is slightly lower. Overall, we observe that there is no clear best-performing model. Each model has at least one setting in which it performs significantly worse than another model, which indicates that performance may be dataset dependent. Results when using two-tower models for position are similar to results for other bias estimation methods, which indicates that in this setting, two-tower models can be applied for bias estimations.

Previously discussed results have focused on the effect of bias estimation on performance under a frequency-based logging policy estimate. To study the effect of propensity estimation methods under different logging policy estimates, we evaluate performance under the feature-based logging policy. The results of these experiments, which follow the same structure as frequency-based results, are shown in Table 10.2. Looking at results, we observe that in general, differences between models are larger and more significant, where in the frequency-based results 19 results were found to be strongly significant, we now find 26 strongly significant results. Looking at two-tower models for bias estimation, we again observe performance that is generally similar to other models. We find that the additive two-tower model is significantly worse than the best model in six settings, whereas it performs the best in two settings. For the multiplicative two-tower model we find similar results to previous experiments, with occasionally high fluctuation in performance, while performance may be better in other settings. Overall, we observe that the CTR leads to the best performance when using feature-based propensity estimation, with significantly worse results in only a single setting, and best results in nine settings. This is an unexpected result, given that the CTR is the most simple technique, which may suffer from placement biases due to the logging policy placing more relevant documents higher, introducing a confounding effect between CTR and position that is independent of relevance. The improved performance suggests a bias-variance trade-off, as CTR does not require joint observation of query-document pairs for bias estimation, and may therefore

Table 10.2: Comparison of feature-based propensity estimation under different position bias estimation methods for counterfactual LTR on MSLR-WEB30K, Yahoo!, and Istella-S. Rows are grouped by dataset, and columns are grouped first by counterfactual learning objective and then by sample count. Cells show mean (SD) NDCG. Bold marks the best non-oracle model within each comparison block. Significance markers compare each other model with this best model using a paired two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Bias Source	IPS		DM		DR	
		N = 10^4	N = 10^5	N = 10^4	N = 10^5	N = 10^4	N = 10^5
MSLR-WEB30K	True Bias ^{oracle}	0.437 (0.014)	0.500 (0.002)	0.481 (0.007)	0.496 (0.006)	0.476 (0.008)	0.499 (0.001)
	Add. Two-Tower Bias	0.416 (0.020)	0.499 (0.003)	0.470 (0.007) $\blacktriangledown$	0.503 (0.002)	0.466 (0.001) $\triangledown$	0.488 (0.014)
	Mul. Two-Tower Bias	0.401 (0.008) $\triangledown$	0.489 (0.014)	0.416 (0.076)	0.502 (0.004)	0.447 (0.013)	0.308 (0.017) $\blacktriangledown$
	Adjacent Chain	0.428 (0.010) $\blacktriangledown$	0.487 (0.004) $\triangledown$	0.439 (0.018) $\triangledown$	0.480 (0.006) $\triangledown$	0.268 (0.016) $\blacktriangledown$	0.447 (0.041)
	CTR	0.440 (0.006)	0.501 (0.002)	0.483 (0.006)	0.502 (0.003)	0.473 (0.001)	0.478 (0.014)
	Global All Pairs	0.442 (0.008)	0.499 (0.003)	0.481 (0.009)	0.499 (0.003)	0.459 (0.006) $\triangledown$	0.496 (0.008)
	Pivot One	0.426 (0.009) $\blacktriangledown$	0.502 (0.001)	0.449 (0.006) $\triangledown$	0.505 (0.003)	0.267 (0.020) $\blacktriangledown$	0.483 (0.022)
Yahoo!	True Bias ^{oracle}	0.764 (0.001)	0.784 (0.002) $\blacktriangledown$	0.784 (0.002)	0.784 (0.007)	0.780 (0.001)	0.781 (0.003)
	Add. Two-Tower Bias	0.734 (0.001) $\blacktriangledown$	0.786 (0.005)	0.754 (0.003) $\blacktriangledown$	0.785 (0.001) $\triangledown$	0.754 (0.007) $\triangledown$	0.781 (0.001)
	Mul. Two-Tower Bias	0.734 (0.008) $\triangledown$	0.794 (0.001)	0.750 (0.006) $\blacktriangledown$	0.797 (0.003)	0.751 (0.014)	0.677 (0.006) $\blacktriangledown$
	Adjacent Chain	0.717 (0.003) $\blacktriangledown$	0.775 (0.000) $\blacktriangledown$	0.749 (0.003) $\blacktriangledown$	0.765 (0.002) $\blacktriangledown$	0.703 (0.006) $\blacktriangledown$	0.766 (0.003) $\blacktriangledown$
	CTR	0.770 (0.007)	0.789 (0.005)	0.788 (0.003)	0.790 (0.002) $\triangledown$	0.779 (0.003)	0.781 (0.002)
	Global All Pairs	0.766 (0.005)	0.786 (0.004)	0.781 (0.002) $\blacktriangledown$	0.784 (0.003) $\blacktriangledown$	0.776 (0.005)	0.777 (0.002)
	Pivot One	0.721 (0.002) $\blacktriangledown$	0.784 (0.004)	0.760 (0.010) $\triangledown$	0.785 (0.002) $\triangledown$	0.743 (0.007) $\blacktriangledown$	0.778 (0.004)
Istella-S	True Bias ^{oracle}	0.699 (0.007)	0.749 (0.002)	0.739 (0.005)	0.755 (0.004)	0.737 (0.009)	0.758 (0.009)
	Add. Two-Tower Bias	0.666 (0.007)	0.750 (0.009)	0.716 (0.014)	0.757 (0.003)	0.723 (0.009)	0.728 (0.033)
	Mul. Two-Tower Bias	0.657 (0.028)	0.741 (0.010)	0.698 (0.046)	0.640 (0.049) $\triangledown$	0.687 (0.031)	0.226 (0.156) $\triangledown$
	Adjacent Chain	0.667 (0.017)	0.748 (0.007)	0.616 (0.011) $\blacktriangledown$	0.756 (0.006)	0.074 (0.006) $\blacktriangledown$	0.473 (0.182)
	CTR	0.681 (0.007)	0.745 (0.002)	0.727 (0.004)	0.762 (0.010)	0.712 (0.026)	0.697 (0.042)
	Global All Pairs	0.678 (0.006)	0.746 (0.003)	0.725 (0.008)	0.749 (0.003)	0.719 (0.010)	0.761 (0.008)
	Pivot One	0.661 (0.019)	0.746 (0.002)	0.656 (0.013) $\blacktriangledown$	0.749 (0.003)	0.031 (0.001) $\blacktriangledown$	0.760 (0.009)

estimate bias effectively from fewer samples. Additionally, the combination of a weak logging policy trained on a small subset of data, and a partially random logging policy may reduce the effect of confounding bias.

The results in this section not only allow us to evaluate effectiveness of two-tower models for position bias estimation, but also the effectiveness of feature-based propensity estimation under different bias estimation strategies. To study these effects, we compute the difference between frequency-based propensity estimation and feature-based propensity estimation, as shown in Table 10.3. Here a positive number indicates an increase in performance for feature-based propensity estimation. Looking at results, we observe improvements in performance in the majority of settings, with over 50 statistically significant performance increases across all model-propensity estimation strategies. Looking at performance for models which perform well in bias estimation, we observe that for both CTR and the additive two-tower bias model performance is increased in all but a single setting. Additionally, when we look at the best models in absolute performance marked in bold, we observe that performance is improved in all but two settings, where performance decrease is non-significant, and very small. These results suggest that the feature-based propensity estimator still performs well when we estimate position bias across all estimation methods. Importantly, we find that performance increases are also consistently found in the models which perform best, which suggests feature-based propensity estimation can increase top performance across datasets, models and sample sizes.

We also evaluate results for different bias sources when the oracle propensity estimation described in Chapter 7 is used. These results are very similar to those of experiments shown here, and are included for completeness in Appendix F.

To conclude, we find a partially positive answer to the first part of **RQ5**: the bias tower of an optimized two-tower model can be used as position bias estimate, leading to similar performance as existing intervention harvesting methods. We do find that performance is not

Table 10.3: Paired difference between the feature-based MLP propensity estimator and the frequency-based propensity estimator on MSLR-WEB30K, Yahoo!, and Istella-S for different position bias estimates. Rows are grouped by dataset, and columns are grouped first by counterfactual learning objective and then by sample count. Cells show $\Delta = \text{MLP} - \text{frequency-based}$ in NDCG, with the standard deviation of paired differences in parentheses. Bold marks the row whose highest absolute mean NDCG across the two propensity estimators is largest within each dataset-objective-sample block. Significance markers use a paired two-sided Student t-test against zero difference: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Bias Source	IPS		DM		DR	
		N = 10^4	N = 10^5	N = 10^4	N = 10^5	N = 10^4	N = 10^5
MSLR-WEB30K	True Bias ^{oracle}	+0.040 (0.010) $\triangle$	+0.017 (0.010)	+0.030 (0.014)	+0.032 (0.009) $\triangle$	+0.029 (0.016)	+0.036 (0.014)
	Add. Two-Tower Bias	+0.019 (0.009)	+0.016 (0.003) $\blacktriangle$	+0.050 (0.024)	+0.033 (0.002) $\blacktriangle$	+0.044 (0.008) $\triangle$	+0.019 (0.018)
	Mul. Two-Tower Bias	+0.005 (0.012)	+0.060 (0.009) $\blacktriangle$	+0.000 (0.074)	+0.095 (0.081)	+0.028 (0.023)	-0.100 (0.035) $\triangledown$
	Adjacent Chain	+0.014 (0.019)	+0.142 (0.005) $\blacktriangle$	+0.111 (0.025) $\triangle$	+0.107 (0.006) $\blacktriangle$	-0.140 (0.035) $\triangledown$	+0.111 (0.041) $\triangle$
	CTR	+0.051 (0.030)	+0.019 (0.001) $\blacktriangle$	+0.043 (0.019)	+0.030 (0.002) $\blacktriangle$	+0.040 (0.013)$\triangle$	+0.010 (0.011)
	Global All Pairs	+0.092 (0.027)$\triangle$	+0.013 (0.002) $\triangle$	+0.119 (0.031) $\triangle$	+0.025 (0.006) $\triangle$	+0.112 (0.016) $\blacktriangle$	+0.025 (0.017)
	Pivot One	+0.015 (0.014)	+0.105 (0.025)$\triangle$	+0.121 (0.011) $\blacktriangle$	+0.104 (0.027)$\triangle$	-0.144 (0.041) $\triangledown$	+0.118 (0.059)
Yahoo!	True Bias ^{oracle}	+0.019 (0.004) $\triangle$	+0.008 (0.002) $\triangle$	+0.016 (0.004) $\triangle$	+0.006 (0.005)	+0.013 (0.002) $\blacktriangle$	+0.001 (0.002)
	Add. Two-Tower Bias	+0.001 (0.001)	+0.009 (0.003) $\triangle$	+0.012 (0.002) $\blacktriangle$	+0.016 (0.001) $\blacktriangle$	+0.011 (0.007)	+0.002 (0.001)
	Mul. Two-Tower Bias	+0.006 (0.006)	+0.024 (0.011)	+0.009 (0.007)	+0.015 (0.002)$\blacktriangle$	+0.012 (0.015)	-0.093 (0.008) $\blacktriangledown$
	Adjacent Chain	-0.005 (0.007)	+0.025 (0.005) $\triangle$	+0.034 (0.005) $\blacktriangle$	+0.024 (0.004) $\triangle$	-0.026 (0.010) $\triangledown$	+0.016 (0.000) $\blacktriangle$
	CTR	+0.031 (0.017)	+0.008 (0.000) $\blacktriangle$	+0.025 (0.012)	+0.012 (0.006)	+0.015 (0.010)	+0.007 (0.003)
	Global All Pairs	+0.048 (0.006) $\blacktriangle$	+0.009 (0.002) $\triangle$	+0.058 (0.003) $\blacktriangle$	+0.010 (0.010)	+0.057 (0.002) $\blacktriangle$	+0.003 (0.004)
	Pivot One	+0.002 (0.003)	+0.009 (0.002) $\triangle$	+0.037 (0.018)	+0.016 (0.002) $\blacktriangle$	+0.022 (0.012)	+0.003 (0.005)
Istella-S	True Bias ^{oracle}	+0.048 (0.021)	+0.006 (0.021)	+0.020 (0.017)	+0.001 (0.010)	+0.009 (0.008)	+0.003 (0.009)
	Add. Two-Tower Bias	+0.031 (0.007) $\triangle$	+0.008 (0.013)	+0.027 (0.009) $\triangle$	+0.002 (0.007)	+0.021 (0.006)$\triangle$	-0.023 (0.031)
	Mul. Two-Tower Bias	+0.035 (0.019)	+0.130 (0.027) $\triangle$	+0.023 (0.034)	+0.177 (0.249)	+0.005 (0.009)	-0.390 (0.165)
	Adjacent Chain	+0.094 (0.049)	+0.202 (0.043) $\triangle$	+0.382 (0.026) $\blacktriangle$	+0.211 (0.045) $\triangle$	-0.561 (0.019) $\blacktriangledown$	-0.015 (0.167)
	CTR	+0.091 (0.002)$\blacktriangle$	+0.024 (0.008) $\triangle$	+0.027 (0.016)	+0.003 (0.018)	+0.017 (0.014)	-0.041 (0.066)
	Global All Pairs	+0.225 (0.018) $\blacktriangle$	-0.006 (0.009)	+0.297 (0.078) $\triangle$	-0.013 (0.002) $\blacktriangledown$	+0.340 (0.121) $\triangle$	-0.003 (0.014)
	Pivot One	+0.093 (0.031) $\triangle$	+0.002 (0.006)	+0.422 (0.017) $\blacktriangle$	-0.005 (0.003)	-0.600 (0.025) $\blacktriangledown$	+0.007 (0.006)

improved generally, and may be worse depending on setting. We find that the additive two-tower model architecture performs more consistently than the multiplicative two-tower model architecture.

10.2 Two-Tower Models for Relevance Estimation for Unbiased Learning-to-Rank Methods

In this section, we answer the second part of **RQ5**, investigating to what extent relevance estimation for DM and DR can be performed using two-tower models. We evaluate performance for feature-based propensity estimation, where IPS is omitted from results, as no relevance estimate is used in IPS.

Results of these experiments are shown in Table 10.4, where we compare with the original relevance model described in Section 3.4.1, where different types of positional bias estimation are applied. We first discuss results for DM, where we observe that performance is generally not improved, with significantly decreased performance for two datasets for both two-tower models. We do find performance increases for the Yahoo! dataset for DM. Looking at DR, we do find an improvement in results, where the best results are reached by two-tower models in five out of six settings. We observe that the two-tower model relevance estimate model significantly outperforms the relevance model trained with an oracle position bias baseline. These results suggest that applying the two-tower relevance estimate may be an effective strategy for DR estimation. A possible explanation for why performance may be increased for DR but not for DM is that DR may benefit from having an alternative estimation strategy that is less similar to IPS. This could allow for a more complementary correction term, reducing the extent to which errors in the relevance estimate reinforce errors made by the propensity-based component. In

Table 10.4: Comparison of Relevance estimation techniques for DM and DR on MSLR-WEB30K, Yahoo!, and Istella-S under frequency-based propensity estimation. Rows are grouped by dataset. Cells show mean (SD) NDCG. Bold marks the best non-oracle model within each comparison block. Significance markers compare each other model with this best model using a paired two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Relevance Model	DM		DR	
		N = 10^4	N = 10^5	N = 10^4	N = 10^5
MSLR-WEB30K	True Bias ^{oracle}	0.451 (0.007)	0.464 (0.012)	0.446 (0.008) $\triangledown$	0.462 (0.010) $\blacktriangledown$
	Add. Two-Tower	0.415 (0.005) $\triangledown$	0.406 (0.012) $\blacktriangledown$	0.453 (0.006)	0.496 (0.007)
	Mul. Two-Tower	0.404 (0.036)	0.415 (0.004) $\blacktriangledown$	0.419 (0.022)	0.459 (0.014) $\triangledown$
	Adjacent Chain	0.328 (0.008) $\blacktriangledown$	0.373 (0.012) $\blacktriangledown$	0.408 (0.029)	0.336 (0.006) $\blacktriangledown$
	CTR	0.440 (0.013)	0.472 (0.002)	0.434 (0.014)	0.468 (0.010) $\triangledown$
	Global All Pairs	0.362 (0.022) $\blacktriangledown$	0.474 (0.009)	0.346 (0.015) $\triangledown$	0.471 (0.016)
	Pivot One	0.328 (0.008) $\blacktriangledown$	0.400 (0.025) $\triangledown$	0.411 (0.030)	0.365 (0.038) $\triangledown$
Yahoo!	True Bias ^{oracle}	0.768 (0.005) $\triangledown$	0.777 (0.002)	0.767 (0.002) $\triangledown$	0.780 (0.004)
	Add. Two-Tower	0.783 (0.002)	0.774 (0.004)	0.770 (0.001)	0.784 (0.004)
	Mul. Two-Tower	0.779 (0.004)	0.775 (0.003)	0.774 (0.003)	0.779 (0.005)
	Adjacent Chain	0.715 (0.002) $\blacktriangledown$	0.741 (0.003) $\triangledown$	0.729 (0.006) $\blacktriangledown$	0.751 (0.003) $\blacktriangledown$
	CTR	0.763 (0.010) $\triangledown$	0.779 (0.006)	0.764 (0.009)	0.774 (0.004)
	Global All Pairs	0.723 (0.002) $\blacktriangledown$	0.774 (0.008)	0.719 (0.007) $\triangledown$	0.775 (0.003) $\blacktriangledown$
	Pivot One	0.723 (0.010) $\blacktriangledown$	0.768 (0.005)	0.721 (0.008) $\blacktriangledown$	0.775 (0.005) $\blacktriangledown$
Istella-S	True Bias ^{oracle}	0.719 (0.013) $\triangle$	0.754 (0.005)	0.728 (0.007)	0.755 (0.016)
	Add. Two-Tower	0.630 (0.021) $\triangledown$	0.677 (0.017) $\blacktriangledown$	0.710 (0.008)	0.744 (0.004)
	Mul. Two-Tower	0.643 (0.015)	0.673 (0.014) $\triangledown$	0.714 (0.005)	0.744 (0.004)
	Adjacent Chain	0.234 (0.016) $\blacktriangledown$	0.545 (0.050) $\triangledown$	0.635 (0.024) $\triangledown$	0.487 (0.023) $\blacktriangledown$
	CTR	0.700 (0.017)	0.759 (0.009)	0.695 (0.027)	0.737 (0.024)
	Global All Pairs	0.428 (0.071) $\triangledown$	0.762 (0.005)	0.379 (0.114) $\triangledown$	0.764 (0.007)
	Pivot One	0.234 (0.016) $\blacktriangledown$	0.754 (0.001)	0.631 (0.025) $\triangledown$	0.753 (0.004)

contrast, DM relies directly on the accuracy and calibration of the relevance model, making it more sensitive to misspecification in the estimated relevance scores. The two-tower architecture may therefore provide useful additional signal for DR, while still being insufficiently calibrated to serve as a standalone relevance target for DM.

Overall, these findings indicate that the usefulness of two-tower relevance estimates depends strongly on how they are incorporated into the estimator, with DR appearing better suited to exploit their strengths.

10.3 Combining Two-Tower Models with Doubly Robust Estimation

Finally, we study the extent to which both the position bias and relevance can be estimated using two-tower models for DR estimation. Results of experiments in this setting are shown in Table 10.5, where we compare all previous combinations with a setting where both position bias and relevance are estimated using two-tower models for DR. To isolate the effect of model depth, we also apply a deep two-tower model (5 layers, 128 neurons). Comparing performance across all models, we observe that for both two-tower architectures, estimating both bias and relevance using two-tower models does not lead to improved performance. We find that this model type is consistently outperformed by the regression-only model architecture. We find that differences between shallow and deep model architectures are small, with only minimal

Table 10.5: Comparison of different bias and relevance estimation techniques for DR on MSLR-WEB30K, Yahoo!, and Istella-S under frequency-based propensity estimation. The table includes the oracle-bias baseline in the position-bias estimation group and groups additive, multiplicative, and CSV position-bias variants separately. Cells show mean (SD) NDCG. Bold marks the best non-oracle model within each comparison block. Significance markers compare each other model with this best model using a paired two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Comparison Group	Model	MSLR-WEB30K		Yahoo!		Istella-S	
		N = 10 ⁴	N = 10 ⁵	N = 10 ⁴	N = 10 ⁵	N = 10 ⁴	N = 10 ⁵
Position-Bias Estimation	True Bias ^{oracle}	0.446 (0.008) $\triangledown$	0.462 (0.010) $\blacktriangledown$	0.767 (0.002) $\triangledown$	0.780 (0.004)	0.728 (0.007)	0.755 (0.016)
	Adjacent Chain	0.408 (0.029)	0.336 (0.006) $\blacktriangledown$	0.729 (0.006) $\blacktriangledown$	0.751 (0.003) $\blacktriangledown$	0.635 (0.024) $\triangledown$	0.487 (0.023) $\blacktriangledown$
	CTR	0.434 (0.014)	0.468 (0.010) $\triangledown$	0.764 (0.009)	0.774 (0.004)	0.695 (0.027)	0.737 (0.024)
	Global All Pairs	0.346 (0.015) $\triangledown$	0.471 (0.016)	0.719 (0.007) $\triangledown$	0.775 (0.003) $\blacktriangledown$	0.379 (0.114) $\triangledown$	0.764 (0.007)
	Pivot One	0.411 (0.030)	0.365 (0.038) $\triangledown$	0.721 (0.008) $\blacktriangledown$	0.775 (0.005) $\blacktriangledown$	0.631 (0.025) $\triangledown$	0.753 (0.004)
Additive Two-Tower	Bias Only	0.421 (0.009)	0.469 (0.012)	0.742 (0.002) $\blacktriangledown$	0.779 (0.002) $\triangledown$	0.701 (0.014)	0.751 (0.012)
	Regression Only	0.453 (0.006)	0.496 (0.007)	0.770 (0.001)	0.784 (0.004)	0.710 (0.008)	0.744 (0.004)
	Full Two-Tower	0.414 (0.011) $\triangledown$	0.490 (0.011)	0.738 (0.005) $\triangledown$	0.779 (0.005)	0.686 (0.019)	0.752 (0.010) $\triangledown$
	Full Two-Tower (Shallow)	0.411 (0.035)	0.492 (0.013)	0.743 (0.005) $\triangledown$	0.779 (0.004)	0.683 (0.019)	0.756 (0.003)
Multiplicative Two-Tower	Bias Only	0.418 (0.012)	0.408 (0.024) $\triangledown$	0.739 (0.004) $\triangledown$	0.770 (0.007)	0.682 (0.023)	0.615 (0.021) $\triangledown$
	Regression Only	0.419 (0.022)	0.459 (0.014) $\triangledown$	0.774 (0.003)	0.779 (0.005)	0.714 (0.005)	0.744 (0.004)
	Full Two-Tower	0.389 (0.005) $\blacktriangledown$	0.434 (0.012) $\triangledown$	0.738 (0.003) $\blacktriangledown$	0.778 (0.002) $\triangledown$	0.596 (0.062)	0.627 (0.019) $\blacktriangledown$
	Full Two-Tower (Shallow)	0.392 (0.005) $\blacktriangledown$	0.442 (0.013) $\blacktriangledown$	0.742 (0.003) $\blacktriangledown$	0.770 (0.005) $\triangledown$	0.658 (0.031)	0.624 (0.013) $\blacktriangledown$

increases or decreases, which are not consistent across datasets. These findings align with the suggestion from previous section that it may be useful to apply different types of modelling strategies, such that relevance estimation errors can be corrected more effectively. Here, it is important to note that the regression only model for DR is informed by the true position bias for the IPS estimate for the position bias, which may increase performance compared to settings where a different position bias estimate is used. Looking at the multiplicative two-tower model, we find that results are generally not better than additive two-tower models, with lower performance in most settings, and non-significantly better performance in two cases. This suggests that the multiplicative two-tower model may not be a more effective strategy for bias or relevance modelling than the existing additive structure.

Overall, we provide a nuanced answer to **RQ5**: while bias and relevance estimates from two-tower models can be used to reach performance comparable to existing methods, the improvements are very limited and dataset-dependent. The largest gains are found when applying the two-tower regression estimate for DR estimation. However, because the regression target is partly shaped by position-dependent exposure patterns, these gains may reflect the model’s ability to capture position bias rather than purely improved relevance estimation. Further research is therefore needed to determine how much improvement can be expected across different settings. Finally, we find that the additive two-tower strategy performs consistently well, providing stable results across settings, whereas the proposed multiplicative two-tower model performs more erratically and is slightly worse than the additive strategy overall.

Chapter 11

Discussion: Limitations and Future Work

In this chapter, we discuss the limitations, practical implications, and future directions of this work. The thesis introduced a feature-based propensity estimation paradigm for counterfactual LTR, supported by theoretical analysis, propensity recovery experiments, and semi-synthetic ranking experiments. It also studied the use of two-tower click models for estimating position bias and relevance within counterfactual LTR. To the best of our knowledge, this work is the first to investigate these topics in this specific setting; nevertheless, several limitations remain.

11.1 Feature-Based Propensity Estimation

This thesis has motivated the support for feature-based propensity through three complementary forms of inquiry: theoretical analysis, practical evidence from propensity estimation experiments, and empirical evidence from semi-synthetic experiments.

Our work provides several implications for the field of counterfactual LTR. First, we have shown that frequency-based propensity estimators introduce bias into reward estimates depending on the variance of the logging policy and the number of observations per document. This provides an important practical insight: in sparse-data settings, frequency-based estimates may be unreliable. We propose feature-based propensity estimation as an alternative approach to propensity estimation. Across the evaluated settings, this method showed favorable theoretical properties, strong propensity recovery, and improved downstream ranking performance. In some downstream settings, it even outperformed a ground-truth propensity baseline, likely because the learned estimates reduced variance during optimization. This suggests that feature-based propensity estimation may be useful not only when the original logging policy is unavailable, but also when it is available and variance reduction is beneficial. Feature-based propensity estimation provide practitioners with a method for effective propensity estimation in long-tail settings, resolving an important issue in real-world settings.

While we conducted extensive empirical ablations to verify our findings, several open directions remain. First, our evaluation is limited to an epsilon-greedy logging policy. While this method is easy to compute and has been applied in related work to isolate theoretical contributions [23, 32], future work could investigate more realistic sampling settings, such as Plackett-Luce sampling. Furthermore, our experiments are conducted in a setting that aligns closely with the theoretical assumptions of feature-based propensity estimation: the logging policy was computed using the same features that were available to the downstream models. This is a central limitation, as the method depends on the assumption that the available features are sufficiently informative of the logging policy. Improvements may therefore not hold when relevant logging policy features are unknown or unobserved, which is a known source of bias

also explored in Hager et al. [32]. Future work should explore the robustness of feature-based propensity estimation under partial feature observability.

Our experiments for feature-based propensity estimation rely on the regression-based neural model, for which the empirical results show strong performance recovery across the evaluated settings. It is possible, however, that clustering methods may be more effective in settings where little feature information is available. Future work may explore more extensively how to fine-tune and select propensity estimation methods, which could lead to further improvements in performance and robustness.

In addition to this, we conducted experiments in semi-synthetic dataset settings, which are commonly used to evaluate theoretical contributions [55, 32, 27, 57, 56]. Future work may explore the effectiveness of feature-based propensity estimation in real-world click datasets, as these datasets may be particularly affected by long-tail query distributions (see Figure 1.1).

Our experiments have also explored a sparse data setting where each query-document pair included only a single observation. To the best of our knowledge, this effect has not previously been studied for the LTR DR estimator, although similar effects have been reported in reinforcement learning [41]. These experiments highlight an important failure mode of DR estimation that future work could explore. Specifically, the poor performance in the single-sample setting may require intervention when DR methods are applied in settings with sparse observations, such as real-world web search. Possible solutions could include regularizing or weighting the IPS estimate based on uncertainty, similar to the approach of Zhang et al. [81].

11.2 Two-Tower Models for Counterfactual Learning-to-Rank

In our experiments with two-tower models for use in counterfactual LTR, we have explored a new application of click models. Current experimental results on semi-synthetic data have shown limited improvement, but one promising direction is to investigate more expressive bias models that go beyond position-based assumptions.

Experiments with click models have primarily focused on the commonly used two-tower model, which relies on the original ideas of the position-based model (PBM). This is, however, not the only method of modelling bias in clicks and user browsing behavior. Recent work by Hager et al. [31] has shown that other browsing models can remain effective in modern neural architectures and lead to performance improvements in real-world click datasets. This motivates further research into combining sophisticated click models with counterfactual LTR methods, particularly when using Plackett-Luce ranking models that allow for different optimization targets, such as fairness and diversity [79, 68, 51].

Furthermore, our experiments have explored a setting in which the bias tower was implemented as an embedding-based component that captures bias as a single vector of values. We used this design to keep the comparison between different bias estimation methods fair. A large advantage of two-tower models, however, is that they are able to take multiple bias features into account, which has been shown to be an important factor in settings such as web search [16, 78, 82]. Future work could explore more complex bias features or architectures, which may improve effectiveness by allowing counterfactual LTR models to capture richer sources of bias without relying only on position-based information.

Chapter 12

Conclusion

In this work, we introduced propensity estimation based on query-document features for counterfactual LTR. We also proposed applying click-based two-tower models to counterfactual LTR, studying how two established approaches to bias mitigation can be combined. In this section, we answer the two main research questions and discuss their implications.

Does feature-based propensity estimation lead to more effective learning to rank in settings with long-tail query distributions?

We first analyzed the theoretical and empirical problems of applying frequency-based propensity estimation methods in long-tail query distributions. We showed that frequency-based propensity estimates become more biased when fewer observations are available, directly affecting downstream optimization through poor reward estimates. We then identified conditions under which propensity estimation can use query-document features, showing that this is applicable when logging policies are based on such features. We evaluated feature-based propensity methods across diverse experiments with large semi-synthetic datasets for IPS, DM, and DR estimators, and found strong performance improvements across all models, with especially large gains for IPS-based methods. We find results generalize across dataset sizes and levels of logging policy randomness, and in some cases even outperformed the oracle propensity baseline, likely due to reduced variance in downstream optimization. Our proposed method addresses an often-overlooked long-tail query setting and provides an effective solution. With feature-based propensity estimation, practitioners can obtain accurate and tractable estimates, even when individual query-document pairs have few observations.

How can two-tower models be used in conjunction with doubly robust estimation for more effective learning-to-rank methods?

We then studied two-tower models for counterfactual LTR, investigating bias and relevance estimation separately and in combination. We showed that two-tower models can be used to estimate position bias and document relevance in these settings. However, these estimates do not generally improve ranking performance over existing methods across dataset settings. This suggests that, although two-tower models are compatible with counterfactual LTR, more advanced approaches, such as complex browsing models or including multiple bias features, may be needed for improved performance.

Overall, our work offers theoretical and practical insight into sparse click data. The results show that long-tail query distributions pose fundamental challenges for frequency-based propensity estimation, but that these challenges can be mitigated when propensities are estimated from query-document features. At the same time, our study of two-tower models shows that while they can be applied to counterfactual estimation, this does not necessarily translate into improved ranking performance. Together, these findings highlight the importance of accurately estimating the components of counterfactual LTR, and provide a foundation for more robust learning-to-rank methods under sparse observations.

Appendix A

Bias of Inverse Propensity Scoring

Here, we partially repeat a proof of the bias from Oosterhuis [55] to show the expected value of the IPS estimator.

Theorem A.1. *The IPS estimator (Eq. 3.11) has the following bias:*

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{IPS}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d. \quad (\text{A.1})$$

Proof. Using Eqs. 3.1, 3.6, 3.7, and 3.11 we have:

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{IPS}(\pi)] = \sum_{d \in \mathcal{D}} \frac{\hat{\omega}_d}{\hat{\rho}_d} \rho_d R_d \quad (\text{A.2})$$

$$= R(\pi) + \sum_{d \in \mathcal{D}} \frac{\hat{\omega}_d}{\hat{\rho}_d} \left(\rho_d - \hat{\rho}_d \frac{\omega_d}{\hat{\omega}_d} \right) R_d \quad (\text{A.3})$$

$$= R(\pi) + \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d. \quad (\text{A.4})$$

□

Lemma A.2. *By the definitions of ω (Eq. 3.9) and $\hat{\omega}$ (Eq. 3.10):*

$$(\forall k \in K, \alpha_k = \hat{\alpha}_k) \longrightarrow (\forall d \in D, \hat{\omega} = \omega). \quad (\text{A.5})$$

Corollary A.3. *When $\hat{\alpha}$ is correctly estimated, the IPS has the bias:*

$$(\forall k \in K, \alpha_k = \hat{\alpha}_k) \longrightarrow \mathbb{E}_{c,y \sim \pi_0} [\hat{R}_{IPS}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} R_d \omega_d \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right). \quad (\text{A.6})$$

Proof. Follows from applying Lemma A.2 to Theorem A.1. □

Theorem A.4. *The IPS estimator Eq. 3.11 is unbiased when $\hat{\alpha}$ and $\hat{\rho}$ are correctly estimated:*

$$((\forall d \in D, \rho_d = \hat{\rho}_d), (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\hat{R}_{IPS}] = R(\pi_\theta). \quad (\text{A.7})$$

Proof. Follows from applying Lemma A.2 to Theorem A.1. □

Theorem A.5. *The IPS estimator (Eq. 3.11) has the following gradient:*

$$\nabla_\theta \hat{R}_{IPS}(\pi_\theta) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} c_i(d) \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_\theta \pi_\theta(k \mid d). \quad (\text{A.8})$$

Proof. Using Eqs. 3.1, 3.6, 3.7, and 3.11 we have:

$$\nabla_{\theta} \hat{R}_{\text{IPS}}(\pi_{\theta}) = \nabla_{\theta} \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} c_i(d) \quad (\text{A.9})$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} \nabla_{\theta} \frac{\sum_{k=1}^K \pi_{\theta}(k | d) \hat{\alpha}_k}{\hat{\rho}_d} c_i(d) \quad (\text{A.10})$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} c_i(d) \frac{\sum_{k=1}^K \hat{\alpha}_k \nabla_{\theta} \pi_{\theta}(k | d)}{\hat{\rho}_d} \quad (\text{A.11})$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d \in D} c_i(d) \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.12})$$

□

Theorem A.6. *The reward (Eq. 3.6) has the following gradient:*

$$\nabla_{\theta} R(\pi_{\theta}) = \sum_{d \in \mathcal{D}} R_d \sum_{k=1}^K \alpha_k \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.13})$$

Proof. Using Eqs. 3.6 and 3.9 we have:

$$\nabla_{\theta} R(\pi_{\theta}) = \nabla_{\theta} \sum_{d \in \mathcal{D}} \omega_d R_d \quad (\text{A.14})$$

$$= \sum_{d \in \mathcal{D}} R_d \nabla_{\theta} \sum_{k=1}^K \pi_{\theta}(k | d) \alpha_k \quad (\text{A.15})$$

$$= \sum_{d \in \mathcal{D}} R_d \sum_{k=1}^K \alpha_k \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.16})$$

□

Theorem A.7. *The bias of the gradient of the IPS estimator is as follows:*

$$\mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) = \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\alpha}_k - \alpha_k \right) \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.17})$$

Proof. Using Eqs. A.13, 3.1, 3.7, and 3.11 we have:

$$\mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] = \sum_{d \in D} R_d \rho_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_{\theta} \pi_{\theta}(k | d) \quad (\text{A.18})$$

$$= \sum_{d \in D} \left(R_d \rho_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_{\theta} \pi_{\theta}(k | d) \right) - \nabla_{\theta} R(\pi_{\theta}) + \nabla_{\theta} R(\pi_{\theta}) \quad (\text{A.19})$$

$$= \nabla_{\theta} R(\pi_{\theta}) + \sum_{d \in D} R_d \rho_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \nabla_{\theta} \pi_{\theta}(k | d) - R_d \sum_{k=1}^K \alpha_k \nabla_{\theta} \pi_{\theta}(k | d) \quad (\text{A.20})$$

$$= \nabla_{\theta} R(\pi_{\theta}) + \sum_{d \in D} R_d \left(\sum_{k=1}^K \frac{\hat{\alpha}_k \rho_d}{\hat{\rho}_d} \nabla_{\theta} \pi_{\theta}(k | d) - \sum_{k=1}^K \alpha_k \nabla_{\theta} \pi_{\theta}(k | d) \right) \quad (\text{A.21})$$

$$= \nabla_{\theta} R(\pi_{\theta}) + \sum_{d \in D} R_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_d} \left(\rho_d - \hat{\rho}_d \frac{\alpha_k}{\hat{\alpha}_k} \right) \nabla_{\theta} \pi_{\theta}(k | d) \quad (\text{A.22})$$

$$= \nabla_{\theta} R(\pi_{\theta}) + \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\alpha}_k - \alpha_k \right) \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.23})$$

□

Corollary A.8. *When $\hat{\alpha}$ is correct, the gradient of IPS has the bias:*

$$(\forall k \in K, \alpha_k = \hat{\alpha}_k) \longrightarrow \mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) = \sum_{d \in D} R_d \alpha_k \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right) \sum_{k=1}^K \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.24})$$

Proof. Follows from Theorem A.7. □

Corollary A.9. *When $\hat{\rho}$ is correct, the gradient of IPS has the bias:*

$$(\forall d \in D, \rho_d = \hat{\rho}_d) \longrightarrow \mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] - \nabla_{\theta} R(\pi_{\theta}) = \sum_{d \in D} R_d \sum_{k=1}^K (\hat{\alpha}_k - \alpha_k) \nabla_{\theta} \pi_{\theta}(k | d). \quad (\text{A.25})$$

Proof. Follows from Theorem A.7. □

Theorem A.10. *The gradient of the IPS estimator Eq. 3.11 is unbiased when $\hat{\alpha}$ and $\hat{\rho}$ are correctly estimated:*

$$((\forall d \in D, \rho_d = \hat{\rho}_d), (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\nabla_{\theta} \hat{R}_{\text{IPS}}] = \nabla_{\theta} R(\pi_{\theta}). \quad (\text{A.26})$$

Proof. Follows from Theorem A.7. □

Appendix B

Bias of Doubly Robust Estimator

Lemma B.1. *The CV estimator (Eq. 3.23) has the expected value [55]:*

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{\mathcal{R}}_{CV}(\pi)] = \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}] \hat{R}_d. \quad (\text{B.1})$$

Proof. Follows from Eq. 3.23 □

Theorem B.2. *The CV estimator (Eq. 3.23) is an unbiased estimate of DM (Eq. 3.18) if $\hat{\pi}_0$ is correctly estimated.*

$$(\forall d \in D, \hat{\pi}_0(d) = \pi_0(d)) \longrightarrow \mathbb{E}_{c,y \sim \pi_0} [\hat{\mathcal{R}}_{CV}(\pi)] = \hat{\mathcal{R}}_{DM}(\pi). \quad (\text{B.2})$$

Proof. Follows from assuming $(\forall d \in D, \hat{\pi}_0(d) = \pi_0(d))$, and applying the definition of $\hat{\rho}_d$ (Eq. 3.8) to Lemma B.1.

Theorem B.3. *The DR estimator (Eq. 3.25) has the following bias [55]:*

$$\begin{aligned} \mathbb{E}_{c,y \sim \pi_0} [\hat{\mathcal{R}}_{DR}(\pi)] - \mathcal{R}(\pi) \\ = \sum_{d \in D} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d + \left(\frac{\mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}]}{\hat{\rho}_d} - 1 \right) \hat{\omega}_d \hat{R}_d. \end{aligned} \quad (\text{B.3})$$

Proof.

$$\mathbb{E}_{c,y \sim \pi_0} [\hat{\mathcal{R}}_{DR}(\pi)] = \hat{\mathcal{R}}_{DM}(\pi) + \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \left(\rho_d R_d - \mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}] \hat{R}_d \right) \quad (\text{B.4})$$

$$= \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \left(\rho_d R_d - (\mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}] - \hat{\rho}_d) \hat{R}_d \right) \quad (\text{B.5})$$

$$= \mathcal{R}(\pi) + \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \left(\left(\rho_d - \frac{\hat{\rho}_d}{\hat{\omega}_d} \omega_d \right) R_d + (\mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}] - \hat{\rho}_d) \hat{R}_d \right) \quad (\text{B.6})$$

$$= \mathcal{R}(\pi) + \sum_{d \in D} \left(\frac{\rho_d}{\hat{\rho}_d} \hat{\omega}_d - \omega_d \right) R_d + \left(\frac{\mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}]}{\hat{\rho}_d} - 1 \right) \hat{\omega}_d \hat{R}_d. \quad (\text{B.7})$$

□

Lemma B.4. *By the definition of ρ_d (Eq. 3.7):*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \longrightarrow (\forall d \in D, \mathbb{E}_{y \sim \pi_0} [\hat{\alpha}_{k(d)}] = \rho_d). \quad (\text{B.8})$$

Corollary B.5. *The bias of the DR estimator when position bias is correctly estimated is:*

$$(\forall k, \hat{\alpha}_k = \alpha_k) \longrightarrow \mathbb{E}_{c, y \sim \pi_0}[\hat{\mathcal{R}}_{DR}(\pi)] - \mathcal{R}(\pi) = \sum_{d \in D} \omega_d \left(\frac{\rho_d}{\hat{\rho}_d} - 1 \right) (R_d - \hat{R}_d). \quad (\text{B.9})$$

Proof. Follows from applying Lemma A.2 to Theorem B.3. □

Theorem B.6. *The DR estimator (Eq. 3.25) is unbiased if the position bias is correctly estimated and per item we have that either the propensity is correctly estimated, or $\hat{R}_d$ is correct:*

$$\begin{aligned} ((\forall k, \hat{\alpha}_k = \alpha_k) \wedge (\forall d \in D, (\hat{\rho}_d = \rho_d \vee \hat{R}_d = R_d))) \\ \longrightarrow \mathbb{E}_{c, y \sim \pi_0}[\hat{\mathcal{R}}_{DR}] = \mathcal{R}_d. \end{aligned} \quad (\text{B.10})$$

Proof. Follows from applying Corollary A.3 to Corollary B.5. □

Appendix C

Bias of Feature-Based Inverse Propensity Scoring

Theorem C.1. *When the behavior of the logging policy is determined exactly by the features, the feature and document propensities are equal:*

$$(\forall(k, d) \in K \times D, \pi_0(k \mid d) = \pi_0(k \mid x_d)) \longrightarrow (\forall d, \rho_d = \rho_{x_d}). \quad (\text{C.1})$$

Proof. Using Eqs. 3.7 and 5.2, and assuming $\forall(k, d) \in K \times D, \pi_0(k \mid d) = \pi_0(k \mid x_d)$ we have that for any document:

$$\rho_{x_d} = \mathbb{E}[\rho_d \mid x_d] \quad (\text{C.2})$$

$$= \mathbb{E}\left[\sum_{k=1}^K \pi_0(k \mid d) \alpha_k \mid x_d\right] \quad (\text{C.3})$$

$$= \sum_{k=1}^K \mathbb{E}[\pi_0(k \mid d) \mid x_d] \alpha_k \quad (\text{C.4})$$

$$= \sum_{k=1}^K \pi_0(k \mid x_d) \alpha_k \quad (\text{C.5})$$

$$= \rho_d. \quad (\text{C.6})$$

□

Theorem C.2. *The feature-based IPS estimator Eq. 5.6 has the following bias:*

$$\mathbb{E}_{c, y \sim \pi_0}[\hat{R}_{IPS}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_{x_d}} \hat{\omega}_d - \omega_d \right) R_d. \quad (\text{C.7})$$

Proof. Using Eqs. 3.1, 3.6, 3.7, and 5.6 we have:

$$\mathbb{E}_{c, y \sim \pi_0}[\hat{R}_{IPS}(\pi)] = \sum_{d \in \mathcal{D}} \frac{\hat{\omega}_d}{\hat{\rho}_d} \rho_d R_d \quad (\text{C.8})$$

$$= R(\pi) + \sum_{d \in \mathcal{D}} \frac{\hat{\omega}_d}{\hat{\rho}_{x_d}} \left(\rho_d - \hat{\rho}_{x_d} \frac{\omega_d}{\hat{\omega}_d} \right) R_d \quad (\text{C.9})$$

$$= R(\pi) + \sum_{d \in \mathcal{D}} \left(\frac{\rho_d}{\hat{\rho}_{x_d}} \hat{\omega}_d - \omega_d \right) R_d. \quad (\text{C.10})$$

□

Corollary C.3. When $\hat{\alpha}$ is correctly estimated, the IPS has the bias:

$$(\forall k \in K, \alpha_k = \hat{\alpha}_k) \longrightarrow \mathbb{E}_{c, y \sim \pi_0} [\hat{R}_{IPS}(\pi)] - R(\pi) = \sum_{d \in \mathcal{D}} R_d \left(\frac{\rho_d}{\hat{\rho}_{x_d}} - 1 \right). \quad (\text{C.11})$$

Proof. Follows from applying Lemma A.2 to Theorem C.2. $\square$

Theorem C.4. The Feature-based IPS estimator Eq. 5.6 is unbiased when $\hat{\alpha}$ and $\hat{\rho}_{x_d}$ are correctly estimated, and the logging policy is determined by the features:

$$((\forall d \in D, \rho_d = \hat{\rho}_{x_d}), (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\hat{R}_{IPS}] = R(\pi_\theta). \quad (\text{C.12})$$

Proof. Follows from applying Theorem C.1 and Corollary C.3 to Theorem C.2. $\square$

Theorem C.5. The bias of the gradient of the feature-based IPS estimator is as follows:

$$\mathbb{E}[\nabla_\theta \hat{R}_{IPS}] - \nabla_\theta R(\pi_\theta) = \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_{x_d}} \hat{\alpha}_k - \alpha_k \right) \nabla_\theta \pi_\theta(k | d). \quad (\text{C.13})$$

Proof. Using Eqs. A.13, 3.1, 3.7, and 3.11 we have:

$$\mathbb{E}[\nabla_\theta \hat{R}_{IPS}] = \nabla_\theta \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_{x_d}} c_i(d) \quad (\text{C.14})$$

$$= \sum_{d \in D} \nabla_\theta \frac{\sum_{k=1}^K \pi_\theta(k | d) \hat{\alpha}_k}{\hat{\rho}_{x_d}} c_i(d) \quad (\text{C.15})$$

$$= \sum_{d \in D} R_d \rho_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_{x_d}} \nabla_\theta \pi_\theta(k | d) \quad (\text{C.16})$$

$$= \sum_{d \in D} \left(R_d \rho_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_{x_d}} \nabla_\theta \pi_\theta(k | d) \right) - \nabla_\theta R(\pi_\theta) + \nabla_\theta R(\pi_\theta) \quad (\text{C.17})$$

$$= \nabla_\theta R(\pi_\theta) + \sum_{d \in D} R_d \sum_{k=1}^K \frac{\hat{\alpha}_k}{\hat{\rho}_{x_d}} \left(\rho_d - \hat{\rho}_{x_d} \frac{\alpha_k}{\hat{\alpha}_k} \right) \nabla_\theta \pi_\theta(k | d) \quad (\text{C.18})$$

$$= \nabla_\theta R(\pi_\theta) + \sum_{d \in D} R_d \sum_{k=1}^K \left(\frac{\rho_d}{\hat{\rho}_{x_d}} \hat{\alpha}_k - \alpha_k \right) \nabla_\theta \pi_\theta(k | d). \quad (\text{C.19})$$

$\square$

Corollary C.6. When $\hat{\alpha}$ is correctly estimated, the gradient of IPS has the bias:

$$(\forall k \in K, \alpha_k = \hat{\alpha}_k) \longrightarrow \mathbb{E}[\nabla_\theta \hat{R}_{IPS}] - \nabla_\theta R(\pi_\theta) = \sum_{d \in D} R_d \left(\frac{\rho_d}{\hat{\rho}_{x_d}} - 1 \right) \sum_{k=1}^K \nabla_\theta \pi_\theta(k | d). \quad (\text{C.20})$$

Proof. Follows from applying Lemma A.2 to Theorem C.5. $\square$

Theorem C.7. The gradient of the feature-based IPS estimator Eq. 5.2 is unbiased when $\hat{\alpha}$ and $\hat{\rho}_{x_d}$ are correctly estimated, and the behavior of the logging policy is determined exactly by the features:

$$((\forall d \in D, \rho_d = \hat{\rho}_{x_d}), (\forall k \in K, \alpha_k = \hat{\alpha}_k)) \longrightarrow \mathbb{E}[\nabla_\theta \hat{R}_{IPS}] = \nabla_\theta R(\pi_\theta). \quad (\text{C.21})$$

Proof. Follows from applying Theorem C.1 and Corollary C.6 to Theorem C.5. $\square$

Appendix D

Propensity Estimation Comparison Plots

This appendix collects the propensity-estimation comparison plots for the feature-grouping and neural propensity estimators discussed in the main text. Each plot compares the deviation of the estimated propensity from the oracle propensity as the number of observations changes.

D.1 Cosine Similarity Grouping

Cosine Similarity Grouping forms neighborhoods by applying thresholds to cosine similarity between document feature vectors. Results for the evaluated thresholds are shown in Figure D.1.

D.2 k -Means Grouping

k -Means Grouping uses mini-batch k -Means to assign documents to feature-space clusters, after which propensity estimates are computed from grouped observations. Results for different values of K are shown in Figure D.2.

D.3 KNN Grouping

KNN Grouping estimates propensities by using observations from the K nearest neighbors in feature space. Results for different values of K are shown in Figure D.3.

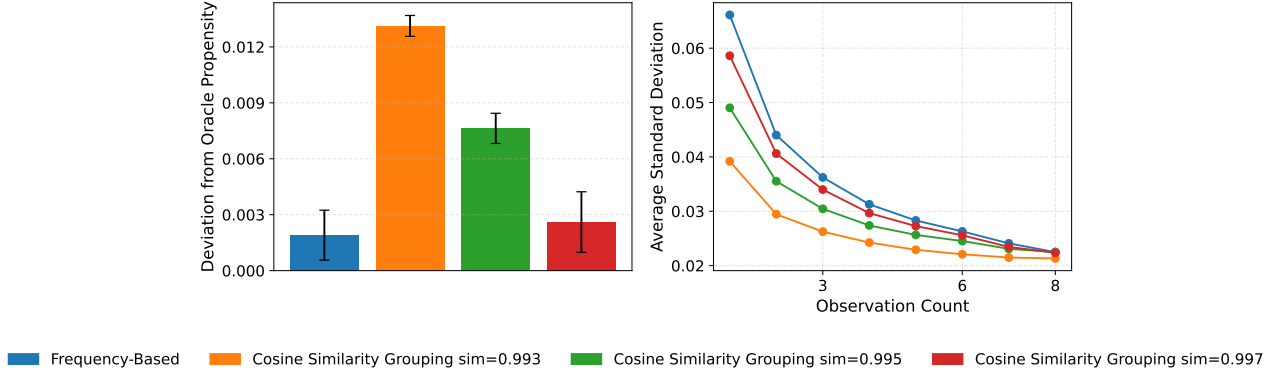


Figure D.1: Cosine-similarity grouping propensity comparison over similarity thresholds.

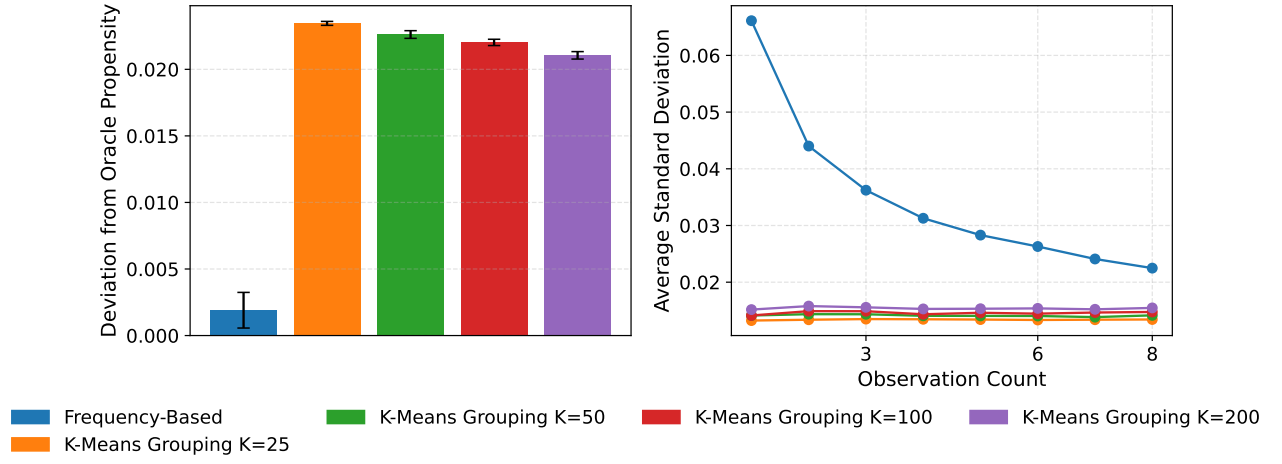


Figure D.2: k -Means grouping propensity comparison over K .

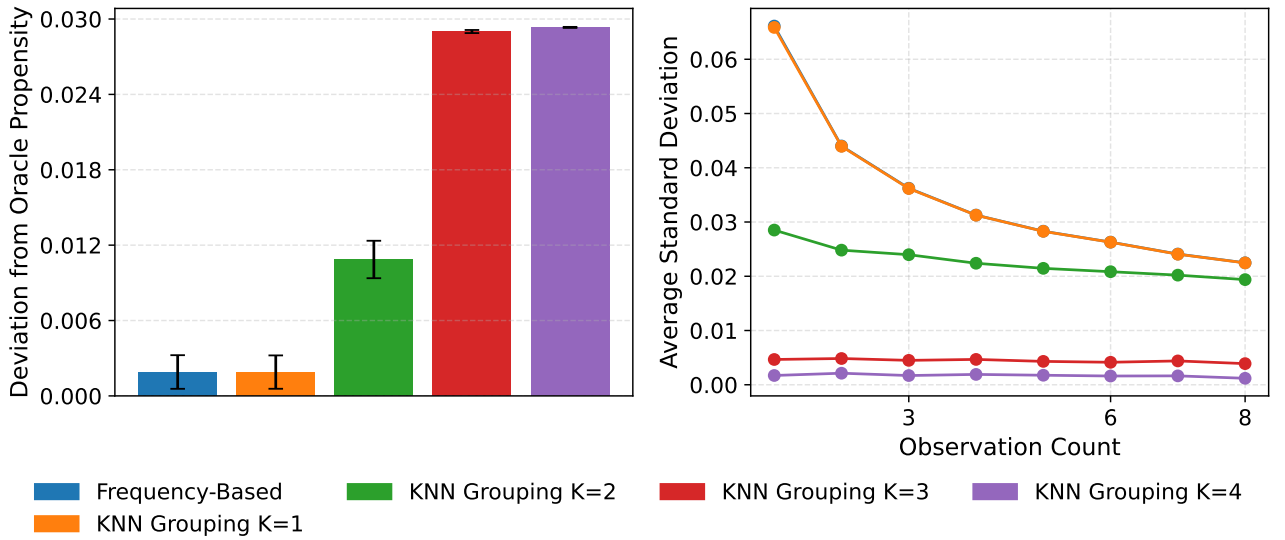


Figure D.3: KNN grouping propensity comparison over K .

Appendix E

Temperature-Specific Results for Inverse Propensity Scoring, Direct Method, and Doubly Robust

In this appendix, we report the results for experiments with different logging policy temperatures at a fixed training size of $N = 10^4$ sessions. As in the main analysis, we compare the frequency-based propensity estimator, the feature-based MLP estimator, and the oracle propensity baseline.

E.1 Inverse Propensity Scoring

The IPS temperature results are shown in Table E.1. The clearest pattern is that increasing the temperature generally reduces performance across all datasets and propensity estimators. This effect is strongest on MSLR-WEB30K and Istella-S, where the frequency-based baseline deteriorates steadily as the policy becomes more exploratory. The MLP estimator remains strongest or tied strongest in most intermediate-temperature settings, which indicates that feature-based propensity estimation is especially useful when the logging policy introduces enough randomness to make document-level counts noisy, but still preserves enough structure for the model to generalize across feature-similar documents.

E.2 Direct Method

The DM temperature results are shown in Table E.2. Compared with IPS, DM is more stable as temperature increases. The MLP estimator remains clearly beneficial for MSLR-WEB30K and Yahoo! at low and moderate temperatures, and it stays competitive with the oracle on Istella-S.

At the same time, the tables show that the gains from better propensity estimation become smaller at the highest temperature. For MSLR-WEB30K, the MLP is strongest from $T = 0$ through $T = 0.5$, but this advantage disappears at $T = 0.75$ and reverses at $T = 1.0$. A similar weakening appears on Istella-S and Yahoo!, where differences between methods narrow at the most exploratory setting. This indicates that once the logging policy becomes fully random, there is less informative exposure structure left for improved propensity estimation to exploit in the DM objective.

Table E.1: IPS results for propensity estimation methods across policy temperatures. Columns are grouped by temperature, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-temperature-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	0		0.25		0.5		0.75		1.0	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.434 (0.010)	0.620 (0.019)	0.426 (0.028) $\triangle$	0.605 (0.053) $\triangle$	0.417 (0.026) $\triangle$	0.590 (0.049) $\triangle$	0.374 (0.027)	0.511 (0.050)	0.338 (0.013)	0.456 (0.018)
	Frequency-Based	0.435 (0.008)	0.732 (0.250)	0.419 (0.028)	0.589 (0.052)	0.383 (0.022)	0.620 (0.266)	0.355 (0.010)	0.480 (0.015)	0.332 (0.008)	0.448 (0.011)
	MLP	0.449 (0.001)	0.652 (0.001)	0.454 (0.009)$\blacktriangle$	0.659 (0.013)$\blacktriangle$	0.440 (0.008)$\blacktriangle$	0.671 (0.128)$\triangle$	0.392 (0.020)$\triangle$	0.540 (0.043)$\triangle$	0.338 (0.014)	0.456 (0.019)
Istella-S	Oracle Propensity	0.681 (0.008)	0.970 (0.012)	0.677 (0.015) $\triangle$	0.964 (0.026) $\blacktriangle$	0.681 (0.012) $\blacktriangle$	0.971 (0.020) $\blacktriangle$	0.651 (0.019) $\blacktriangle$	0.921 (0.031) $\blacktriangle$	0.524 (0.031) $\blacktriangle$	0.697 (0.052) $\blacktriangle$
	Frequency-Based	0.683 (0.007)	0.973 (0.011)	0.668 (0.021)	0.947 (0.036)	0.649 (0.020)	0.916 (0.033)	0.609 (0.015)	0.847 (0.025)	0.472 (0.026)	0.614 (0.042)
	MLP	0.711 (0.011)$\blacktriangle$	1.023 (0.017)$\blacktriangle$	0.698 (0.010)$\triangle$	1.001 (0.016)$\triangle$	0.688 (0.010)$\blacktriangle$	0.983 (0.018)$\blacktriangle$	0.665 (0.010)$\blacktriangle$	0.945 (0.017)$\blacktriangle$	0.514 (0.031)$\triangle$	0.682 (0.050)$\triangle$
Yahoo!	Oracle Propensity	0.766 (0.007)	0.840 (0.011)	0.763 (0.005) $\blacktriangle$	0.836 (0.009) $\blacktriangle$	0.768 (0.004) $\blacktriangle$	0.845 (0.006) $\blacktriangle$	0.758 (0.002) $\blacktriangle$	0.829 (0.005) $\blacktriangle$	0.733 (0.004) $\triangle$	0.784 (0.007) $\triangle$
	Frequency-Based	0.763 (0.004)	0.836 (0.006)	0.755 (0.004)	0.822 (0.007)	0.754 (0.009)	0.820 (0.017)	0.736 (0.002)	0.789 (0.004)	0.728 (0.002)	0.776 (0.004)
	MLP	0.775 (0.004)$\blacktriangle$	0.853 (0.006)$\blacktriangle$	0.772 (0.006)$\blacktriangle$	0.848 (0.008)$\blacktriangle$	0.765 (0.005)$\triangle$	0.839 (0.008)$\triangle$	0.751 (0.005)$\blacktriangle$	0.816 (0.008)$\blacktriangle$	0.728 (0.004)	0.774 (0.006)

Table E.2: DM results for propensity estimation methods across policy temperatures. Columns are grouped by temperature, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-temperature-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	0		0.25		0.5		0.75		1.0	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.438 (0.007)	0.624 (0.008)	0.450 (0.010) $\triangle$	0.649 (0.014) $\blacktriangle$	0.455 (0.022) $\blacktriangle$	0.657 (0.029) $\blacktriangle$	0.450 (0.016)	0.654 (0.017)	0.434 (0.016)	0.611 (0.025)
	Frequency-Based	0.439 (0.006)	0.625 (0.007)	0.445 (0.010)	0.636 (0.015)	0.440 (0.021)	0.629 (0.029)	0.449 (0.011)	0.642 (0.018)	0.433 (0.016)	0.610 (0.027)
	MLP	0.488 (0.004)$\triangle$	0.707 (0.005)$\blacktriangle$	0.487 (0.010)$\blacktriangle$	0.706 (0.011)$\blacktriangle$	0.481 (0.007)$\blacktriangle$	0.696 (0.010)$\blacktriangle$	0.462 (0.025)	0.659 (0.053)	0.421 (0.017) $\triangledown$	0.594 (0.022)
Istella-S	Oracle Propensity	0.720 (0.011)	1.032 (0.019)	0.726 (0.012)	1.045 (0.018) $\triangle$	0.736 (0.015)	1.060 (0.025)	0.730 (0.010) $\blacktriangle$	1.052 (0.017) $\blacktriangle$	0.686 (0.043)	0.976 (0.065)
	Frequency-Based	0.721 (0.011)	1.034 (0.019)	0.723 (0.009)	1.038 (0.014)	0.728 (0.016)	1.047 (0.026)	0.702 (0.015)	1.003 (0.026)	0.689 (0.012)	0.976 (0.019)
	MLP	0.741 (0.008)$\triangle$	1.068 (0.014)$\triangle$	0.738 (0.010)$\blacktriangle$	1.065 (0.018)$\blacktriangle$	0.732 (0.008)	1.055 (0.011)	0.736 (0.008)$\blacktriangle$	1.062 (0.015)$\blacktriangle$	0.678 (0.032)	0.959 (0.051)
Yahoo!	Oracle Propensity	0.771 (0.005)	0.846 (0.010)	0.774 (0.002) $\blacktriangle$	0.853 (0.002) $\blacktriangle$	0.779 (0.004) $\blacktriangle$	0.861 (0.007) $\blacktriangle$	0.777 (0.005) $\blacktriangle$	0.855 (0.010) $\blacktriangle$	0.770 (0.005) $\triangle$	0.843 (0.010) $\triangle$
	Frequency-Based	0.769 (0.006)	0.843 (0.010)	0.768 (0.002)	0.841 (0.004)	0.768 (0.007)	0.840 (0.013)	0.764 (0.003)	0.832 (0.007)	0.764 (0.004)	0.832 (0.008)
	MLP	0.782 (0.003)$\blacktriangle$	0.866 (0.006)$\blacktriangle$	0.783 (0.001)$\blacktriangle$	0.866 (0.004)$\blacktriangle$	0.782 (0.004)$\blacktriangle$	0.864 (0.007)$\blacktriangle$	0.776 (0.007)$\blacktriangle$	0.854 (0.013)$\blacktriangle$	0.765 (0.008)	0.834 (0.014)

E.3 Doubly Robust

The temperature results for the Doubly-Robust estimator are shown in Table E.3. The overall pattern is similar to DM, but the dependence on propensity quality remains slightly more visible. On MSLR-WEB30K and Yahoo!, the MLP estimator delivers the strongest DR performance for most low-to-moderate temperatures, and several of these gains are statistically significant relative to the frequency-based baseline. This matches the expectation that DR can benefit from better propensity estimates both in its IPS correction term and in the regression component.

As with DM, the advantage of the MLP weakens when the temperature approaches 1. On MSLR-WEB30K, the MLP remains strongest up to $T = 0.75$, but drops below both the frequency-based and oracle baselines at $T = 1.0$. On Istella-S and Yahoo!, the high-temperature differences are smaller, with the oracle often matching or slightly exceeding the learned model. Overall, these tables reinforce the main finding from the chapter: feature-based propensity estimation is most helpful in partially stochastic regimes, where propensities are difficult to estimate from counts alone, but are still structured enough to be predicted from document features.

Table E.3: DR results for propensity estimation methods across policy temperatures. Columns are grouped by temperature, with NDCG and ECP reported for each. The oracle propensity baseline appears first for each dataset, and bold marks the best non-oracle value within each dataset-temperature-metric block. Cells show mean (SD). Significance markers compare each non-frequency-based method with the frequency-based method using a two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Model	0		0.25		0.5		0.75		1.0	
		NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP	NDCG	ECP
MSLR-WEB30K	Oracle Propensity	0.430 (0.006)	0.613 (0.005)	0.452 (0.012) $\blacktriangle$	0.654 (0.015) $\blacktriangle$	0.450 (0.021) $\triangle$	0.650 (0.029) $\triangle$	0.448 (0.012)	0.644 (0.023)	0.431 (0.017)	0.609 (0.028)
	Frequency-Based	0.431 (0.007)	0.615 (0.006)	0.445 (0.011)	0.640 (0.015)	0.437 (0.022)	0.627 (0.031)	0.445 (0.010)	0.637 (0.016)	0.425 (0.014)	0.600 (0.025)
	MLP	0.485 (0.011)$\triangle$	0.701 (0.016)$\triangle$	0.482 (0.010)$\blacktriangle$	0.699 (0.014)$\blacktriangle$	0.473 (0.011)$\blacktriangle$	0.685 (0.015)$\blacktriangle$	0.464 (0.022)$\triangle$	0.667 (0.045)$\blacktriangle$	0.408 (0.019) $\triangledown$	0.575 (0.030) $\triangledown$
Istella-S	Oracle Propensity	0.729 (0.009)	1.051 (0.015)	0.733 (0.011)	1.058 (0.018)	0.735 (0.011)	1.061 (0.018)	0.728 (0.008) $\blacktriangle$	1.049 (0.013) $\triangle$	0.693 (0.033)	0.989 (0.051)
	Frequency-Based	0.729 (0.013)	1.051 (0.020)	0.729 (0.007)	1.050 (0.011)	0.732 (0.009)	1.057 (0.016)	0.707 (0.015)	1.013 (0.027)	0.688 (0.016)	0.976 (0.024)
	MLP	0.742 (0.006)	1.072 (0.011)	0.739 (0.008)$\blacktriangle$	1.067 (0.013)$\blacktriangle$	0.735 (0.007)	1.062 (0.010)	0.737 (0.007)$\blacktriangle$	1.065 (0.013)$\triangle$	0.680 (0.012)	0.966 (0.019)
Yahoo!	Oracle Propensity	0.773 (0.005)	0.851 (0.008)	0.775 (0.002) $\blacktriangle$	0.854 (0.003) $\blacktriangle$	0.777 (0.003) $\blacktriangle$	0.859 (0.005) $\blacktriangle$	0.771 (0.002) $\blacktriangle$	0.848 (0.005) $\blacktriangle$	0.767 (0.005)	0.840 (0.009) $\triangle$
	Frequency-Based	0.773 (0.005)	0.851 (0.009)	0.769 (0.003)	0.844 (0.006)	0.769 (0.005)	0.845 (0.009)	0.763 (0.003)	0.833 (0.005)	0.763 (0.003)	0.832 (0.006)
	MLP	0.783 (0.005)$\blacktriangle$	0.868 (0.008)$\blacktriangle$	0.780 (0.003)$\blacktriangle$	0.862 (0.006)$\blacktriangle$	0.777 (0.004)$\triangle$	0.858 (0.006)$\triangle$	0.775 (0.004)$\blacktriangle$	0.854 (0.007)$\blacktriangle$	0.762 (0.007)	0.832 (0.011)

Appendix F

Oracle-Propensity Two-Tower Bias Results

For completeness, we also report the matched two-tower bias comparison under the oracle propensity baseline setting. The qualitative picture is the same as in the main chapter: the two-tower bias estimates remain competitive with the CSV baselines, and the oracle-propensity table does not change the main conclusions.

Table F.1: Comparison of feature-based propensity estimation under different position bias estimation methods for counterfactual LTR on MSLR-WEB30K, Yahoo!, and Istella-S. Rows are grouped by dataset, and columns are grouped first by counterfactual learning objective and then by sample count. Cells show mean (SD) NDCG. Bold marks the best non-oracle model within each comparison block. Significance markers compare each other model with this best model using a paired two-sided Student t-test: $p < 0.01$ ($\blacktriangle$ and $\blacktriangledown$), $p < 0.05$ ($\triangle$ and $\triangledown$).

Dataset	Bias Source	IPS		DM		DR	
		N = 10^4	N = 10^5	N = 10^4	N = 10^5	N = 10^4	N = 10^5
MSLR-WEB30K	True Bias ^{oracle}	0.420 (0.012)	0.496 (0.002)	0.468 (0.008)	0.469 (0.013)	0.461 (0.008)	0.463 (0.033)
	Add. Two-Tower Bias	0.411 (0.015)	0.498 (0.006)	0.423 (0.005) $\triangledown$	0.476 (0.007)	0.429 (0.010)	0.463 (0.022)
	Mul. Two-Tower Bias	0.401 (0.011)	0.445 (0.004) $\blacktriangledown$	0.412 (0.003) $\triangledown$	0.389 (0.061)	0.414 (0.003)	0.352 (0.015) $\triangledown$
	Adjacent Chain	0.415 (0.008) $\blacktriangledown$	0.448 (0.008) $\triangledown$	0.328 (0.008) $\blacktriangledown$	0.430 (0.005) $\triangledown$	0.356 (0.024) $\blacktriangledown$	0.356 (0.012) $\triangledown$
	CTR	0.427 (0.008)	0.498 (0.006)	0.453 (0.012)	0.476 (0.011)	0.445 (0.017)	0.448 (0.025) $\triangledown$
	Global All Pairs	0.383 (0.032)	0.490 (0.006)	0.443 (0.019)	0.474 (0.005)	0.412 (0.018) $\blacktriangledown$	0.467 (0.020)
	Pivot One	0.414 (0.007) $\triangledown$	0.487 (0.011)	0.328 (0.008) $\blacktriangledown$	0.471 (0.019)	0.384 (0.025)	0.433 (0.040)
Yahoo!	True Bias ^{oracle}	0.765 (0.003)	0.784 (0.002)	0.781 (0.003)	0.780 (0.002) $\blacktriangledown$	0.777 (0.004)	0.777 (0.004)
	Add. Two-Tower Bias	0.739 (0.006) $\blacktriangledown$	0.784 (0.003)	0.749 (0.004) $\blacktriangledown$	0.776 (0.002) $\blacktriangledown$	0.746 (0.005) $\triangledown$	0.771 (0.004)
	Mul. Two-Tower Bias	0.734 (0.002) $\blacktriangledown$	0.786 (0.008)	0.741 (0.001) $\blacktriangledown$	0.796 (0.002)	0.741 (0.004) $\triangledown$	0.718 (0.005) $\blacktriangledown$
	Adjacent Chain	0.736 (0.008) $\triangledown$	0.768 (0.001) $\blacktriangledown$	0.726 (0.010) $\blacktriangledown$	0.758 (0.003) $\blacktriangledown$	0.682 (0.006) $\blacktriangledown$	0.759 (0.001) $\blacktriangledown$
	CTR	0.770 (0.007)	0.789 (0.000)	0.778 (0.002)	0.786 (0.004)	0.776 (0.006)	0.770 (0.006)
	Global All Pairs	0.756 (0.019)	0.785 (0.002)	0.773 (0.005)	0.779 (0.003) $\blacktriangledown$	0.765 (0.007)	0.775 (0.002)
	Pivot One	0.727 (0.012) $\triangledown$	0.784 (0.002) $\triangledown$	0.738 (0.019)	0.780 (0.003) $\blacktriangledown$	0.706 (0.005) $\blacktriangledown$	0.775 (0.002)
Istella-S	True Bias ^{oracle}	0.685 (0.016)	0.751 (0.006)	0.719 (0.006)	0.759 (0.005)	0.723 (0.006)	0.762 (0.008)
	Add. Two-Tower Bias	0.662 (0.009)	0.749 (0.001)	0.709 (0.013)	0.763 (0.002)	0.719 (0.014)	0.722 (0.015) $\triangledown$
	Mul. Two-Tower Bias	0.638 (0.026)	0.707 (0.013) $\triangledown$	0.696 (0.028)	0.375 (0.255)	0.696 (0.036)	0.641 (0.010) $\blacktriangledown$
	Adjacent Chain	0.581 (0.034)	0.745 (0.014)	0.234 (0.016) $\blacktriangledown$	0.749 (0.012)	0.595 (0.002) $\blacktriangledown$	0.379 (0.086) $\triangledown$
	CTR	0.659 (0.012)	0.748 (0.001)	0.728 (0.012)	0.767 (0.003)	0.725 (0.005)	0.649 (0.021) $\blacktriangledown$
	Global All Pairs	0.630 (0.025)	0.748 (0.001)	0.690 (0.010)	0.760 (0.007)	0.680 (0.014)	0.759 (0.010)
	Pivot One	0.588 (0.022) $\triangledown$	0.747 (0.001) $\triangledown$	0.318 (0.130) $\triangledown$	0.755 (0.006) $\triangledown$	0.445 (0.042) $\blacktriangledown$	0.749 (0.009)

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